Solutions to Select Problems From *Goldstein. Classical Mechanics, 2nd ed.*

Roan Arendtsz

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Preface

This document is a compilation of exercises I have completed from the second edition of Herbert Goldstein's *Classical Mechanics*. There are several motivations behind this project. First, I was unable to take the advanced classical mechanics course at my undergraduate institution. Second, many of the mathematical tools developed in classical mechanics extend into other topics, such as quantum mechanics, and can be constructed with the most intuition in the classical case. Moreover, mechanics was my first formal introduction to physics and retains especially interesting to me.

The textbook was chosen due to its reputation as the canonical graduate classical mechanics textbook. Very little outside knowledge is assumed, which was particularly important when I first started this project in my second year of my undergraduate degree.

As of this writing, the project remains in progress.

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1 Survey of the Elemental Particles

1.1 Solutions to Select Problems

Problem 1.2

Problem Statement

The escape velocity of a particle on the earth is the minimum velocity required at the surface of the earth in order that the particle can escape from the earth's gravitational field. Neglecting the resistance of the atmosphere, the system is conservative. From the conservation theorem for potential plus kinetic energy show that the escape velocity for the earth. ignoring the presence of the moon, is 6.95 mi/sec.

The only force acting on the particle is F_g, which is conservative. The conservation theorem for potential plus kinetic energy thereby yields

$$T_1 + V_{g_1} = T_2 + V_{g_2} \tag{1-1}$$

for any two states 1 and 2. Defining $V_g = 0$ at $r = \infty$ with respect to the center of the Earth's gravitational field yields

$$V(r) = -\int_\infty^r F_g(r') \cdot \mathrm{d}\mathbf{r}' = \int_\infty^r \frac{GMm}{r'^2}\hat{\mathbf{r}} \cdot \mathrm{d}\mathbf{r}' = -\int_r^\infty \frac{GMm}{r'^2}\,\mathrm{d}r' = \frac{GMm}{r'}\bigg|_{r'=r}^{r'=\infty} = -\frac{GMm}{r}, \tag{1-2}$$

while

$$T(\dot{\boldsymbol{r}}) = \frac{1}{2}m\dot{\boldsymbol{r}} \cdot \dot{\boldsymbol{r}} = \frac{1}{2}m|\dot{\boldsymbol{r}}|^2. \tag{1-3}$$

To escape the gravitational field I must therefore have $r = \infty$. To achieve this in the minimal case, I therefore have $T_f = 0$. Thus,

$$\frac{1}{2}m|\dot{\boldsymbol{r}}|^2 - \frac{GMm}{r} = T_i + V_{g_i} = T_f + V_{g_f} = 0 + 0 = 0. \tag{1-4}$$

Hence,

$$\frac{1}{2}m|\dot{\boldsymbol{r}}|^2 = \frac{GMm}{r} \tag{1-5}$$

which implies

$$|\dot{\boldsymbol{r}}| = \sqrt{\frac{2GM}{R}} \tag{1-6}$$

$$= \sqrt{\frac{2 \cdot 6.67 \times 10^{-11}\,\mathrm{m^3\,kg^{-1}\,s^{-2}} \cdot 5.972 \times 10^{24}\,\mathrm{kg}}{6.371 \times 10^{6}\,\mathrm{m}}} \tag{1-7}$$

$$= 1.12 \times 10^{4}\,\mathrm{m\,s^{-1}} \cdot \frac{1\,\mathrm{mi}}{1609.34\,\mathrm{m}} \tag{1-8}$$

$$= 6.95\,\mathrm{mi\,s^{-1}}\,. \tag{1-9}$$

as desired.

Problem 1.4

Problem Statement

Show that for a single particle with constant mass the equation of motion implies the following differential equation for the kinetic energy:

$$\frac{\mathrm{d}T}{\mathrm{d}t} = \boldsymbol{F} \cdot \boldsymbol{v} \tag{1-10}$$

while if the mass varies with time the corresponding equation is

$$\frac{\mathrm{d}(mT)}{\mathrm{d}t} = \boldsymbol{F} \cdot \boldsymbol{p}. \tag{1-11}$$

In the general case,

$$T = \frac{1}{2} m \boldsymbol{v} \cdot \boldsymbol{v} \tag{1-12}$$

so

$$\begin{aligned} mT =& \frac{1}{2} m^2 \boldsymbol{v} \cdot \boldsymbol{v} && (1\text{-}13)\\ =& \frac{1}{2} (m\boldsymbol{v}) \cdot (m\boldsymbol{v}) && (1\text{-}14)\\ =& \frac{1}{2} \boldsymbol{p} \cdot \boldsymbol{p}. && (1\text{-}15) \end{aligned}$$

Taking the time derivative yields

$$\begin{aligned} \frac{\mathrm{d}(mT)}{\mathrm{d}t} =& \frac{\mathrm{d}}{\mathrm{d}t} \left(\frac{1}{2} \boldsymbol{p} \cdot \boldsymbol{p} \right) && (1\text{-}16)\\ =& \frac{1}{2} \frac{\mathrm{d}}{\mathrm{d}t} (\boldsymbol{p} \cdot \boldsymbol{p}) && (1\text{-}17)\\ =& \frac{1}{2} \cdot 2 \cdot \frac{\mathrm{d}\boldsymbol{p}}{\mathrm{d}t} \cdot \boldsymbol{p} && (1\text{-}18)\\ =& \boldsymbol{F} \cdot \boldsymbol{p}, && (1\text{-}19) \end{aligned}$$

as desired. In the case that m is constant,

$$\frac{\mathrm{d}T}{\mathrm{d}t} = \frac{\mathrm{d}(mT)}{\mathrm{d}t} \cdot \frac{1}{m} = \frac{1}{m} \boldsymbol{F} \cdot \boldsymbol{p} = \boldsymbol{F} \cdot \boldsymbol{v}. \tag{1-20}$$

Problem 1.5

Problem Statement

Prove that the magnitude R of the position vector for the center of mass from an arbitrary origin is given by the equation

$$M^2 R^2 = M \sum_i m_i r_i^2 - \frac{1}{2} \sum_{i,j} m_i m_j r_{ij}^2. \tag{1-21}$$

By definition,

$$M^2 R^2 = M\boldsymbol{R} \cdot M\boldsymbol{R} = \sum_i m_i \boldsymbol{r_i} \cdot \sum_i m_i \boldsymbol{r_i}. \tag{1-22}$$

I can expand by grouping terms to get

$$M^2R^2 = \sum_i m_i^2 r_i^2 + \sum_{i \neq j} m_i m_j \boldsymbol{r_i} \cdot \boldsymbol{r_j} \tag{1-23}$$

Substituting Equation 1-23 into Equation 1-21, I have

$$\begin{aligned} M^2R^2 - M\sum_i m_i r_i^2 + \frac{1}{2}\sum_{i,j} m_i m_j r_{ij}^2 = &\sum_i m_i^2 r_i^2 + \sum_{i \neq j} m_i m_j \boldsymbol{r}_i \cdot \boldsymbol{r}_j \\ &- M\sum_i m_i r_i^2 + \frac{1}{2}\sum_{i,j} m_i m_j r_{ij}^2. \end{aligned} \tag{1-24}$$

I can write

$$M\sum_i m_i r_i^2 = \sum_i m_i^2 r_i^2 + \frac{1}{2}\sum_{i \neq j} m_i m_j \left(r_i^2 + r_j^2\right) \tag{1-25}$$

and

$$\frac{1}{2}\sum_{i,j} m_i m_j r_{ij}^2 = \frac{1}{2}\sum_{i \neq j} m_i m_j r_{ij}^2 \tag{1-26}$$

owing to $r_{ii} = 0$ for all i. Therefore,

$$\begin{aligned} M^2R^2 - M\sum_i m_i r_i^2 + \frac{1}{2}\sum_{i,j} m_i m_j r_{ij}^2 = &\sum_{i \neq j} m_i m_j \boldsymbol{r}_i \cdot \boldsymbol{r}_j - \frac{1}{2}\sum_{i \neq j} m_i m_j \left(r_i^2 + r_j^2\right) \\ &+ \frac{1}{2}\sum_{i \neq j} m_i m_j r_{ij}^2 \end{aligned} \tag{1-27}$$

$$= \frac{1}{2}\sum_{i \neq j} m_i m_j \left(2\left(\boldsymbol{r}_i \cdot \boldsymbol{r}_j\right) - r_i^2 - r_j^2 + r_{ij}^2\right). \tag{1-28}$$

By the law of cosines,

$$r_{ij}^2 - r_i^2 - r_j^2 + 2r_i r_j \cos(\theta) = 0, \tag{1-29}$$

where θ is the angle formed between $\boldsymbol{r}_i$ and $\boldsymbol{r}_j$ (see Figure 1.1). But since

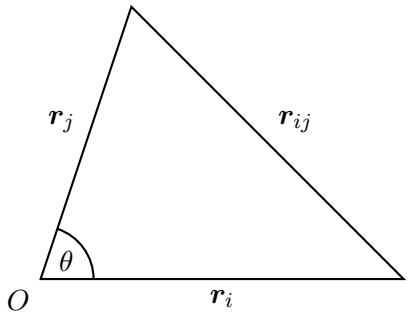


Figure 1: Geometric interpretation of the law of cosines with respect to the position vectors of interest.

$$\boldsymbol{r}_i \cdot \boldsymbol{r}_j = r_i r_j \cos(\theta), \tag{1-30}$$

I have

$$2\left(\boldsymbol{r}_i \cdot \boldsymbol{r}_j\right) - r_i^2 - r_j^2 + r_{ij}^2 = 2r_i r_j \cos(\theta) - r_i^2 - r_j^2 + r_{ij}^2 = 0 \tag{1-31}$$

by Equation 1-29. Substituting into Equation 1-27 thereby yields

$$M^2R^2 - M\sum_i m_i r_i^2 + \frac{1}{2}\sum_{i,j} m_i m_j r_{ij}^2 = 0 \tag{1-32}$$

and hence

$$M^2R^2 = M\sum_i m_i r_i^2 - \frac{1}{2}\sum_{i,j} m_i m_j r_{ij}^2, \tag{1-33}$$

as desired.

Problem 1.6

Problem Statement

Suppose a system of two particles is known to obey the equations of motion given by

$$M\frac{\mathrm{d}^2\boldsymbol{R}}{\mathrm{d}t^2} = \sum_i \boldsymbol{F}_i^{(e)} \equiv \boldsymbol{F}^{(e)} \tag{1-34}$$

and

$$\frac{\mathrm{d}\boldsymbol{L}}{\mathrm{d}t} = \boldsymbol{N}^{(e)}, \tag{1-35}$$

where e denotes that the vector of interest is external in nature. Then from the equations of the motion of the individual particles show that the internal forces between particles satisfy both the weak and the strong laws of action and reaction.

Consider particles 1 and 2. Separating internal and external contributions in their equation of motion yields

$$\dot{\boldsymbol{p}}_1 = \boldsymbol{F}_1^{(e)} + \boldsymbol{F}_{21}, \tag{1-36}$$

$$\dot{\boldsymbol{p}}_2 = \boldsymbol{F}_2^{(e)} + \boldsymbol{F}_{12}. \tag{1-37}$$

I can now re-write Equation 1-34 using the definition of $\boldsymbol{R}$ as

$$\begin{aligned}\boldsymbol{F}^{(e)} = M\frac{\mathrm{d}^2\boldsymbol{R}}{\mathrm{d}t^2} = \frac{\mathrm{d}^2}{\mathrm{d}t^2}\sum_i m_i\boldsymbol{r}_i = \frac{\mathrm{d}^2}{\mathrm{d}t^2}\left(m_1\boldsymbol{r}_1 + m_2\boldsymbol{r_2}\right) \\ = \dot{\boldsymbol{p}}_1 + \dot{\boldsymbol{p}}_2 = \boldsymbol{F}_1^{(e)} + \boldsymbol{F}_{21} + \boldsymbol{F}_2^{(e)} + \boldsymbol{F}_{12} = \boldsymbol{F}^{(e)} + \boldsymbol{F}_{21} + \boldsymbol{F}_{12},\end{aligned} \tag{1-38}$$

which implies

$$\boldsymbol{F}_{12} = -\boldsymbol{F}_{21}, \tag{1-39}$$

equivalent to the weak law of action and reaction (Definition **??**). Similarly,

$$\dot{\boldsymbol{L}}_1 = \boldsymbol{N}_1^{(e)} + \boldsymbol{N}_{21}, \tag{1-40}$$

$$\dot{\boldsymbol{L}}_2 = \boldsymbol{N}_2^{(e)} + \boldsymbol{N}_{12}. \tag{1-41}$$

Equation 1-35 thereby gives

$$\boldsymbol{N}^{(e)} = \frac{\mathrm{d}\boldsymbol{L}}{\mathrm{d}t} = \dot{\boldsymbol{L}}_1 + \dot{\boldsymbol{L}}_2 = \boldsymbol{N}_1^{(e)} + \boldsymbol{N}_{21} + \boldsymbol{N}_2^{(e)} + \boldsymbol{N}_{12} = \boldsymbol{N}^{(e)} + \boldsymbol{N}_{12} + \boldsymbol{N}_{21} \tag{1-42}$$

which implies

$$\boldsymbol{N}_{12} = \boldsymbol{r}_1 \times \boldsymbol{F}_{21} = -\boldsymbol{r}_2 \times \boldsymbol{F}_{12} = -\boldsymbol{N}_{21}. \tag{1-43}$$

Substituting Equation 1-39 yields

$$0 = (\boldsymbol{r}_1 \times \boldsymbol{F}_{21}) + (\boldsymbol{r}_2 \times \boldsymbol{F}_{12}) = (\boldsymbol{r}_1 \times \boldsymbol{F}_{21}) - (\boldsymbol{r}_2 \times \boldsymbol{F}_{21}) = \boldsymbol{r}_{21} \times \boldsymbol{F}_{21}. \tag{1-44}$$

Disallowing the possibility of both particles occupying the same physical space yields that any internal forces between the two particles must be parallel to the line joining the particles; since the force contact point is at $\boldsymbol{r}_1$, the force must therefore be directed along this line.

Problem 1.7

Problem Statement

The equations of constraint for the rolling disk on a horizontal plane are given by

$$\mathrm{d}x - a\sin(\theta)\,\mathrm{d}\phi = 0, \tag{1-45}$$
$$\mathrm{d}y - a\cos(\theta)\,\mathrm{d}\phi = 0, \tag{1-46}$$

where a is the radius of the disk, (x, y) denotes the spatial projection of the center of the disk on the plane, ϕ denotes the angle of rotation about the axis of the disk, and θ denotes the angle between the axis of the disk and the x-axis. These equations are special cases of general linear differential equations of the form

$$\sum_{i=1}^{n} g_i(x_1, \ldots, x_n)\,\mathrm{d}x_i = 0. \tag{1-47}$$

A constraint condition of this type is holonomic (see Definition **??**) only if an integrating function $f(x_1, \ldots, x_n)$ can be found that turns it into an exact differential. The function must be such that

$$\frac{\partial(f g_i)}{\partial x_j} = \frac{\partial(f g_j)}{\partial x_i} \tag{1-48}$$

for all $i \neq j$. Show that no such integrating factor can be found for Equations 1-45 or 1-46.

Consider Equation 1-45. The relevant g_i are given by

$$\begin{aligned} g_x &= 1, \\ g_y &= 0, \\ g_\theta &= 0, \\ g_\phi &= -a\sin(\theta). \end{aligned}$$

To satisfy Equation 1-48 I must therefore have

$$\frac{\partial(f g_x)}{\partial y} = \frac{\partial f}{\partial y} = 0 = \frac{\partial(f \cdot 0)}{\partial x} = \frac{\partial(f g_y)}{\partial x}, \tag{1-49}$$
$$\frac{\partial(f g_x)}{\partial \phi} = \frac{\partial f}{\partial \phi} = 0 = \frac{\partial(f \cdot 0)}{\partial x} = \frac{\partial(f g_\phi)}{\partial x}, \tag{1-50}$$

which implies that any integrating function f is given by $f := f(x, \phi)$. Therefore, the only relevant pair of differentials yields

$$\frac{\partial(fg_x)}{\partial\phi} = \frac{\partial f}{\partial x} = \frac{\partial(f \cdot -a\sin(\theta))}{\partial x} = \frac{\partial(fg_\phi)}{\partial x}. \tag{1-51}$$

Since g_ϕ is not dependent on x, the relevant equation is therefore

$$\frac{\partial f}{\partial \phi} = -a\sin(\theta)\frac{\partial f}{\partial x}. \tag{1-52}$$

Since f is not dependent on θ, the left-hand side of Equation 1-52 cannot have any θ dependence to match the right-hand side. Therefore, such a function f cannot exist and the constraint given by Equation 1-45 cannot be holonomic in nature. Turning to Equation 1-46, the relevant g_i are similarly given by

$$\begin{aligned} g_x &= 0, \\ g_y &= 1, \\ g_\theta &= 0, \\ g_\phi &= -a\cos(\theta). \end{aligned}$$

By similarly leveraging $g_x = 0 = g_\theta$, it must be the case that any integrating function f is given by $f := f(y, \phi)$. The relevant equation gives

$$\frac{\partial f}{\partial \phi} = -a\sin(\theta)\frac{\partial f}{\partial y}, \tag{1-53}$$

which again shows that f cannot exist due to the $\cos(\theta)$ discrepancy.

Problem 1.8

Problem Statement

Two wheels of radius a are mounted on the ends of a common axle of length b such that the wheels rotate independently. The whole combination rolls without slipping on a plane. Show that there are two nonholonomic equations of constraint

$$\cos(\theta)\,\mathrm{d}x + \sin(\theta)\,\mathrm{d}y = 0, \tag{1-54}$$
$$\sin(\theta)\,\mathrm{d}x - \cos(\theta)\,\mathrm{d}y = a(\mathrm{d}\phi_1 + \mathrm{d}\phi_2), \tag{1-55}$$

and one holonomic equation of constraint

$$\theta = C - \frac{a}{b}(\phi_1 - \phi_2), \tag{1-56}$$

where (x, y) denotes the spatial projection of the coordinates of the point on the axle midway between the wheels, ϕ_1 and ϕ_2 denote the angle of rotation about the axis of the two disks, respectively, θ denotes the angle formed between the axis of the axle and the x-axis, and C is a constant. [a]

[a] I have chosen to substitute Goldstein's notation of ϕ and ϕ' for the labeling of the individual wheels to ϕ_1 and ϕ_2 for clarity in distinguishing between the wheels and the center.

I denote the positional centers of each wheel by $\boldsymbol{r}_1 := \langle x_1, y_1\rangle$ and $\boldsymbol{r}_2 := \langle x_2, y_2\rangle$, respectively. The center of the wheel, $\boldsymbol{r}$, is therefore defined by

$$\boldsymbol{r} := \langle x, y\rangle = \frac{\boldsymbol{r}_1 + \boldsymbol{r}_2}{2} = \left\langle \frac{x_1 + x_2}{2}, \frac{y_1 + y_2}{2} \right\rangle = \frac{1}{2}\langle x_1 + x_2, y_1 + y_2\rangle. \tag{1-57}$$

The differential motion of the wheel can be described by its velocity vector, which depends on the magnitude of the linear velocity v of the wheel and its angle θ with respect to the x-axis. Since the

rotations of the two wheels are independent, there are separate equations to dictate the motions of the individual wheels. For wheel 1, these equations are

$$v_1 = a\dot{\phi}_1, \tag{1-58}$$
$$\dot{x}_1 = v_1 \sin(\theta), \tag{1-59}$$
$$\dot{y}_1 = -v_1 \cos(\theta), \tag{1-60}$$

where the discrepancy in signs results from the choice of the definition of θ. Similarly, for wheel 2

$$v_2 = a\dot{\phi}_2, \tag{1-61}$$
$$\dot{x}_2 = v_2 \sin(\theta), \tag{1-62}$$
$$\dot{y}_2 = -v_2 \cos(\theta). \tag{1-63}$$

Substituting Equations 1-58 and 1-61 and explicitly writing the time derivatives gives

$$\frac{\mathrm{d}x_1}{\mathrm{d}t} = a\frac{\mathrm{d}\phi_1}{\mathrm{d}t}\sin(\theta), \tag{1-64}$$
$$\frac{\mathrm{d}y_1}{\mathrm{d}t} = -a\frac{\mathrm{d}\phi_1}{\mathrm{d}t}\cos(\theta), \tag{1-65}$$
$$\frac{\mathrm{d}x_2}{\mathrm{d}t} = a\frac{\mathrm{d}\phi_2}{\mathrm{d}t}\sin(\theta), \tag{1-66}$$
$$\frac{\mathrm{d}y_2}{\mathrm{d}t} = -a\frac{\mathrm{d}\phi_2}{\mathrm{d}t}\cos(\theta). \tag{1-67}$$

Since the variables of interest within the time derivatives are inherently differentiable with respect to time, and continuity is assumed, I can apply the useful differential identity whose form for x_1, for instance, is given by

$$\mathrm{d}x_1 = \frac{\mathrm{d}x_1}{\mathrm{d}t}\,\mathrm{d}t. \tag{1-68}$$

Dividing out the time differential $\mathrm{d}t$ (which is non-zero) on both sides gives

$$\mathrm{d}x_1 = a\,\mathrm{d}\phi_1 \sin(\theta), \tag{1-69}$$
$$\mathrm{d}y_1 = -a\,\mathrm{d}\phi_1 \cos(\theta), \tag{1-70}$$
$$\mathrm{d}x_2 = a\,\mathrm{d}\phi_2 \sin(\theta), \tag{1-71}$$
$$\mathrm{d}y_2 = -a\,\mathrm{d}\phi_2 \cos(\theta). \tag{1-72}$$

Since by Equation 1-57 I have $x = \frac{1}{2}(x_1 + x_2)$ and $y = \frac{1}{2}(y_1 + y_2)$, the corresponding differentials are

$$\mathrm{d}x = \frac{1}{2}\left(\mathrm{d}x_1 + \mathrm{d}x_2\right) = \frac{a\sin(\theta)}{2}\left(\mathrm{d}\phi_1 + \mathrm{d}\phi_2\right), \tag{1-73}$$
$$\mathrm{d}y = \frac{1}{2}\left(\mathrm{d}y_1 + \mathrm{d}y_2\right) = -\frac{a\cos(\theta)}{2}\left(\mathrm{d}\phi_1 + \mathrm{d}\phi_2\right). \tag{1-74}$$

Solving for $\frac{a}{2}\left(\mathrm{d}\phi_1 + \mathrm{d}\phi_2\right)$ gives

$$\frac{a}{2}\left(\mathrm{d}\phi_1 + \mathrm{d}\phi_2\right) = \frac{\mathrm{d}x}{\sin(\theta)} = \frac{-\,\mathrm{d}y}{\cos(\theta)} \tag{1-75}$$

and thus

$$\cos(\theta)\,\mathrm{d}x + \sin(\theta)\,\mathrm{d}y = 0, \tag{1-76}$$

as with Equation 1-54. It can be readily shown through the methods developed in Problem 1.1 that since the only differentials involved are $\mathrm{d}x$ and $\mathrm{d}y$, any integrating function f must be defined as $f := f(x, y)$. Applying the condition imposed by Equation 1-48 yields that

$$\frac{\partial(fg_x)}{\partial y} = \frac{\partial(f\cos(\theta))}{\partial y} = \cos(\theta)\frac{\partial f}{\partial y} = \sin(\theta)\frac{\partial f}{\partial x} = \frac{\partial(f\sin(\theta))}{\partial x} = \frac{\partial(fg_y)}{\partial x}. \tag{1-77}$$

However, since $f = f(x, y)$, there can be no θ dependence in the partial derivatives and thus the disagreement between $\cos(\theta)$ and $\sin(\theta)$ on the left- and right-hand sides cannot be resolved. Thus, f cannot exist and the constraint is nonholonomic. It also follows from Equations 1-73 and 1-74 that

$$\begin{aligned}\sin(\theta)\,\mathrm{d}x - \cos(\theta)\,\mathrm{d}y &= \sin(\theta)\frac{a\sin(\theta)}{2}\left(\mathrm{d}\phi_1 + \mathrm{d}\phi_2\right) + \cos(\theta)\frac{a\cos(\theta)}{2}\left(\mathrm{d}\phi_1 + \mathrm{d}\phi_2\right) \\ &= \frac{a}{2}\left(\mathrm{d}\phi_1 + \mathrm{d}\phi_2\right),\end{aligned} \tag{1-78}$$

which is in agreement with Equation 1-55 up to the scalar multiple of $1/2$. [1]. I can now similarly show that this constraint is non-holonomic by showing that an integrating function f is given by $f := f(x, y, \phi_1, \phi_2)$ (crucially with no θ dependence) and thus show that the condition

$$\frac{\partial(fg_x)}{\partial y} = \frac{\partial(f\sin(\theta))}{\partial y} = \sin(\theta)\frac{\partial f}{\partial y} = -\cos(\theta)\frac{\partial f}{\partial x} = \frac{\partial(f\cdot -\sin(\theta))}{\partial x} = \frac{\partial(fg_y)}{\partial x}. \tag{1-79}$$

admits no such function f. I have now produced the two velocity constraints for the system produced by the rolling without slipping condition; we also have to consider the fixed geometry of the axle. The equations of constraint thus far have not involved differentials with respect to θ. Changes in θ result from differences in the differential distance traveled between the two wheels. To account for the rigidity of the axle, the whole system must turn, therefore imparting a $\mathrm{d}\theta$. The mathematical description of this change is given by

$$\tan(\mathrm{d}\theta) = \frac{|\,\mathrm{d}\boldsymbol{r}_1| - |\,\mathrm{d}\boldsymbol{r}_2|}{b} = \frac{v_1\,\mathrm{d}t - v_2\,\mathrm{d}t}{b} = \frac{a}{b}\left(\frac{\mathrm{d}\phi_1}{\mathrm{d}t}\,\mathrm{d}t - \frac{\mathrm{d}\phi_2}{\mathrm{d}t}\,\mathrm{d}t\right) = \frac{a}{b}\left(\mathrm{d}\phi_1 - \mathrm{d}\phi_2\right) \tag{1-80}$$

via Equation 1-68 and is depicted geometrically in Figure 1.1. Note that the absolute value is used to denote the fact that we care only about velocity perpendicular to the axle axis, which in the differential case is exactly the same as the magnitude of the velocity. Note also that since we are dealing with differentials, we can treat this path length difference observed between the two wheels as an arc length such that $\tan(\mathrm{d}\theta) = \mathrm{d}\theta$.[2] Substituting into Equation 1-80 thereby gives

$$\mathrm{d}\theta - \frac{a}{b}\,\mathrm{d}\phi_1 + \frac{a}{b}\,\mathrm{d}\phi_2 = 0. \tag{1-81}$$

Consider the function $f = 1$. All of the partials in Equation 1-48 vanish so that f is an integrating factor. Therefore, there exists a function $F := F(\theta, \phi_1, \phi_2)$ such that

$$\mathrm{d}F = f\left(\mathrm{d}\theta - \frac{a}{b}\,\mathrm{d}\phi_1 + \frac{a}{b}\,\mathrm{d}\phi_2\right) = \mathrm{d}\theta - \frac{a}{b}\,\mathrm{d}\phi_1 + \frac{a}{b}\,\mathrm{d}\phi_2 = 0, \tag{1-82}$$

is an exact differential. I can construct F through a series of partial integrations. Since $\mathrm{d}\theta$ has a coefficient of 1, it follows from the definition of a differential that $\frac{\partial F}{\partial \theta} = 1$, $\frac{\partial F}{\partial \phi_1} = -\frac{a}{b}$ and , $\frac{\partial F}{\partial \phi_2} = \frac{a}{b}$. Integrating with respect to θ yields

[1] It seems that Goldstein has adopted some generalized coordinate to absorb this multiplicative constant which requires re-defining some variables.

[2] For those not satisfied with my mathematical rigor here, my knowledge of differentials and differential equations is not sufficient to provide a more thorough explanation of the final arc length argument. I approached this result through the small-angle approximation, which I believe is employed in differentials in calculus of multiple variables, for instance. However, it may simply be the case that the differential is "curvy" in nature – note that if the velocities are constant and different from each other, the path will trace out a perfect circle – and hence that no such approximation is necessary. I would welcome any insight here :)

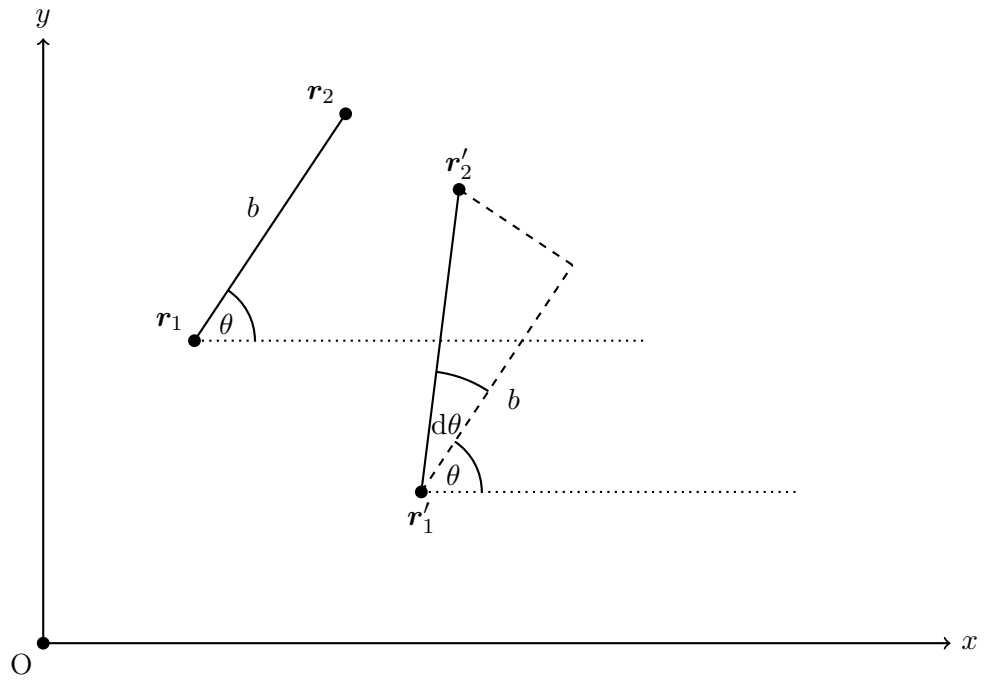


Figure 2: Geometrical interpretation of the differential change in θ in Problem 1.8. The $\boldsymbol{r}_1$ and $\boldsymbol{r}_2$ vectors denote the positions of the two wheels at an instant of time and the $\boldsymbol{r}_1'$ and $\boldsymbol{r}_2'$ vectors denote their positions after some differential time. Discrepancies in velocities of the two wheels impart a differential $\mathrm{d}\theta$ to the system.

$$F = \int \left(\frac{\partial F}{\partial \theta} \, \mathrm{d}\theta \right) = \int \mathrm{d}\theta = \theta + \gamma(\phi_1, \phi_2), \tag{1-83}$$

where $\gamma(\phi_1, \phi_2$ is some function of ϕ_1 and ϕ_2. Now since $\frac{\partial \gamma}{\partial \phi_1} = \frac{\partial F}{\partial \phi_1} = -\frac{a}{b}$, it follows that

$$\gamma = \int \left(\frac{\partial \gamma}{\partial \phi_1} \, \mathrm{d}\phi_1 \right) = \int \left(\frac{-a}{b} \, \mathrm{d}\phi_1 \right) = -\frac{a}{b}\phi_1 + h(\phi_2), \tag{1-84}$$

where h is some function of ϕ_2. Finally, since $\frac{\mathrm{d}h}{\mathrm{d}\phi_2} = \frac{\partial \gamma}{\partial \phi_2} = \frac{\partial F}{\partial \phi_2} = \frac{a}{b}$, it follows that

$$h = \int \left(\frac{\mathrm{d}h}{\mathrm{d}\phi_2} \, \mathrm{d}\phi_2 \right) = \int \frac{a}{b} \, \mathrm{d}\phi_2 = \frac{a}{b}\phi_2 + C_1, \tag{1-85}$$

where C_1 is a constant. Finally, since $\mathrm{d}F = 0$, it must be that $F = C_2$, where C_2 is a constant. Therefore, I have

$$F = \theta + \gamma(\phi_1, \phi_2) = \theta - \frac{a}{b}\phi_1 + h(\phi_2) = \theta - \frac{a}{b}\phi_1 + \frac{a}{b}\phi_2 + C_1 = C_2, \tag{1-86}$$

so that

$$\theta = C - \frac{a}{b}(\phi_2 - \phi_1), \tag{1-87}$$

where $C := C_2 - C_1$. Note that this result differs marginally from Equation 1-56 – this difference results from a mismatch in labeling of wheels 1 and 2 and is not an essential detail.

Problem 1.9 *

Problem Statement

A particle moves in the x-y plane under the constraint that its velocity vector is always directed towards a point on the x-axis whose abscissa (see Definition **??**) is some given function of time $f(t)$. Show that for differentiable $f(t)$, but otherwise arbitrary, the constraint is nonholonomic.

Consider an instant of time t in which the anchor abscissa points is at $(f(t), 0)$ and the particle is at (x, y). I can define an angle θ formed with the x-axis, the anchor point, and the position of the particle such that counter-clockwise is positive. Given some velocity v, the differentials are given by

$$\mathrm{d}x = v\cos(\theta), \tag{1-88}$$
$$\mathrm{d}y = v\sin(\theta). \tag{1-89}$$

I thus have from the geometry of the constraint that

$$\frac{\mathrm{d}x}{\mathrm{d}y} = \tan(\theta) = \frac{y}{x - f(t)}. \tag{1-90}$$

The corresponding equation is

$$y\,\mathrm{d}x + (f(t) - x)\,\mathrm{d}y = 0. \tag{1-91}$$

with coefficients

$$g_x = y,$$
$$g_y = f(t) - x,$$
$$g_t = 0,$$

as in Equation 1-47 where I have explicitly wrote out the time variable t. Since $g_t = 0$, I can conclude as in Problems 1.1 and 1.1 that an integrating function h must be defined by $h := h(x, y)$. The condition imposed by Equation 1-48 yields

$$0 = \frac{\partial(hg_t)}{\partial y} = \frac{\partial(hg_y)}{\partial t} = \frac{\partial\left(hf(t) - hx\right)}{\partial t} = \frac{\partial\left(h(x,y)f(t)\right)}{\partial t} - \frac{\partial\left(h(x,y)x\right)}{\partial t} = h(x,y)\frac{\partial f}{\partial t}. \tag{1-92}$$

Since $h = 0$ permits no constraint, it must be that $f = C$, where C is a constant. Applying Equation 1-48 with x and y yields

$$\frac{\partial\left(hy\right)}{\partial y} = \frac{\partial\left(h(C - x)\right)}{\partial x} \tag{1-93}$$

which evaluates to

$$y\frac{\partial h}{\partial y} + h = C\frac{\partial h}{\partial x} - h - x\frac{\partial h}{\partial x} \tag{1-94}$$

and thus that

$$2h + y\frac{\partial h}{\partial y} + (x - C)\frac{\partial h}{\partial x} = 0. \tag{1-95}$$

It is at this point that I do not know how to proceed but it seems likely that such an h cannot exist ... One can also approach this from the standard definition of a holonomic constraint (see Definition **??**). I have concluded that the anchor point must be static – the particle then can only move in a straight line towards the anchor point. While the velocity of the particle is not defined in this problem, and can

depend on time, it seems the case that it must continuously move towards that point (i.e. $v \neq 0$). Since there is only one position vector, and the time variable, I must find an equation $k(x, y, t) = 0$ which applies to the system to establish the constraint as holonomic. I still cannot provide an explanation for why k cannot exist, however.

Problem 1.10

Problem Statement

Two points of mass m are joined by a rigid weightless rod of length l, the center of which is constrained to move on a circle of radius a. Set up the kinetic energy in generalized coordinates.

I define the angle θ to be that formed by the line connecting particle 2 to particle 1 with the x-axis; I add π radians to the angle if particle 2 is lower than particle 1, by convention. I moreover define ϕ as the angle formed by line connecting the center of the circle of radius a (center (x_a, y_a) with the center of the rod, and the horizontal line running through the circle of radius a. The system is fully describable by these two variables and thus there are two degrees of freedom, implying two holonomic constraints on the four-dimensional phase space (x_1, x_2, y_1, y_2). The relevant set of equations is given by

$$x_1 = x_a + a\cos(\phi) - \frac{l}{2}\cos(\theta), \tag{1-96}$$

$$y_1 = y_a + a\sin(\phi) - \frac{l}{2}\sin(\theta), \tag{1-97}$$

$$x_2 = x_a + a\cos(\phi) + \frac{l}{2}\cos(\theta), \tag{1-98}$$

$$y_2 = y_a + a\sin(\phi) + \frac{l}{2}\sin(\theta), \tag{1-99}$$

from which the two holonomic equations of constraint

$$(x_1 - x_2)^2 + (y_1 - y_2)^2 = 2l^2, \tag{1-100}$$

$$\left(\frac{x_1 + x_2}{2} - x_a\right)^2 + \left(\frac{y_1 + y_2}{2} - y_a\right)^2 = a^2, \tag{1-101}$$

can be obtained. Note that I do not explicitly derive Equations 1-96–1-99 from the constrains defined in Equations 1-100 and 1-101 through a definition of my generalized coordinates θ and ϕ, but this in principle could be done. The kinetic energy in general is defined by

$$T = T_1 + T_2 = \frac{m}{2}(\dot{x}_1^2 + \dot{y}_1^2) + \frac{m}{2}(\dot{x}_2^2 + \dot{y}_2^2) = \frac{m}{2}(\dot{x}_1^2 + \dot{y}_1^2 + \dot{x}_2^2 + \dot{y}_2^2). \tag{1-102}$$

Computing the derivatives gives

$$\begin{aligned} T = \frac{m}{2}\Bigg(& a^2\dot{\phi}^2\sin^2(\phi) - al\dot{\phi}\dot{\theta}\sin(\phi)\sin(\theta) + \frac{l^2}{4}\dot{\phi}^2\sin^2(\theta) \\ & + a^2\dot{\phi}^2\cos^2(\phi) - al\dot{\phi}\dot{\theta}\cos(\phi)\cos(\theta) + \frac{l^2}{4}\dot{\phi}^2\cos^2(\theta) \\ & + a^2\dot{\phi}^2\sin^2(\phi) + al\dot{\phi}\dot{\theta}\sin(\phi)\sin(\theta) + \frac{l^2}{4}\dot{\phi}^2\sin^2(\theta) \\ & + a^2\dot{\phi}^2\cos^2(\phi) + al\dot{\phi}\dot{\theta}\cos(\phi)\cos(\theta) + \frac{l^2}{4}\dot{\phi}^2\cos^2(\theta)\Bigg) \\ = m\Bigg(& a^2\dot{\phi}^2 + \frac{l^2}{4}\dot{\theta}^2\Bigg), \end{aligned} \tag{1-103}$$

as desired.

Problem 1.11

Problem Statement

Show that Lagrange's Equations as defined in Equation **??** can be rewritten as

$$\frac{\partial \dot{T}}{\partial \dot{q_j}} - 2\frac{\partial T}{\partial q_j} = Q_j. \tag{1-104}$$

I aim to show that

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{q_j}}\right) = \frac{\partial \dot{T}}{\partial \dot{q_j}} - \frac{\partial T}{\partial q_j}. \tag{1-105}$$

Note that since $T := T(q_1, \dot{q_1}, q_2, \dot{q_2}, \ldots, q_n, \dot{q_n})$, for a specified q_j the partial $\frac{\partial T}{\partial \dot{q_j}}$ is given by

$$\frac{\partial T}{\partial \dot{q_j}} := \frac{\partial T}{\partial \dot{q_j}}\left(q_1, \dot{q_1}, q_2, \dot{q_2}, \ldots, q_n, \dot{q_n}\right).$$

Evaluating the time derivative requires abiding by the normal rules of partial differentiation; each phase variable is potentially dependent on time, so

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{q_j}}\right) = \frac{\partial}{\partial q_1}\left(\frac{\partial T}{\partial \dot{q_j}}\right)\frac{\mathrm{d}q_1}{\mathrm{d}t} + \frac{\partial}{\partial \dot{q_1}}\left(\frac{\partial T}{\partial \dot{q_j}}\right)\frac{\mathrm{d}\dot{q_1}}{\mathrm{d}t} + \cdots + \frac{\partial}{\partial q_j}\left(\frac{\partial T}{\partial \dot{q_j}}\right)\frac{\mathrm{d}q_j}{\mathrm{d}t} + \frac{\partial}{\partial \dot{q_j}}\left(\frac{\partial T}{\partial \dot{q_j}}\right)\frac{\mathrm{d}\dot{q_j}}{\mathrm{d}t} + \ldots \tag{1-106}$$

$$+ \frac{\partial}{\partial q_n}\left(\frac{\partial T}{\partial \dot{q_j}}\right)\frac{\mathrm{d}q_n}{\mathrm{d}t} + \frac{\partial}{\partial \dot{q_n}}\left(\frac{\partial T}{\partial \dot{q_j}}\right)\frac{\mathrm{d}\dot{q_n}}{\mathrm{d}t}$$

$$= \frac{\partial^2 T}{\partial q_1 \partial \dot{q_j}}\dot{q_1} + \frac{\partial^2 T}{\partial \dot{q_1}\partial \dot{q_j}}\ddot{q_1} + \cdots + \frac{\partial^2 T}{\partial q_j \partial \dot{q_j}}\dot{q_j} + \frac{\partial^2 T}{\partial \dot{q_j}\partial \dot{q_j}}\ddot{q_j} + \cdots + \frac{\partial^2 T}{\partial q_n \partial \dot{q_j}}\dot{q_n} + \frac{\partial^2 T}{\partial \dot{q_n}\partial \dot{q_j}}\ddot{q_n} \tag{1-107}$$

$$= \sum_{i\neq j}^{n}\frac{\partial^2 T}{\partial q_i \partial \dot{q_j}}\dot{q_i} + \sum_{i\neq j}^{n}\frac{\partial^2 T}{\partial \dot{q_i}\partial \dot{q_j}}\ddot{q_i} + \frac{\partial^2 T}{\partial q_j \partial \dot{q_j}}\dot{q_j} + \frac{\partial^2 T}{\partial \dot{q_j}\partial \dot{q_j}}\ddot{q_j}. \tag{1-108}$$

On the other side, I have

$$\frac{\partial \dot{T}}{\partial \dot{q_j}} = \frac{\partial}{\partial \dot{q_j}}\left(\frac{\mathrm{d}T}{\mathrm{d}t}\right) = \frac{\partial}{\partial \dot{q_j}}\left(\frac{\partial T}{\partial q_1}\frac{\mathrm{d}q_1}{\mathrm{d}t} + \frac{\partial T}{\partial \dot{q_1}}\frac{\mathrm{d}\dot{q_1}}{\mathrm{d}t} + \cdots + \frac{\partial T}{\partial q_j}\frac{\mathrm{d}q_j}{\mathrm{d}t} + \frac{\partial T}{\partial \dot{q_j}}\frac{\mathrm{d}\dot{q_j}}{\mathrm{d}t} + \cdots + \frac{\partial T}{\partial q_n}\frac{\mathrm{d}q_n}{\mathrm{d}t} + \frac{\partial T}{\partial \dot{q_n}}\frac{\mathrm{d}\dot{q_n}}{\mathrm{d}t}\right) \tag{1-109}$$

$$= \frac{\partial}{\partial \dot{q_j}}\left(\frac{\partial T}{\partial q_1}\dot{q_1} + \frac{\partial T}{\partial \dot{q_1}}\ddot{q_1} + \cdots + \frac{\partial T}{\partial q_j}\dot{q_j} + \frac{\partial T}{\partial \dot{q_j}}\ddot{q_j} + \cdots + \frac{\partial T}{\partial q_n}\dot{q_n} + \frac{\partial T}{\partial \dot{q_n}}\ddot{q_n}\right) \tag{1-110}$$

$$= \frac{\partial^2 T}{\partial \dot{q_j}\partial q_1}\dot{q_1} + \frac{\partial^2 T}{\partial \dot{q_j}\partial \dot{q_1}}\ddot{q_1} + \cdots + \frac{\partial^2 T}{\partial \dot{q_j}\partial q_j}\dot{q_j} + \frac{\partial T}{\partial q_j}\frac{\partial \dot{q_j}}{\partial \dot{q_j}} + \frac{\partial^2 T}{\partial \dot{q_j}\partial \dot{q_j}}\ddot{q_j} + \cdots + \frac{\partial^2 T}{\partial \dot{q_j}\partial q_n}\dot{q_n} + \frac{\partial^2 T}{\partial \dot{q_j}\partial \dot{q_n}}\ddot{q_n} \tag{1-111}$$

$$= \sum_{i\neq j}^{n}\frac{\partial^2 T}{\partial q_i \partial \dot{q_j}}\dot{q_i} + \sum_{i\neq j}^{n}\frac{\partial^2 T}{\partial \dot{q_i}\partial \dot{q_j}}\ddot{q_i} + \frac{\partial^2 T}{\partial q_j \partial \dot{q_j}}\dot{q_j} + \frac{\partial T}{\partial q_j} + \frac{\partial^2 T}{\partial \dot{q_j}\partial \dot{q_j}}\ddot{q_j}, \tag{1-112}$$

where the swapping of the second-order partial derivatives from Equation 1-111 to Equation 1-112 follows from the independence of $(q_1, \dot{q_1}, \ldots q_n, \dot{q_n})$. Therefore,

$$\frac{\partial \dot{T}}{\partial \dot{q_j}} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{q_j}}\right) + \frac{\partial T}{\partial q_j}, \tag{1-113}$$

as with Equation 1-105, and so

$$\frac{\partial \dot{T}}{\partial \dot{q}_j} - 2\frac{\partial T}{\partial q_j} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{q}_j}\right) + \frac{\partial T}{\partial q_j} - 2\frac{\partial T}{\partial q_j} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{q}_j}\right) - \frac{\partial T}{\partial q_j} = Q_j \tag{1-114}$$

by Equation **??**, as desired.

Problem 1.12

Problem Statement

A point particle moves in space under the influence of a force derivable from a generalized potential U of the form

$$U(\boldsymbol{r}, \boldsymbol{v}) = V(r) + \boldsymbol{\sigma} \cdot \boldsymbol{L}, \tag{1-115}$$

where $\boldsymbol{r}$ is the radius vector from a fixed point, $\boldsymbol{L}$ is the angular momentum about that point, and $\boldsymbol{\sigma}$ is a fixed vector in space.

(a) Find the components of the force on the particle in both Cartesian and spherical polar coordinates, on the basis of the following equation:

$$Q_j = -\frac{\partial U}{\partial q_j} + \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial U}{\partial \dot{q}_j}\right), \tag{1-116}$$

where U is the generalized potential.

(b) Show that the components in the two coordinate systems are related to each other as in Equation **??**.

(c) Obtain the equations of motion in spherical polar coordinates.

(a) Using the definition of $\boldsymbol{L}$, the generalized potential can be rewritten as

$$\begin{aligned} U(x, y, z, \dot{x}, \dot{y}, \dot{z}) &= V\left(\sqrt{x^2 + y^2 + z^2}\right) + \langle \sigma_x, \sigma_y, \sigma_z \rangle \cdot (\langle x, y, z \rangle \times \langle m\dot{x}, m\dot{y}, m\dot{z} \rangle) \\ &= V\left(\sqrt{x^2 + y^2 + z^2}\right) + m\langle \sigma_x, \sigma_y, \sigma_z \rangle \cdot \langle y\dot{z} - z\dot{y}, z\dot{x} - x\dot{z}, x\dot{y} - y\dot{x} \rangle \\ &= V\left(\sqrt{x^2 + y^2 + z^2}\right) + m\left(\sigma_x(y\dot{z} - z\dot{y}) + \sigma_y(z\dot{x} - x\dot{z}) + \sigma_z(x\dot{y} - y\dot{x})\right). \end{aligned} \tag{1-117}$$

The corresponding generalized forces (and hence the standard notion of force from Newtonian mechanics) can therefore be derived using Equation 1-116 as

$$F_x = Q_x = -\frac{\mathrm{d}V}{\mathrm{d}r}\frac{\partial r}{\partial x} + m\sigma_y\dot{z} - m\sigma_z\dot{y} + \frac{\mathrm{d}}{\mathrm{d}t}(m\sigma_y z - m\sigma_z y) = -\frac{x}{\sqrt{x^2 + y^2 + z^2}}\frac{\mathrm{d}V}{\mathrm{d}r} + 2m(\sigma_y\dot{z} - \sigma_z\dot{y}), \tag{1-118}$$

$$F_y = Q_y = -\frac{\mathrm{d}V}{\mathrm{d}r}\frac{\partial r}{\partial y} + m\sigma_z\dot{x} - m\sigma_x\dot{z} + \frac{\mathrm{d}}{\mathrm{d}t}(m\sigma_z x - m\sigma_x z) = -\frac{y}{\sqrt{x^2 + y^2 + z^2}}\frac{\mathrm{d}V}{\mathrm{d}r} + 2m(\sigma_z\dot{x} - \sigma_x\dot{z}), \tag{1-119}$$

$$F_z = Q_z = -\frac{\mathrm{d}V}{\mathrm{d}r}\frac{\partial r}{\partial z} + m\sigma_x\dot{y} - m\sigma_y\dot{x} + \frac{\mathrm{d}}{\mathrm{d}t}(m\sigma_x y - m\sigma_y x) = -\frac{z}{\sqrt{x^2 + y^2 + z^2}}\frac{\mathrm{d}V}{\mathrm{d}r} + 2m(\sigma_x\dot{y} - \sigma_y\dot{x}). \tag{1-120}$$

Note that it is physically required for σ to have units of t^{-1} in order for the units of the generalized potential to have units of energy and thus the generalized force to have units of force. In the spherical polar case, I compute

$$\boldsymbol{L} = \boldsymbol{r} \times \boldsymbol{p} = m(\boldsymbol{r} \times \boldsymbol{v}) = m\left(\boldsymbol{r} \times \frac{\mathrm{d}\boldsymbol{r}}{\mathrm{d}t}\right) = mr^2\left(\hat{\boldsymbol{r}} \times \frac{\mathrm{d}\hat{\boldsymbol{r}}}{\mathrm{d}t}\right). \tag{1-121}$$

To derive $\frac{\mathrm{d}\hat{\boldsymbol{r}}}{\mathrm{d}t}$ in spherical polar coordinates, I appeal to the linkage between the cartesian and spherical polar coordinate systems in writing

$$\frac{\mathrm{d}\hat{\boldsymbol{r}}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{r}\left(x\hat{\boldsymbol{x}} + y\hat{\boldsymbol{y}} + z\hat{\boldsymbol{z}}\right)\right) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{1}{r}\left(r\sin(\theta)\cos(\phi)\hat{\boldsymbol{x}} + r\sin(\theta)\sin(\phi)\hat{\boldsymbol{y}} + r\cos(\theta)\hat{\boldsymbol{z}}\right)\right) \tag{1-122}$$

$$= \frac{\mathrm{d}}{\mathrm{d}t}(\sin(\theta)\cos(\phi)\hat{\boldsymbol{x}} + \sin(\theta)\sin(\phi)\hat{\boldsymbol{y}} + \cos(\theta)\hat{\boldsymbol{z}}) \tag{1-123}$$

$$= \frac{\partial}{\partial\theta}(\sin(\theta)\cos(\phi)\hat{\boldsymbol{x}} + \sin(\theta)\sin(\phi)\hat{\boldsymbol{y}} + \cos(\theta)\hat{\boldsymbol{z}})\frac{\mathrm{d}\theta}{\mathrm{d}t} \tag{1-124}$$
$$+ \frac{\partial}{\partial\phi}(\sin(\theta)\cos(\phi)\hat{\boldsymbol{x}} + \sin(\theta)\sin(\phi)\hat{\boldsymbol{y}} + \cos(\theta)\hat{\boldsymbol{z}})\frac{\mathrm{d}\phi}{\mathrm{d}t}$$

$$= \dot{\theta}(\cos(\theta)\cos(\phi)\hat{\boldsymbol{x}} + \cos(\theta)\sin(\phi)\hat{\boldsymbol{y}} - \sin(\theta)\hat{\boldsymbol{z}}) + \dot{\phi}(-\sin(\theta)\sin(\phi)\hat{\boldsymbol{x}} + \sin(\theta)\cos(\phi)\hat{\boldsymbol{y}}) \tag{1-125}$$

$$= \dot{\theta}\hat{\boldsymbol{\theta}} + \sin(\theta)\dot{\phi}\hat{\boldsymbol{\phi}}. \tag{1-126}$$

Note that there are no partials with respect to the cartesian unit vectors since they are unchanging. Substituting into Equation 1-121 yields

$$\boldsymbol{L} = mr^2(\hat{\boldsymbol{r}}\times(\dot{\theta}\hat{\boldsymbol{\theta}}+\sin(\theta)\dot{\phi}\hat{\boldsymbol{\phi}})) = mr^2\dot{\theta}(\hat{\boldsymbol{r}}\times\hat{\boldsymbol{\theta}})+mr^2\sin(\theta)\dot{\phi}(\hat{\boldsymbol{r}}\times\hat{\boldsymbol{\phi}}) = mr^2(\dot{\theta}\hat{\boldsymbol{\phi}}+\sin(\theta)\dot{\phi}\hat{\boldsymbol{\theta}}). \tag{1-127}$$

Finally, I dot $\boldsymbol{\sigma}$ with $\boldsymbol{L}$. Note that in my definition of the spherical polar coordinates, I have not assumed a particular set of cartesian unit vectors relative to which it is defined. Therefore I have liberality in my definition of $\hat{\boldsymbol{x}}, \hat{\boldsymbol{y}}$, and $\hat{\boldsymbol{z}}$ (assuming they are mutually perpendicular). In particular, a convenient definition has $\hat{\boldsymbol{z}}$ parallel to $\hat{\boldsymbol{\sigma}}_z$. [3] Therefore, $\boldsymbol{\sigma}$ can be broken down as $\boldsymbol{\sigma} = \sigma\cos(\theta)\hat{\boldsymbol{r}} + \sigma\sin(\theta)\hat{\boldsymbol{\theta}}$ and thus

$$\boldsymbol{\sigma}\cdot\boldsymbol{L} = (\sigma\cos(\theta)\hat{\boldsymbol{r}} + \sigma\sin(\theta)\hat{\boldsymbol{\theta}})\cdot(mr^2(\dot{\theta}\hat{\boldsymbol{\phi}} + \sin(\theta)\dot{\phi}\hat{\boldsymbol{\theta}}) = mr^2\sigma\dot{\phi}\sin^2(\theta). \tag{1-128}$$

I thus finally have

$$U = U(r,\theta,\phi,\dot{r},\dot{\theta},\dot{\phi}) = V(r) + mr^2\sigma\dot{\phi}\sin^2(\theta) \tag{1-129}$$

The corresponding generalized force equations are given by

$$F_r = Q_r = -\frac{\mathrm{d}V}{\mathrm{d}r} - \frac{\partial}{\partial r}(mr^2\sigma\dot{\phi}\sin^2(\theta)) = -\frac{\mathrm{d}V}{\mathrm{d}r} - 2mr\sigma\dot{\phi}\sin^2(\theta), \tag{1-130}$$

$$F_\theta = Q_\theta = -\frac{\partial}{\partial\theta}(mr^2\sigma\dot{\phi}\sin^2(\theta)) = -2mr^2\sigma\dot{\phi}\sin(\theta)\cos(\theta), \tag{1-131}$$

$$F_\phi = Q_\phi = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial}{\partial\dot{\phi}}(mr^2\sigma\dot{\phi}\sin^2(\theta))\right) = \frac{\mathrm{d}}{\mathrm{d}t}\left(mr^2\sigma\sin^2(\theta)\right) = \tag{1-132}$$

$$= m\sigma\left(2r\dot{r}\sin^2(\theta) + 2r^2\sin(\theta)\cos(\theta)\dot{\theta}\right)$$

$$= 2mr\dot{r}\sigma\sin^2(\theta) + 2mr^2\dot{\theta}\sigma\sin(\theta)\cos(\theta). \tag{1-133}$$

[3] I originally derived this in the more general case, but upon finding a nice solution by Michael Goode redefined my coordinate system to greatly reduce the necessary algebra. The link is solution

Note that with our new definition of the cartesian coordinate system, $\sigma = \sigma_z$ and Equations 1-118–1-120 simplify to

$$F_x = -\frac{x}{r}\frac{\mathrm{d}V}{\mathrm{d}r} - 2m\sigma\dot{y}, \tag{1-134}$$

$$F_y = -\frac{y}{r}\frac{\mathrm{d}V}{\mathrm{d}r} + 2m\sigma\dot{x}, \tag{1-135}$$

$$F_z = -\frac{z}{r}\frac{\mathrm{d}V}{\mathrm{d}r}. \tag{1-136}$$

(b) Noting that $\frac{\partial \boldsymbol{r}}{\partial r} = \hat{\boldsymbol{r}}$, I compute

$$\boldsymbol{Q}_r = \boldsymbol{F}\cdot\frac{\partial \boldsymbol{r}}{\partial r} = \boldsymbol{F}\cdot\hat{\boldsymbol{r}} \tag{1-137}$$

$$= \left(\left(-\frac{x}{r}\frac{\mathrm{d}V}{\mathrm{d}r} - 2m\sigma\dot{y}\right)\hat{\boldsymbol{x}} + \left(-\frac{y}{r}\frac{\mathrm{d}V}{\mathrm{d}r} + 2m\sigma\dot{x}\right)\hat{\boldsymbol{y}} + \left(-\frac{z}{r}\frac{\mathrm{d}V}{\mathrm{d}r}\right)\hat{\boldsymbol{z}}\right)\cdot\left(\frac{1}{r}x\hat{\boldsymbol{x}} + y\hat{\boldsymbol{y}} + z\hat{\boldsymbol{z}}\right) \tag{1-138}$$

$$= -\frac{1}{r^2}\left(\frac{x^2+y^2+z^2}{r^2}\frac{\mathrm{d}V}{\mathrm{d}r}\right) + \frac{2m\sigma}{r}\left(y\dot{x} - x\dot{y}\right) \tag{1-139}$$

$$= -\frac{\mathrm{d}V}{\mathrm{d}r} + \frac{2m\sigma}{r}[r\sin(\theta)\sin(\phi)[\dot{r}\sin(\theta)\cos(\phi) + \dot{\theta}r\cos(\theta)\cos(\phi) - \dot{\phi}r\sin(\theta)\sin(\phi)] \tag{1-140}$$

$$- r\sin(\theta)\cos(\phi)[\dot{r}\sin(\theta)\sin(\phi) + \dot{\theta}r\cos(\theta)\sin(\phi) + \dot{\theta}r\sin(\theta)\cos(\phi)]]$$

$$= -\frac{\mathrm{d}V}{\mathrm{d}r} + \frac{2m\sigma}{r}(-r^2\dot{\phi}\sin^2(\theta)) = -\frac{\mathrm{d}V}{\mathrm{d}r} - 2mr\sigma\dot{\phi}\sin^2(\theta), \tag{1-141}$$

as expected. Next, I compute

$$\frac{\partial \boldsymbol{r}}{\partial \theta} = \frac{\partial}{\partial\theta}(r\sin(\theta)\cos(\phi)\hat{\boldsymbol{x}} + r\sin(\theta)\sin(\phi)\hat{\boldsymbol{y}} + r\cos(\theta)\hat{\boldsymbol{z}}) \tag{1-142}$$

$$= r\cos(\theta)\cos(\phi)\hat{\boldsymbol{x}} + r\cos(\theta)\sin(\phi)\hat{\boldsymbol{y}} - r\sin(\theta)\hat{\boldsymbol{z}} \tag{1-143}$$

so that

$$\boldsymbol{Q}_\theta = \boldsymbol{F}\cdot\frac{\partial \boldsymbol{r}}{\partial \theta} = \left(\left(-\frac{x}{r}\frac{\mathrm{d}V}{\mathrm{d}r} - 2m\sigma\dot{y}\right)\hat{\boldsymbol{x}} + \left(-\frac{y}{r}\frac{\mathrm{d}V}{\mathrm{d}r} + 2m\sigma\dot{x}\right)\hat{\boldsymbol{y}} + \left(-\frac{z}{r}\frac{\mathrm{d}V}{\mathrm{d}r}\right)\hat{\boldsymbol{z}}\right)\cdot$$

$$(r\cos(\theta)\cos(\phi)\hat{\boldsymbol{x}} + r\cos(\theta)\sin(\phi)\hat{\boldsymbol{y}} - r\sin(\theta)\hat{\boldsymbol{z}}) \tag{1-144}$$

$$= -\frac{\mathrm{d}V}{\mathrm{d}r}(r\sin(\theta)\cos(\theta)\cos^2(\phi) + r\sin(\theta)\cos(\phi)\sin^2(\phi) - r\sin(\theta)\cos(\theta)) \tag{1-145}$$

$$+ 2m\sigma r(\dot{r}\sin(\theta)\cos(\theta)\sin(\phi)\cos(\phi) + \dot{\theta}r\cos^2(\theta)\sin(\phi)\cos(\phi) - \dot{\phi}r\sin(\theta)\cos(\theta)\sin^2(\phi))$$

$$- 2m\sigma r(\dot{r}\sin(\theta)\cos(\theta)\sin(\phi)\cos(\phi) + \dot{\theta}r\cos^2(\theta)\sin(\phi)\cos(\phi) + \dot{\phi}r\sin(\theta)\cos(\theta)\cos^2(\phi))$$

$$= -2mr^2\sigma\dot{\phi}\sin(\theta)\cos(\theta), \tag{1-146}$$

as expected. Finally, I compute

$$\frac{\partial \boldsymbol{r}}{\partial \phi} = -r\sin(\theta)\sin(\phi)\hat{\boldsymbol{x}} + r\sin(\theta)\cos(\phi)\hat{\boldsymbol{y}} \tag{1-147}$$

so that

$$Q_\phi = \boldsymbol{F}\cdot\frac{\partial \boldsymbol{r}}{\partial\phi} = \left(\left(-\frac{x}{r}\frac{\mathrm{d}V}{\mathrm{d}r} - 2m\sigma\dot{y}\right)\hat{\boldsymbol{x}} + \left(-\frac{y}{r}\frac{\mathrm{d}V}{\mathrm{d}r} + 2m\sigma\dot{x}\right)\hat{\boldsymbol{y}} + \left(-\frac{z}{r}\frac{\mathrm{d}V}{\mathrm{d}r}\right)\hat{\boldsymbol{z}}\right)\cdot$$
$$(-r\sin(\theta)\sin(\phi)\hat{\boldsymbol{x}} + r\sin(\theta)\cos(\phi)\hat{\boldsymbol{y}}) \tag{1-148}$$
$$= \frac{\mathrm{d}V}{\mathrm{d}r}(r\sin^2(\theta)\sin(\phi)\cos(\phi) - r\sin^2(\theta)\sin(\phi)\cos(\phi)) \tag{1-149}$$
$$+ 2mr\sigma(\dot{r}\sin^2(\theta)\sin^2(\phi) + \dot{\theta}r\sin(\theta)\cos(\theta)\sin^2(\phi) + \dot{\phi}r\sin^2(\theta)\sin(\phi)\cos(\phi))$$
$$+ 2mr\sigma(\dot{r}\sin^2(\theta)\cos^2(\phi) + \dot{\theta}r\sin(\theta)\cos(\theta)\cos^2(\phi) - \dot{\phi}r\sin^2(\theta)\sin(\phi)\cos(\phi))$$
$$= 2mr\dot{r}\sigma\sin^2(\theta) + 2mr^2\dot{\theta}\sigma\sin(\theta)\cos(\theta), \tag{1-150}$$

as expected.

(c) Note that since $x := x(r,\theta,\phi)$, the generic form of $\dot{x}$ is given by $\dot{x} = \frac{\partial x}{\partial r}\dot{r} + \frac{\partial x}{\partial\theta}\dot{\theta} + \frac{\partial x}{\partial\phi}\dot{\phi}$ and similarly for $\dot{y}$ and $\dot{z}$. The kinetic energy is thus given by

$$T = \frac{m}{2}(\dot{x}^2+\dot{y}^2+\dot{z}^2) = \frac{m}{2}[(\dot{r}\sin(\theta)\cos(\phi) + \dot{\theta}\cos(\theta)\cos(\phi) - \dot{\phi}r\sin(\theta)\sin(\phi))^2$$
$$+ (\dot{r}\sin(\theta)\sin(\phi) + \dot{\theta}\cos(\theta)\sin(\phi) + \dot{\phi}r\sin(\theta)\sin(\phi))^2 + (\dot{r}\cos(\theta) - \dot{\theta}r\sin(\theta))^2] \tag{1-151}$$
$$= \frac{m}{2}(\dot{r}^2 + \dot{\theta}^2r^2 + \dot{\phi}^2r^2\sin^2(\theta)). \tag{1-152}$$

Applying Lagrange's equations for r thereby yields

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial\dot{r}}\right) - \frac{\partial T}{\partial r} = \frac{\mathrm{d}}{\mathrm{d}t}(m\dot{r}) - mr\dot{\theta}^2 - mr\sin^2(\theta)\dot{\phi}^2 = m\ddot{r} - mr\dot{\theta}^2 - mr\sin^2(\theta)\dot{\phi}^2 \tag{1-153}$$
$$= Q_r = -\frac{\mathrm{d}V}{\mathrm{d}r} - 2mr\sigma\dot{\phi}\sin^2(\theta) \tag{1-154}$$

which gives

$$\ddot{r} = -\frac{1}{m}\frac{\mathrm{d}V}{\mathrm{d}r} - 2r\sigma\dot{\phi}\sin^2(\theta) + r\dot{\theta}^2 + r\dot{\phi}^2\sin^2(\theta) == -\frac{1}{m}\frac{\mathrm{d}V}{\mathrm{d}r} + r[\dot{\theta}^2 + \sin^2(\theta)(\dot{\phi}^2 - 2\sigma\dot{\phi})]. \tag{1-155}$$

For θ,

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial\dot{\theta}}\right) - \frac{\partial T}{\partial\theta} = \frac{\mathrm{d}}{\mathrm{d}t}(mr^2\dot{\theta}) - mr^2\dot{\phi}^2\sin(\theta)\cos(\theta) = 2mr\dot{r}\dot{\theta} + mr^2\ddot{\theta} - mr^2\dot{\phi}^2\sin(\theta)\cos(\theta) \tag{1-156}$$
$$= Q_\theta = -2mr^2\sigma\dot{\phi}\sin(\theta)\cos(\theta) \tag{1-157}$$

which gives

$$\ddot{\theta} = -2\sigma\dot{\phi}\sin(\theta)\cos(\theta) + \dot{\phi}^2\sin(\theta)\cos(\theta) - \frac{2\dot{r}\dot{\theta}}{r} = \sin(\theta)\cos(\theta)\dot{\phi}(\dot{\phi} - 2\sigma) - \frac{2\dot{r}\dot{\theta}}{r}. \tag{1-158}$$

Finally,

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{\phi}}\right) - \frac{\partial T}{\partial \phi} = \frac{\mathrm{d}}{\mathrm{d}t}(m\dot{\phi}r^2\sin^2(\theta)) = m[\ddot{\phi}r^2\sin^2(\theta) + 2\dot{r}\dot{\phi}r\sin^2(\theta) + 2\dot{\theta}\dot{\phi}r^2\sin(\theta)\cos(\theta)] \tag{1-159}$$

$$= Q_\phi = 2mr\dot{r}\sigma\sin^2(\theta) + 2mr^2\dot{\theta}\sigma\sin(\theta)\cos(\theta) \tag{1-160}$$

which gives

$$\ddot{\phi} = \frac{2\dot{r}}{r}(\sigma - \dot{\phi}) + 2\dot{\theta}\cot(\theta)(\sigma - \dot{\phi}) = 2(\sigma - \dot{\phi})\left(\frac{\dot{r}}{r} + \cot(\theta)\dot{\theta}\right). \tag{1-161}$$

Problem 1.13

Problem Statement

A particle moves in a plane under the influence of a force, acting toward a center of force, whose magnitude is

$$F = \frac{1}{r^2}\left(1 - \frac{\dot{r}^2 - 2r\dot{r}}{c^2}\right), \tag{1-162}$$

where r is the distance of the particle to the center of force. Find the generalized potential that will result in such a force, and from that that the Lagrangian for the motion in a plane.

Using the polar coordinate system, I have the generalized forces

$$Q_r = F, \tag{1-163}$$

$$Q_\theta = 0. \tag{1-164}$$

The generalized potential which satisfies these forces is given by

$$U(r, \theta, \dot{r}, \dot{\theta}) = \frac{\dot{r}^2}{rc^2} + \frac{1}{r} \tag{1-165}$$

up to an additive constant, which I take to be 0. For T, I compute

$$T = \frac{m}{2}(\dot{x}^2 + \dot{y}^2) = \frac{m}{2}[(\dot{r}\cos(\theta) - \dot{\theta}r\sin(\theta))^2 + (\dot{r}\sin(\theta) + \dot{\theta}r\cos(\theta))^2] = \frac{m}{2}(\dot{r}^2 + r^2\dot{\theta}^2) \tag{1-166}$$

so that

$$L = T - U = \frac{m}{2}\dot{r}^2 + \frac{m}{2}r^2\dot{\theta}^2 - \frac{\dot{r}^2}{rc^2} - \frac{1}{r} = \dot{r}^2\left(\frac{m}{2} - \frac{1}{rc^2}\right) + \frac{m}{2}r^2\dot{\theta}^2 - \frac{1}{r}, \tag{1-167}$$

as desired.

Problem 1.14

Problem Statement

If L is a Lagrangian for a system of n degrees of freedom satisfying Lagrange's Equations, show by direct substitution that

$$L' = L + \frac{\mathrm{d}F(q_1, \ldots, q_n, t)}{\mathrm{d}t} \tag{1-168}$$

also satisfies Lagrange's Equations where F is an arbitrary, but differentiable, function of its arguments.

I first compute, using the linearity of the partial derivative of L' and Equation **??**, that for a fixed j,

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L'}{\partial \dot{q}_j}\right) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}_j}\right) + \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial}{\partial \dot{q}_j}\left(\frac{\mathrm{d}F(q_1,\ldots,q_n)}{\mathrm{d}t}\right)\right) \tag{1-169}$$

$$= \frac{\partial L}{\partial q_j} + \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial}{\partial \dot{q}_j}\left(\frac{\partial F}{\partial q_1}\frac{\mathrm{d}q_1}{\mathrm{d}t} + \cdots + \frac{\partial F}{\partial q_n}\frac{\mathrm{d}q_n}{\mathrm{d}t}\right)\right) = \frac{\partial L}{\partial q_j} + \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial}{\partial \dot{q}_j}\left(\sum_i^n \frac{\partial F}{\partial q_i}\dot{q}_i\right)\right) \tag{1-170}$$

$$= \frac{\partial L}{\partial q_j} + \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial}{\partial \dot{q}_j}\left(\frac{\partial F}{\partial q_j}\dot{q}_j\right)\right) = \frac{\partial L}{\partial q_j} + \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial F}{\partial q_j}\right) = \frac{\partial L}{\partial q_j} + \sum_i^n \frac{\partial^2 F}{\partial q_j \partial q_i}\dot{q}_j. \tag{1-171}$$

Note that in evaluating the partial derivative presented at the end of Equation 1-170, $\frac{\partial F}{\partial q_i} := \frac{\partial F}{\partial q_i}(q_1,\ldots,q_n)$ and hence acts as a constant in the partial differentiation with respect to $\dot{q}_j$. On the other hand,

$$\frac{\partial L'}{\partial q_j} = \frac{\partial L}{\partial q_j} + \frac{\partial}{\partial q_j}\left(\sum_i^n \frac{\partial F}{\partial q_i}\dot{q}_i\right) = \frac{\partial L}{\partial q_j} + \sum_i^n \frac{\partial^2 F}{\partial q_j \partial q_i}\dot{q}_j = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L'}{\partial \dot{q}_j}\right), \tag{1-172}$$

as desired.

Problem 1.15

Problem Statement

Let $q_1,\ldots,q_n$ be a set of independent generalized coordinates for a system of n degrees of freedom, with a Lagrangian $L(q,\dot{q},t)$. Suppose we transform to another set of independent coordinates $s_1,\ldots,s_n$ by means of transformation equations

$$q_i = q_i(s_1,\ldots,s_n,t), \qquad i = 1,\ldots,n. \tag{1-173}$$

These equations are called point transformations. Show that if the Lagrangian function is expressed as a function of $s_j, \dot{s}_j$ and t through the equations of transformation, then L satisfies Lagrange's Equations with respect to the s coordinates:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{s}_j}\right) - \frac{\partial L}{\partial s_j} = 0. \tag{1-174}$$

In other words, show that the form of Lagrange's Equations is invariant under a point transformation.

Taking $L := L(s_1, \dot{s}_1, \ldots, s_n, \ldots s_n)$, and noting that in general $\dot{s}_i = \dot{s}_i(q_1, \dot{q}_1, \ldots, q_n, \dot{q}_n)$, I compute

$$\frac{\partial L}{\partial \dot{s}_j} = \sum_i^n \frac{\partial L}{\partial q_i}\frac{\partial q_i}{\partial \dot{s}_j} + \sum_i^n \frac{\partial L}{\partial \dot{q}_i}\frac{\partial \dot{q}_i}{\partial \dot{s}_j} + \frac{\partial L}{\partial t}\frac{\partial t}{\partial \dot{s}_j} \tag{1-175}$$

which therefore gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{s}_j}\right) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\sum_i^n \frac{\partial L}{\partial q_i}\frac{\partial q_i}{\partial \dot{s}_j} + \sum_i^n \frac{\partial L}{\partial \dot{q}_i}\frac{\partial \dot{q}_i}{\partial \dot{s}_j} + \frac{\partial L}{\partial t}\frac{\partial t}{\partial \dot{s}_j}\right) \tag{1-176}$$

$$= \frac{\mathrm{d}}{\mathrm{d}t}\left(\sum_i^n \frac{\partial L}{\partial q_i}\frac{\partial q_i}{\partial \dot{s}_j}\right) + \frac{\mathrm{d}}{\mathrm{d}t}\left(\sum_i^n \frac{\partial L}{\partial \dot{q}_i}\frac{\partial \dot{q}_i}{\partial \dot{s}_j}\right) + \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial t}\frac{\partial t}{\partial \dot{s}_j}\right) \tag{1-177}$$

$$= \frac{\mathrm{d}}{\mathrm{d}t}\left(\sum_i^n \frac{\partial L}{\partial \dot{q}_i}\frac{\partial \dot{q}_i}{\partial \dot{s}_j}\right) + \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial t}\frac{\partial t}{\partial \dot{s}_j}\right) \tag{1-178}$$

$$= \sum_i^n \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}_i}\right)\frac{\partial \dot{q}_i}{\partial \dot{s}_j} + \sum_i^n \frac{\partial L}{\partial \dot{q}_i}\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \dot{q}_i}{\partial \dot{s}_j}\right) + \frac{\partial L}{\partial t}\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial t}{\partial \dot{s}_j}\right) + \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial t}\right)\frac{\partial t}{\partial \dot{s}_j} \tag{1-179}$$

$$= \sum_i^n \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}_i}\right)\frac{\partial \dot{q}_i}{\partial \dot{s}_j} + \sum_i^n \frac{\partial L}{\partial \dot{q}_i}\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial \dot{q}_i}{\partial \dot{s}_j}\right), \tag{1-180}$$

where the removal of the sum in the step between Equations 1-177 and 1-178 follows directly from the functional dependence of q_i in Equation 1-173 and the removal of the second term in Equation 1-179 follows from t as independent from $\dot{s}_j$. Note that the established Lagrangian with respect to the generalized coordinate system q_i yields

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}_j}\right) = \frac{\partial L}{\partial q_j}, \tag{1-181}$$

and expanding $\frac{\partial \dot{q}_i}{\partial \dot{s}_j}$ gives

$$\frac{\partial \dot{q}_i}{\partial \dot{s}_j} = \frac{\partial}{\partial \dot{s}_j}\left(\frac{\mathrm{d}q_i}{\mathrm{d}t}\right) = \sum_i^n \frac{\partial}{\partial \dot{s}_j}\left(\frac{\partial q_i}{\partial s_i}\dot{s}_i\right) + \frac{\partial}{\partial \dot{s}_j}\left(\frac{\partial q_i}{\partial t}\frac{\mathrm{d}t}{\mathrm{d}t}\right) = \frac{\partial}{\partial \dot{s}_j}\left(\frac{\partial q_j}{\partial s_j}\dot{s}_j\right) = \frac{\partial q_j}{\partial s_j}, \tag{1-182}$$

which can both be substituted into Equation 1-180 to get

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{s}_j}\right) = \sum_i^n \frac{\partial L}{\partial q_j}\frac{\partial q_j}{\partial s_j} + \sum_i^n \frac{\partial L}{\partial \dot{q}_i}\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial q_i}{\partial s_j}\right). \tag{1-183}$$

Next, I compute

$$\frac{\partial L}{\partial s_j} = \sum_i^n \frac{\partial L}{\partial q_i}\frac{\partial q_i}{\partial s_j} + \sum_i^n \frac{\partial L}{\partial \dot{q}_i}\frac{\partial \dot{q}_i}{\partial s_j} + \frac{\partial L}{\partial t}\frac{\partial t}{\partial s_j} = \sum_i^n \frac{\partial L}{\partial q_i}\frac{\partial q_i}{\partial s_j} + \sum_i^n \frac{\partial L}{\partial \dot{q}_i}\frac{\partial \dot{q}_i}{\partial s_j}. \tag{1-184}$$

Subtracting the two expressions gives

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{s}_j}\right) - \frac{\partial L}{\partial s_j} = \sum_i^n \frac{\partial L}{\partial q_j}\frac{\partial q_j}{\partial s_j} + \sum_i^n \frac{\partial L}{\partial \dot{q}_i}\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial q_i}{\partial s_j}\right) - \left(\sum_i^n \frac{\partial L}{\partial q_i}\frac{\partial q_i}{\partial s_j} + \sum_i^n \frac{\partial L}{\partial \dot{q}_i}\frac{\partial \dot{q}_i}{\partial s_j}\right)$$

$$= \sum_i^n \frac{\partial L}{\partial \dot{q}_i}\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial q_i}{\partial s_j}\right) - \sum_i^n \frac{\partial L}{\partial \dot{q}_i}\frac{\partial}{\partial s_j}\left(\frac{\mathrm{d}q_i}{\mathrm{d}t}\right) \tag{1-185}$$

$$= \sum_i^n \frac{\partial L}{\partial \dot{q}_i}\sum_k^n \frac{\partial}{\partial s_k}\left(\frac{\partial q_i}{\partial s_j}\right)\dot{s}_k - \sum_i^n \frac{\partial L}{\partial \dot{q}_i}\frac{\partial}{\partial s_j}\sum_k^n \frac{\partial q_i}{\partial s_k}\dot{s}_k \tag{1-186}$$

$$= \sum_{i,k}^n \frac{\partial L}{\partial \dot{q}_i}\left(\frac{\partial^2 q_i}{\partial s_j \partial s_k}\dot{s}_k - \frac{\partial^2 q_i}{\partial s_j \partial s_k}\dot{s}_k\right) \tag{1-187}$$

$$= 0, \tag{1-188}$$

as desired.

Problem 1.17

Problem Statement

Obtain the Lagrange equations of motion for a spherical pendulum, i.e. a mass point suspended by a rigid weightless rod.

The rod gives us the holonomic constraint $x^2 + y^2 + z^2 = l^2$, where l is the length of the rod. Since spherical polar coordinates are a equivalent way to describe the position of an object in 3-D space, and hence has well-defined 3-D generalized coordinates (l, θ, ϕ), I can take l as constant and hence have only two generalized coordinates to describe the system, θ and ϕ. The kinetic energy in general is given by

$$T = \frac{m}{2}\left(\dot{l}^2 + l^2\dot{\theta}^2 + l^2\sin^2(\theta)\dot{\phi}^2\right) \tag{1-189}$$

and noting that $\dot{l} = 0$ gives

$$T = \frac{m}{2}\left(l^2\dot{\theta}^2 + l^2\sin^2(\theta)\dot{\phi}^2\right). \tag{1-190}$$

The gravitational potential energy is given up to an additive constant by

$$V = -mgz = -mgl\cos(\theta) \tag{1-191}$$

so that

$$L = T - V = \frac{m}{2}\left(l^2\dot{\theta}^2 + l^2\sin^2(\theta)\dot{\phi}^2\right) + mgl\cos(\theta). \tag{1-192}$$

Applying Lagrange's Equations gives

$$ml^2\ddot{\theta} = ml^2\dot{\phi}^2\sin(\theta)\cos(\theta) - mgl\sin(\theta), \tag{1-193}$$

$$ml^2\sin^2(\theta)\ddot{\phi} = 0, \tag{1-194}$$

which can be simplified to

$$\ddot{\theta} = \dot{\phi}^2\sin(\theta)\cos(\theta) + \frac{g\sin(\theta)}{l} = \dot{\phi}^2\sin(2\theta) - \frac{g\sin(\theta)}{l}, \tag{1-195}$$

$$\ddot{\phi} = 0. \tag{1-196}$$

Note that $\ddot{\phi} = 0$ yields $\dot{\phi}$ as constant and hence Equation 1-195 is a differential equation only of the zenith angle θ and not its time derivative. In the case that $\dot{\phi} = 0$, the system reduces to the 1-D pendulum where $\ddot{\theta} = \frac{g\sin(\theta)}{l}$ is akin to the well-known tangential acceleration expression.

Problem 1.18

Problem Statement

A particle of mass m moves in one dimension such that it has the Lagrangian

$$L = \frac{m^2\dot{x}^4}{12} + m\dot{x}^2V(x) - V^2(x) \tag{1-197}$$

where V is some differentiable function of x. Find the equation of motion for $x(t)$ and describe the physical nature of the system on the basis of this equation.

I compute

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{x}}\right) = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{m^2\dot{x}^3}{3} + 2m\dot{x}V\right) = m^2\dot{x}^2\ddot{x} + 2m\ddot{x}V + 2m\dot{x}^2\frac{\mathrm{d}V}{\mathrm{d}x}, \tag{1-198}$$

$$\frac{\partial L}{\partial x} = \frac{\partial L}{\partial V}\frac{\mathrm{d}V}{\mathrm{d}x} = \frac{\mathrm{d}V}{\mathrm{d}x}\left(m\dot{x}^2 - 2V\right). \tag{1-199}$$

Equating the two yields

$$m^2\dot{x}^2\ddot{x} + 2m\ddot{x}V + 2m\dot{x}^2\frac{\mathrm{d}V}{\mathrm{d}x} = \frac{\mathrm{d}V}{\mathrm{d}x}\left(m\dot{x}^2 - 2V\right). \tag{1-200}$$

Rearranging gives

$$\ddot{x}(m^2\dot{x}^2 + 2mV) = -\frac{\mathrm{d}V}{\mathrm{d}x}\left(m\dot{x}^2 + 2V\right), \tag{1-201}$$

which simplifies to

$$\ddot{x} = -\frac{\mathrm{d}V}{\mathrm{d}x}\frac{m\dot{x}^2 - 2V}{m^2\dot{x}^2 + 2mV} = -\frac{\frac{\mathrm{d}V}{\mathrm{d}x}}{m}, \tag{1-202}$$

and hence

$$m\ddot{x} = -\frac{\mathrm{d}V}{\mathrm{d}x}. \tag{1-203}$$

Since $V = V(x)$ is a function of only position, this equation of motion gives a conservative 1-D force $\boldsymbol{F}(x) := -\frac{\mathrm{d}V}{\mathrm{d}x}$ which dictates the motion of the system.

Problem 1.19

Problem Statement

Two points of mass m_1 and m_2 are connected by a string passing through a hole in a smooth table so that m_1 rests on the table surface and m_2 hangs suspended. Assuming m_2 moves only in a vertical line, what are the generalized coordinates for this system? Write down the Lagrange equations for this system and, if possible, discuss the physical significance any of them might have. Reduce the problem to a single second-order differential equation and obtain a first integral of the equation. What is its physical significance? (Consider the motion only so long as neither m_1 nor m_2 passes through the hole).

Let (x_1, y_1, z_1) and (x_2, y_2, z_2) denote the cartesian coordinates of m_1 and m_2, respectively, with positive z defined as upward normal to the table. I transform these coordinates into cylindrical polar coordinates (r_1, θ_1, z_1) and (r_2, θ_2, z_2) as normally defined. The equations of constraint

$$z_1 = 0, \tag{1-204}$$

$$r_2 = 0, \tag{1-205}$$

$$z_2 = 0, \tag{1-206}$$

$$r_1 - z_2 - l = 0, \tag{1-207}$$

enable the reduction to two degrees of freedom $r := r_1$ and $\theta := \theta_1$. Note that the system could be defined alternatively with respect to z_1 instead of r_1 owing to Equation 1-207. As defined in the cylindrical polar coordinate system

$$T = T_1 + T_2 = \frac{m_1}{2}\left(\dot{r}_1^2 + r_1^2\dot{\theta}_1^2 + \dot{z}_1^2\right) + \frac{m_2}{2}\left(\dot{r}_2^2 + r_2^2\dot{\theta}_2^2 + \dot{z}_2^2\right), \tag{1-208}$$

which after substituting my generalized coordinates r and θ and their associated time derivatives gives

$$T = \frac{m_1}{2}\left(\dot{r}^2 + r^2\dot{\theta}^2\right) + \frac{m_2}{2}\left((-\dot{r})^2\right) = \frac{m_1}{2}\left(\dot{r}^2 + r^2\dot{\theta}^2\right) + \frac{m_2}{2}\left(\dot{r}^2\right). \tag{1-209}$$

The potential energy is entirely gravitational and assumed as a constant g model. Its form is given by

$$V = V_1 + V_2 = m_1 g z_1 + m_2 g z_2 = m_2 g(r - l) = m_2 g r - m_2 g l, \tag{1-210}$$

where a potential energy of 0 is defined at the height of the table. The corresponding Lagrangian is therefore

$$L = T - V = \frac{m_1}{2}\left(\dot{r}^2 + r^2\dot{\theta}^2\right) + \frac{m_2}{2}\left(\dot{r}^2\right) - m_2 g r + m_2 g l. \tag{1-211}$$

Note that $m_2 g l$ is simply an additive constant and hence will not affect the equations of motion; I therefore remove it from the Lagrangian to get

$$L = \frac{m_1}{2}\left(\dot{r}^2 + r^2\dot{\theta}^2\right) + \frac{m_2}{2}\left(\dot{r}^2\right) - m_2 g r. \tag{1-212}$$

Lagrange's equations give

$$(m_1 + m_2)\ddot{r} = -m_2 g, \tag{1-213}$$

$$m_1 r^2 \ddot{\theta} = 0, \tag{1-214}$$

which simplify to

$$\ddot{r} = -\frac{m_2}{m_1 + m_2} g, \tag{1-215}$$

$$\ddot{\theta} = 0. \tag{1-216}$$

The physical content of Equations 1-215 and 1-216 shows that all acceleration of m_1 is radial and moreover is constant and always directed inward rather than outward. The relevant second-order differential equation is therefore

$$\frac{\mathrm{d}^2 r}{\mathrm{d}t^2} = c, \tag{1-217}$$

where $c = -\frac{m_2}{m_1 + m_2} g$ is constant. Integrating gives

$$\int_{t_0}^{t} \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}r}{\mathrm{d}t}\right)\, \mathrm{d}t = \int_{t_0}^{t} c\, \mathrm{d}t \tag{1-218}$$

which expanding c gives

$$\dot{r} = ct + \dot{r}_0 = -\frac{m_2}{m_1 + m_2} g t + \dot{r}_0, \tag{1-219}$$

where $\dot{r}_0 = \dot{r}(t_0)$. Therefore, with initial conditions $\dot{r}_0$ and $\dot{\theta}_0$, the velocity of both masses can be described explicitly as a function of time. Note that another integration and initial conditions r_0 and θ_0 gives position explicitly as a function of time.

Problem 1.20 (re-derive w/ theta1-theta2 and theta2 as generalized coordinates

Problem Statement

Obtain the Lagrangian and equations of motion for the double pendulum moving in a plane.

I can use 2-D Cartesian coordinates to describe the system, since it is moving in a plane. Noting that each pendulum reduces a degree of freedom within the system, the double pendulum can be

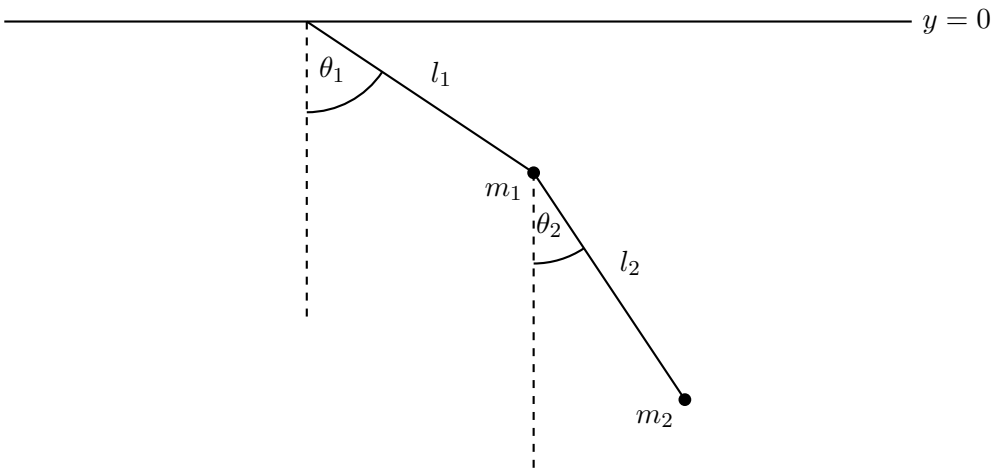


Figure 3: Illustration of the double pendulum. Each θ angle is defined conventionally as increasing counter-clockwise.

described completely by two generalized coordinates, namely θ_1 and θ_2. I physically illustrate the system pendulum in Figure 1.1, with θ_1 and θ_2 defined as increasing counter-clockwise.
Rewriting, I have

$$x_1 = l_1 \sin(\theta_1), \tag{1-220}$$
$$y_1 = -l_1 \cos(\theta_1), \tag{1-221}$$
$$x_2 = x_1 + l_2 \sin(\theta_2) = l_1 \sin(\theta_1) + l_2 \sin(\theta_2), \tag{1-222}$$
$$y_2 = y_1 - l_2 \cos(\theta_2) = -l_1 \cos(\theta_1) - l_2 \cos(\theta_2), \tag{1-223}$$

and

$$\dot{x}_1 = \dot{\theta}_1 l_1 \cos(\theta_1), \tag{1-224}$$
$$\dot{y}_1 = \dot{\theta}_2 l_1 \sin(\theta_1), \tag{1-225}$$
$$\dot{x}_2 = \dot{\theta}_1 l_1 \cos(\theta_1) + \dot{\theta}_2 l_2 \cos(\theta_2), \tag{1-226}$$
$$\dot{x}_2 = \dot{\theta}_1 l_1 \sin(\theta_1) + \dot{\theta}_2 l_2 \sin(\theta_2). \tag{1-227}$$

Solving for T, I get

$$T = \frac{m_1}{2}(\dot{x}_1^2 + \dot{y}_1^2) + \frac{m_2}{2}(\dot{x}_2^2 + \dot{y}_2^2) = \frac{m_1}{2}(\dot{\theta}_1^2 l_1^2 \cos^2(\theta_1) + \dot{\theta}_1^2 l_1^2 \sin^2(\theta_1)) \tag{1-228}$$
$$+ \frac{m_2}{2}(\dot{\theta}_1^2 l_1^2 \cos^2(\theta_1) + \dot{\theta}_2^2 l_2^2 \cos^2(\theta_2) + 2\dot{\theta}_1 \dot{\theta}_2 l_1 l_2 \cos(\theta_1)\cos(\theta_2)) \tag{1-229}$$
$$+ \frac{m_2}{2}(\dot{\theta}_1^2 l_1^2 \sin^2(\theta_1) + \dot{\theta}_2^2 l_2^2 \sin^2(\theta_2) + 2\dot{\theta}_1 \dot{\theta}_2 l_1 l_2 \sin(\theta_1)\sin(\theta_2)) \tag{1-230}$$
$$= \frac{m_1}{2}\dot{\theta}_1^2 l_1^2 + \frac{m_2}{2}\dot{\theta}_1^2 l_1^2 + \frac{m_2}{2}\dot{\theta}_2^2 l_2^2 + m_2 \dot{\theta}_1 \dot{\theta}_2 l_1 l_2 [\cos(\theta_1)\cos(\theta_2) + \sin(\theta_1)\sin(\theta_2)] \tag{1-231}$$
$$= \frac{m_1 + m_2}{2}\dot{\theta}_1^2 l_1^2 + \frac{m_2}{2}\dot{\theta}_2^2 l_2^2 + m_2 \dot{\theta}_1 \dot{\theta}_2 l_1 l_2 \cos(\theta_1 - \theta_2). \tag{1-232}$$

The potential energy is entirely gravitational and is given by

$$V = m_1 g y_1 + m_2 g y_2 = -m_1 g l_1 \cos(\theta_1) - m_2 g l_1 \cos(\theta_1) - m_2 g l_2 \cos(\theta_2) = -(m_1 + m_2)(g l_1 \cos(\theta_1)) - m_2 g l_2 \cos(\theta_2). \tag{1-233}$$

Therefore, the Lagrangian is given by

$$L = T - V = \frac{m_1 + m_2}{2}\dot{\theta}_1^2 l_1^2 + \frac{m_2}{2}\dot{\theta}_2^2 l_2^2 + m_2\dot{\theta}_1\dot{\theta}_2 l_1 l_2 \cos(\theta_1 - \theta_2) + (m_1 + m_2)(g l_1 \cos(\theta_1)) + m_2 g l_2 \cos(\theta_2). \tag{1-234}$$

Differentiating with respect to $\dot{\theta}_1$ gives

$$\frac{\partial L}{\partial \dot{\theta}_1} = (m_1 + m_2)\dot{\theta}_1 l_1^2 + m_2\dot{\theta}_2 l_1 l_2 \cos(\theta_1 - \theta_2), \tag{1-235}$$

so that applying Lagrange's Equations first yields

$$(m_1 + m_2)\ddot{\theta}_1 l_1^2 + m_2\ddot{\theta}_2 l_1 l_2 \cos(\theta_1 - \theta_2) - m_2(\dot{\theta}_1 - \dot{\theta}_2)\dot{\theta}_2 l_1 l_2 \sin(\theta_1 - \theta_2) = \tag{1-236}$$

$$- m_2\dot{\theta}_1\dot{\theta}_2 l_1 l_2 \sin(\theta_1 - \theta_2) - (m_1 + m_2) g l_1 \sin(\theta_1), \tag{1-237}$$

which can be simplified to

$$(m_1 + m_2)\ddot{\theta}_1 l_1^2 + m_2\ddot{\theta}_2 l_1 l_2 \cos(\theta_1 - \theta_2) - m_2\dot{\theta}_2^2 l_1 l_2 \sin(\theta_1 - \theta_2) = -(m_1 + m_2) g l_1 \sin(\theta_1), \tag{1-238}$$

or grouping by mass,

$$\ddot{\theta}_1 l_1 + g \sin(\theta_1) = \frac{m_2 l_2}{m_1 + m_2}[\dot{\theta}_2^2 \sin(\theta_1 - \theta_2) - \ddot{\theta}_2 \cos(\theta_1 - \theta_2)]. \tag{1-239}$$

Turning towards the second equation, I compute

$$\frac{\partial L}{\partial \dot{\theta}_2} = m_2 l_2^2 \dot{\theta}_2 + m_2 l_1 l_2 \dot{\theta}_1 \cos(\theta_1 - \theta_2), \tag{1-240}$$

so that, applying Lagrange's Equations,

$$m_2 l_2^2 \ddot{\theta}_2 + m_2 l_1 l_2 \ddot{\theta}_1 \cos(\theta_1 - \theta_2) - m_2 l_1 l_2 (\dot{\theta}_1 - \dot{\theta}_2)\dot{\theta}_1 \sin(\theta_1 - \theta_2) = m_2 l_1 l_2 \dot{\theta}_1 \dot{\theta}_2 \sin(\theta_1 - \theta_2) - m_2 g l_2 \sin(\theta_2). \tag{1-241}$$

I can factor out m_2 and l_2 and simplify to

$$l_2\ddot{\theta}_2 + g \sin(\theta_2) = l_1[\dot{\theta}_1^2 \sin(\theta_1 - \theta_2) - \ddot{\theta}_1 \cos(\theta_1 - \theta_2)], \tag{1-242}$$

which closely mirrors Equation 1-239.

2 Variational Principles and Lagrange's Equations

2.1 Solutions

Problem 2.1

Problem Statement

Prove that the shortest distance between two points is a straight line.

I first assume that I can parameterize y and z with respect to x, i.e. $y = y(x)$ and $z = z(x)$. Note that this excludes functions which traces out paths which reach a given x value multiple times; however, unless the final point is oriented perpendicular to $\boldsymbol{x}$ (which can be fixed with a redefinition of x), this in general does not pose a problem since these solutions necessitate a looping back on x, which are not physically motivated possibilities for a minimal path length.
Expanding from the 2-D example presented in [1], I take

$$y = y(x,0) + \alpha_y \eta_y(x), \tag{2-1}$$
$$z = z(x,0) + \alpha_z \eta_z(x), \tag{2-2}$$

where the α and η functions for the different variables are allowed to vary independently. Therefore, I can write the paramteric family of curves J as

$$J = \int_{x_1}^{x_2} f(y, \dot{y}, z, \dot{z}, x) = \int_{x_1}^{x_2} f(y(x,\alpha_y), \dot{y}(x,\alpha_y), z(x,\alpha_z), \dot{z}(x,\alpha_z), x)\,\mathrm{d}x. \tag{2-3}$$

The generalized condition for a stationary point is given by

$$\frac{\mathrm{d}J}{\mathrm{d}\alpha_y} = 0, \tag{2-4}$$
$$\frac{\mathrm{d}J}{\mathrm{d}\alpha_z} = 0, \tag{2-5}$$

so that any local variation is accounted for in the α formulation. I have

$$\frac{\mathrm{d}J}{\mathrm{d}\alpha_y} = \int_{x_1}^{x_2} \left[\frac{\partial f}{\partial y}\frac{\partial y}{\partial \alpha_y} + \frac{\partial f}{\partial \dot{y}}\frac{\partial \dot{y}}{\partial \alpha_y}\right] \mathrm{d}x, \tag{2-6}$$
$$\frac{\mathrm{d}J}{\mathrm{d}\alpha_z} = \int_{x_1}^{x_2} \left[\frac{\partial f}{\partial z}\frac{\partial z}{\partial \alpha_z} + \frac{\partial f}{\partial \dot{z}}\frac{\partial \dot{z}}{\partial \alpha_z}\right] \mathrm{d}x, \tag{2-7}$$

where I have omitted several terms due to the independence of their variables with respect to the α terms. I can then apply the arguments made in [1] on Equations 2-6 and 2-7 separately to conclude that J is stationary only if

$$\frac{\partial f}{\partial y} - \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial f}{\partial \dot{y}}\right) = 0, \tag{2-8}$$
$$\frac{\partial f}{\partial z} - \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial f}{\partial \dot{z}}\right) = 0. \tag{2-9}$$

To administer this result to my system, I take the infinitesimal path length $\mathrm{d}s = \sqrt{\mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2}$, I and define

$$J = \int_{s_1}^{s_2} \mathrm{d}s = \int_{s_1}^{s_2} \sqrt{\mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2} = \int_{x_1}^{x_2} \sqrt{1 + \dot{y}^2 + \dot{z}^2}\,\mathrm{d}x, \tag{2-10}$$

where $f = \sqrt{1 + \dot{y}^2 + \dot{z}^2}$ and the overdot denotes differentiation with respect to x. Noting that $f = f(\dot{y}, \dot{z})$, I immediately have $\frac{\partial f}{\partial y} = 0$ and $\frac{\partial f}{\partial z} = 0$. Therefore, Equations 2-8 and 2-9 reduce to

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial f}{\partial \dot{y}}\right) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\dot{y}}{\sqrt{1 + \dot{y}^2 + \dot{z}^2}}\right) = 0, \tag{2-11}$$
$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial f}{\partial \dot{z}}\right) = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\dot{z}}{\sqrt{1 + \dot{y}^2 + \dot{z}^2}}\right) = 0. \tag{2-12}$$

Taking Equation 2-11, I thus have that

$$\frac{\dot{y}}{\sqrt{1 + \dot{y}^2}} = m_1, \tag{2-13}$$

where m_1 is a constant. As in [1], this reduces to $\dot{y} = C$, and, defining the constant b_1 via the y-coordinate of the point corresponding to x_1, reduces in general to

$$y = m_1 x + b_1. \tag{2-14}$$

Similarly, Equation 2-12 reduces to

$$z = m_2 x + b_2, \tag{2-15}$$

so that the paramterezied curve is given by the straight line

$$\langle x, m_1 x + b_1, m_2 x + b_2 \rangle . \tag{2-16}$$

I have thus shown that a straight line is the extremum with respect to curve length in connecting two points; I physically interpret this value as the minimum path length, as desired.

Problem 2.2

Problem Statement

Show that the geodesics of a spherical surface are great circles, i.e. circles whose centers lies at the center of the sphere.

Using spherical polar coordinates, the spherical nature of the surface constrains r to a constant, so that any path can be completely characterized by its θ and ϕ variables. The differential in spherical polar coordinates can be derived as

$$\mathrm{d}s^2 = \mathrm{d}x^2 + \mathrm{d}y^2 + \mathrm{d}z^2, \tag{2-17}$$

and each of the scalar differentials $\mathrm{d}x$, $\mathrm{d}y$, and $\mathrm{d}z$ can be evaluated by considering their dependence on r, θ, and ϕ, i.e.

$$\mathrm{d}x = \frac{\partial x}{\partial r}\,\mathrm{d}r + \frac{\partial x}{\partial \theta}\,\mathrm{d}\theta + \frac{\partial x}{\partial \phi}\,\mathrm{d}\phi = \frac{\partial x}{\partial \theta}\,\mathrm{d}\theta + \frac{\partial x}{\partial \phi}\,\mathrm{d}\phi \tag{2-18}$$

noting that r is constant. Applying as such and performing some trigonometric reductions yields

$$\mathrm{d}s^2 = r^2 \sin^2(\phi)\,\mathrm{d}\theta^2 + r^2\,\mathrm{d}\phi^2, \tag{2-19}$$

so that

$$|\,\mathrm{d}s| = r\sqrt{\sin^2(\phi)\,\mathrm{d}\theta^2 + \mathrm{d}\theta^2} \tag{2-20}$$

For simplicitly, at this point I re-define $\phi \equiv x$ and $\theta \equiv y$. I can manipulate this one step further as

$$|\,\mathrm{d}s| = r\sqrt{\mathrm{d}x^2 + \sin^2(x)\,\mathrm{d}y^2} = r\sqrt{\left(\frac{\mathrm{d}y}{\mathrm{d}x}\right)^2 \sin^2(x) + 1}\,\mathrm{d}x = r\sqrt{\dot{y}^2 \sin^2(x) + 1}\,\mathrm{d}x, \tag{2-21}$$

where the overdot denotes differentiation with resepct to x. The path length between two points a and b on the sphere is given as

$$J = \int_a^b |ds| = r\int_a^b \sqrt{\dot{y}^2 \sin^2(x) + 1}\,\mathrm{d}x. \tag{2-22}$$

At this point I discard the r since it plays no further role in the development of the solution; Hamilton's principle tells us that the integral achieves an extrema when

$$\frac{\partial}{\partial y}\sqrt{\dot{y}^2\sin^2(x)+1} = \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial}{\partial \dot{y}}\sqrt{\dot{y}^2\sin^2(x)+1}\right) \tag{2-23}$$

I note that the left side evaluates to 0, so that $\frac{\partial}{\partial \dot{y}}\sqrt{\dot{y}^2\sin^2(x)+1}$ exhibits no dependence on x. Note that y need not be a well-defined function of x and hence we may not be able to assert this expression as constant – we can argue that a short path will not incessantly loop back on itself, but we can imagine a path in which a value of ϕ may be taken on multiple times. For the problem it hand, it works out that assuming the value as a constant is sufficient, but there may be a more mathematically rigorous way to approach this (differentiating with respect to y may work better, although the math looked more complex at first glance). I then explicitly compute

$$c = \frac{\partial}{\partial \dot{y}}\sqrt{\dot{y}^2\sin^2(x)+1} = \frac{\dot{y}\sin^2(x)}{\sqrt{\dot{y}^2\sin^2(x)+1}}, \tag{2-24}$$

which yields

$$\dot{y}^2 = \frac{c}{\sin^4(x)-c\sin^2(x)} \implies \dot{y} = \pm\frac{c}{\sin(x)}\frac{1}{\sqrt{\sin^2(x)-c^2}}. \tag{2-25}$$

Considering the desired geodesic form of a great circle, a clever guess on c yields $c = \sin(x_0)$, where x_0 denotes the maximal allowed value of x; the derivative becomes singular at this point, where y continues to change while x is kept constant. I associate the distinct parities as divided by these turning points, where the parity of the rate of change of y with resepct to x swaps; I can imagine covering the whole great circle by applying a two-part paramterization at these singular points where our variables cease to make functional sense. We still haven't shown that this minimal path length indeed describes a great circle; the easiest way to do this is graphically, where I observe that a numerical solution to my differential equation in spherical polar coordinates yields a half-circle on the intersection of the plane containing the two specified points and running through the origin with a sphere of constant radius for a variety of initial conditions. Therefore, I am convinced that this equation describes a parameterization of a great circle and hence that the geodesic of a sphere is a great circle.

Problem 2.3

Problem Statement

Complete the solution of the brachistochrone problem begun in Section 2-2 and show that the desired curve is a cycloid with a cusp at the initial point where the particle is released. Show also that if the particle is projected with an initial kinetic energy $\frac{1}{2}mv_0^2$ that the brachistochrone is still a cycloid passing thorugh the two points with a cusp at a height z above the initial point given by $v_0^2 = 2gz$.

We are given that the time taken for the ball to travel from point a to b is defined by

$$T = \int_a^b \sqrt{\frac{1+\dot{y}^2}{2gy}}. \tag{2-26}$$

To find the stationary value, we need

$$\frac{\partial f}{\partial y} - \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial f}{\partial \dot{y}}\right) = 0, \tag{2-27}$$

where $f = \sqrt{\frac{1+\dot{y}^2}{2gy}}$. I first compute

$$\frac{\partial f}{\partial y} = -g(1+\dot{y}^2)^{1/2}(2gy)^{-3/2}. \tag{2-28}$$

I also have

$$\frac{\partial f}{\partial \dot{y}} = \dot{y}(1+\dot{y}^2)^{-1/2}(2gy)^{-1/2}. \tag{2-29}$$

Thus,

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial f}{\partial \dot{y}}\right) = (2gy)^{-1/2}(1+\dot{y}^2)^{-1/2}\left[\ddot{y}\left(1-\frac{\dot{y}^2}{1+\dot{y}^2}\right) - \frac{\dot{y}^2}{2y}\right] \tag{2-30}$$

I can factor Equation 2-28 via

$$\frac{\partial f}{\partial y} = -\frac{g}{2gy}(1+\dot{y}^2)(2gy)^{-1/2}(1+\dot{y}^2)^{-1/2} = -\frac{1+\dot{y}^2}{2y}(2gy)^{-1/2}(1+\dot{y}^2)^{-1/2} \tag{2-31}$$

So that

$$\frac{\partial f}{\partial y} - \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial f}{\partial \dot{y}}\right) = (2gy)^{-1/2}(1+\dot{y}^2)^{-1/2}\left[-\frac{1+\dot{y}^2}{2y} - \ddot{y}\left(1-\frac{\dot{y}^2}{1+\dot{y}^2}\right) + \frac{\dot{y}^2}{2y}\right] = 0. \tag{2-32}$$

I can divide out to find

$$-\frac{1+\dot{y}^2}{2y} - \ddot{y}\left(1-\frac{\dot{y}^2}{1+\dot{y}^2}\right) + \frac{\dot{y}^2}{2y} = 0 \tag{2-33}$$

since our choice of coordinates in equating $mgy = \frac{mv^2}{2}$ implies that $y = 0$ only occurs right at point a (it would not make sense for a short path to repeatedly go back to the top, where it has no velocity), and moreover our equation is not well defined at $y = 0$ to begin with anyway. I can manipulate this to find

$$\ddot{y} = -\frac{1+\dot{y}^2}{2y} \tag{2-34}$$

We have a second-order differential equation, so if I can substitute an ansatz with two degrees of freedom I will have found the general solution. I will do so with the cycloid parameterized by an angle θ as

$$x = R(\theta - \sin(\theta)) + C, \tag{2-35}$$
$$y = R(1-\cos(\theta)), \tag{2-36}$$

where R and C are constants of integration. Note that I have already included the initial condition $y(0) = 0$ since that was assumed in constructing the differential equation. This condition constrains our solution to start at $\theta_a = 0$, i.e. at the cusp. I can compute

$$\dot{y} = \frac{dy/d\theta}{dx/d\theta} = \frac{\sin(\theta)}{1-\cos(\theta)} \tag{2-37}$$

and

$$\ddot{y} = \frac{\mathrm{d}\dot{y}}{\mathrm{d}x} = \frac{\mathrm{d}\theta}{\mathrm{d}x}\frac{\mathrm{d}\dot{y}}{\mathrm{d}\theta} = \frac{1}{R}\frac{1}{1-\cos(\theta)}\frac{\cos(\theta)-1}{(1-\cos(\theta))^2} = -\frac{1}{R(1-\cos(\theta))^2}, \tag{2-38}$$

where the overdot denotes once again differentiation with respect to x. I can then compute

$$-\frac{\dot{y}^2+1}{2y} = -\frac{\frac{\sin^2(\theta)}{(1-\cos(\theta))^2}+1}{2R(1-\cos(\theta))} = -\frac{\frac{\sin^2(\theta)}{1-\cos(\theta)}+1-\cos(\theta)}{2R(1-\cos(\theta))^2} = -\frac{\frac{\sin^2(\theta)(1+\cos(\theta))}{1-\cos^2(\theta)}+1-\cos(\theta)}{2R(1-\cos(\theta))^2} \tag{2-39}$$

$$= -\frac{\frac{\sin^2(\theta)(1+\cos(\theta))}{\sin^2(\theta)}+1-\cos(\theta)}{2R(1-\cos(\theta))^2} = -\frac{1}{R(1-\cos(\theta))^2} = \ddot{y}, \tag{2-40}$$

as desired. We can choose our coordinate system such that the x-axis points in the horizontal direction from a towards b, and without loss of generality take $x(0) = 0$; this constrains $C = 0$, i.e.

$$x = R(\theta - \sin(\theta)), \tag{2-41}$$

$$y = R(1-\cos(\theta)). \tag{2-42}$$

The parameter R is in general more difficult to determine, since we do not know the θ value corresponding to point b; we thus first numerically solve for θ_b. and substitute this into the parameterized equations to determine R. I now let the particle begin with an initial velcoity $\frac{1}{2}mv_0^2$. The relevant energy expression gives $E_0 = \frac{1}{2}mv_0^2 = \frac{1}{2}mv^2 - mgy = E$, which yields $v^2 = v_0^2 + 2gy$. The integrated time expression is much the same, with

$$T = \int_a^b \frac{\mathrm{d}s}{v} = \int_a^b \sqrt{\frac{1+\dot{y}^2}{v_0^2+2gy}}\,\mathrm{d}x. \tag{2-43}$$

The stationary condition proceeds nearly exactly the same, but we are unable to complately simplify the v_0^2 term to find the previous differential equation:

$$\ddot{y} = -\frac{g(1+\dot{y}^2)}{v_0^2+2gy} = -\frac{1+\dot{y}^2}{v_0^2/g+2y} = -\frac{1+\dot{y}^2}{2(v_0^2/2g+y)} \tag{2-44}$$

I must incorporate the constant offset into my parameterization for y in order for the cycloid to remain a solution. I can do so as

$$x = R(\theta - \sin(\theta)) + C, \tag{2-45}$$

$$y = R(1-\cos(\theta)) - \frac{mv_0^2}{2g}, \tag{2-46}$$

which solves the differential equation upon evlaution since the constant offset is not present in higher-order derivatives. Note, however, that in order to satisfy $y(\theta_a) = 0$, I must have $\theta_a > 0$, i.e. we do not start from the cusp of the cycloid. The exact values of R and C will be determined from the initial conditions and may need to be approximated numerically. However, I can state in general that the cusp does indeed start at a height $mv_0^2/2g$ above the release point a and that by construction the cycloid passes thorugh both points.

Problem 2.4

Problem Statement

Find the Euler-Lagrange equation describing the brachisochrone curve for a particle moving *inside* a spherical Earth of uniform mass density. Obtain a first ingegral for this differential equation by analogy to the Jacboi integral h. With the help of this integral show that the desired curve is a hypocycloid. Obtain an express for the time of travel along the brachistocrone between two points on the surface of the Earth. How long would it take to go from New York to Los Angeles (∼3000 miles) along a brachistochrone tunnel (assuming no friction) and how far below the surface would the deepest point fo the tunnel be?

I begin with the expression for time

$$t_{12} = \int_1^2 \frac{\mathrm{d}s}{v}. \tag{2-47}$$

To determine the velocity of the particle within the Earth, I can compute the work done by the gravitational force. Firstly, it is a well-established theorem that in a symmetric mass distribution, a particle feels the gravitational force owing to the mass contained at a radius interior to it. At some radius $r < r_0$, where r is the radius of the Earth, the relevant interior mass M is given as

$$M = \rho V = \frac{M_0 V}{V_0} = M_0 \frac{4\pi r^3/3}{4\pi r_0^3/3} = \frac{M_0 r^3}{r_0^3}, \tag{2-48}$$

where ρ is the mass density. The corresponding gravitational force is thus

$$\mathbf{F}_g = -\frac{GMm}{r^2}\hat{\mathbf{r}} = -\frac{GM_0 mr}{r_0^3}\hat{\mathbf{r}}. \tag{2-49}$$

I can now compute the work done by the gravitational force as

$$W = \int_{r_0}^{r} \mathbf{F}_g \cdot \mathrm{d}\mathbf{r} = \int_r^{r_0} \frac{GM_0 mr}{r_0^3}\,\mathrm{d}r = \frac{GM_0 m}{2r_0^3}\left(r_0^2 - r^2\right). \tag{2-50}$$

With the assumption that the particle begins with no velocity, I can set

$$\frac{mv^2}{2} = K = \Delta K = W = \frac{GM_0 m}{2r_0^3}\left(r_0^2 - r^2\right) \implies v^2 = \frac{GM_0}{r_0^3}\left(r_0^2 - r^2\right) \implies v = \sqrt{\frac{GM_0}{r_0^3}\left(r_0^2 - r^2\right)}. \tag{2-51}$$

I note that my velocity is dependent on the radius r; polar coordinates are therefore a natural coordinate system to describe the brachistochrone. I can write

$$\mathrm{d}s^2 = \mathrm{d}r^2 + r^2\,\mathrm{d}\theta^2, \tag{2-52}$$

so that

$$\mathrm{d}s = \sqrt{\mathrm{d}s^2} = \sqrt{\mathrm{d}r^2 + r^2\,\mathrm{d}\theta^2} = \sqrt{\left(\frac{\mathrm{d}r}{\mathrm{d}\theta}\right)^2 + r^2}\,\mathrm{d}\theta = \sqrt{\dot{r}^2 + r^2}\,\mathrm{d}\theta, \tag{2-53}$$

where the overdot denotes differentiation with respect to θ. I can rewrite the time expression as

$$t_{1,2} = \int_1^2 \frac{\mathrm{d}s}{v} = \sqrt{\frac{r_0^3}{GM_0}} \int_1^2 \sqrt{\frac{\dot{y}^2 + y^2}{y_0^2 - y^2}}\,\mathrm{d}x, \tag{2-54}$$

where I have renamed θ as x and r as y. Therefore, the question we seek is to find the stationary value of

$$f = \sqrt{\frac{\dot{y}^2 + y^2}{y_0^2 - y^2}}. \tag{2-55}$$

To find Euler-Lagrange equation, I first compute

$$\frac{\partial f}{\partial y} = \frac{y(\dot{y}^2 + y^2)^{-1/2}(y_0^2 - y^2)^{1/2} + y(y_0^2 - y^2)^{-1/2}(\dot{y}^2 + y^2)^{1/2}}{y_0^2 - y^2} \tag{2-56}$$

$$= y\left[(\dot{y}^2 + y^2)^{-1/2}(y_0^2 - y^2)^{-1/2} + (\dot{y}^2 + y^2)^{1/2}(y_0^2 - y^2)^{-3/2}\right] \tag{2-57}$$

$$= y(\dot{y}^2 + y^2)^{-1/2}(y_0^2 - y^2)^{-1/2}\left(1 + \frac{\dot{y}^2 + y^2}{y_0^2 - y^2}\right) \tag{2-58}$$

Next,

$$\frac{\partial f}{\partial \dot{y}} = \dot{y}(\dot{y}^2 + y^2)^{-1/2}(y_0^2 - y^2)^{-1/2} \tag{2-59}$$

I compute the total derative

$$\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial f}{\partial \dot{y}}\right) = \left(\ddot{y}(\dot{y}^2 + y^2)^{-1/2} - \frac{1}{2}(\dot{y}^2 + y^2)^{-3/2}\dot{y}(2\dot{y}\ddot{y} + 2y\dot{y})\right)(y_0^2 - y^2)^{-1/2} \tag{2-60}$$

$$+ \frac{1}{2}(y_0^2 - y^2)^{-3/2}2y\dot{y}^2(\dot{y}^2 + y^2)^{-1/2} \tag{2-61}$$

$$= (\dot{y}^2 + y^2)^{-1/2}(y_0^2 - y^2)^{-1/2}\left[\ddot{y}\left(1 - \frac{\dot{y}^2}{\dot{y}^2 + y^2}\right) + \frac{\dot{y}^2 y}{y_0^2 - y^2} - \frac{\dot{y}^2 y}{\dot{y}^2 + y^2}\right], \tag{2-62}$$

So that

$$y\left(1 + \frac{\dot{y}^2 + y^2}{y_0^2 - y^2}\right) = \left[\ddot{y}\left(1 - \frac{\dot{y}^2}{\dot{y}^2 + y^2}\right) + \frac{\dot{y}^2 y}{y_0^2 - y^2} - \frac{\dot{y}^2 y}{\dot{y}^2 + y^2}\right]. \tag{2-63}$$

This expression can be simplified further but we will instead appeal to Jacobi's integral; we note that that the derivation on pp. 60-61 holds exactly for a function $f(x, y, \dot{y})$ satisfying the Euluer-lagrange equations, as we have here, with the substitutions $t \mapsto x$, $q \mapsto y$, and $\dot{q} \mapsto \dot{y}$; since f is not explicitly dependent upon x, I therefore conclude that

$$h = \dot{y}\frac{\partial f}{\partial \dot{y}} - f = C, \tag{2-64}$$

where C is constant. I have already compute the necessary parts for this equality; I thus have

$$h = \dot{y}\dot{y}(\dot{y}^2 + y^2)^{-1/2}(y_0^2 - y^2)^{-1/2} - (\dot{y}^2 + y^2)^{1/2}(y_0^2 - y^2)^{-1/2} \tag{2-65}$$

$$= (y_0^2 - y^2)^{-1/2}\left(\frac{\dot{y}^2}{\sqrt{\dot{y}^2 + y^2}} - \sqrt{\dot{y}^2 + y^2}\right) \tag{2-66}$$

$$= (y_0^2 - y^2)^{-1/2}\left(\frac{\dot{y}^2 - (\dot{y}^2 + y^2)}{\sqrt{\dot{y}^2 + y^2}}\right) \tag{2-67}$$

$$= \frac{y^2}{\sqrt{\dot{y}^2 + y^2}\sqrt{y_0^2 - y^2}} = C. \tag{2-68}$$

I am looking for a general solution to this differential equation with 2 free parameters, since I already have C as a free parameter and will pick up an additional parameter due to the first-order nature of the differential equation. I want to show that the hypocycloid will yield this; I must fix the outer radius as r_0 to satisfy the physical constraints of the problem (assuming the particle starts at rest) since

there is no energetically permissible position for the particle beyond r_0. Introducing the parameter ϕ, and switching my notation back to traditional r and θ, I can write the proposed solution in cartesian coordinates as

$$x(\phi) = (r_0 - b)\cos(\phi - \phi_0) + b\cos\left(\frac{r_0 - b}{b}(\phi - \phi_0)\right), \tag{2-69}$$

$$y(\phi) = (r_0 - b)\sin(\phi - \phi_0) - b\sin\left(\frac{r_0 - b}{b}(\phi - \phi_0)\right), \tag{2-70}$$

where a and ϕ_0 are my free parameters. I can solve for $r(\phi)$ by

$$r(\phi)^2 = x(\phi)^2 + y(\phi)^2 = (r_0 - b)^2 + b^2 \tag{2-71}$$

$$+ 2(r_0 - b)b\left[\cos(\phi - \phi_0)\cos\left(\frac{r_0 - b}{b}(\phi - \phi_0)\right) - \sin(\phi - \phi_0)\sin\left(\frac{r_0 - b}{b}(\phi - \phi_0)\right)\right] \tag{2-72}$$

$$= r_0^2 - 2r_0 b + b^2 + b^2 + 2(r_0 - b)b\cos\left((\phi - \phi_0) + \frac{r_0 - b}{b}(\phi - \phi_0)\right) \tag{2-73}$$

$$= r_0^2 + 2(b^2 - r_0 b)\left(1 - \cos\left(\frac{r_0}{b}(\phi - \phi_0)\right)\right), \tag{2-74}$$

and hence

$$r(\phi) = \left(r_0^2 + 2(b^2 - r_0 b)\left(1 - \cos\left(\frac{r_0}{b}(\phi - \phi_0)\right)\right)\right)^{1/2} \tag{2-75}$$

Next,

$$\theta(\phi) = \arctan\left(\frac{y(\phi)}{x(\phi)}\right) = \arctan\left[\frac{(r_0 - b)\sin(\phi - \phi_0) - b\sin\left(\frac{r_0 - b}{b}(\phi - \phi_0)\right)}{(r_0 - b)\cos(\phi - \phi_0) + b\cos\left(\frac{r_0 - b}{b}(\phi - \phi_0)\right)}\right]. \tag{2-76}$$

I next need to find the derivative $\frac{\mathrm{d}\theta}{\mathrm{d}\phi}$; I can use this to find

$$\dot{r}(\phi) = \frac{\mathrm{d}r}{\mathrm{d}\theta} = \frac{\mathrm{d}r}{\mathrm{d}\phi}\frac{\mathrm{d}\phi}{\mathrm{d}\theta} \tag{2-77}$$

by taking the inverse of $\frac{\mathrm{d}\theta}{\mathrm{d}\phi}$. I compute

$$\frac{\mathrm{d}\theta}{\mathrm{d}\phi} = \frac{1}{1 + (\frac{y}{x})^2}\frac{x\frac{\mathrm{d}y}{\mathrm{d}\phi} - y\frac{\mathrm{d}x}{\mathrm{d}\phi}}{x^2} = \frac{x\frac{\mathrm{d}y}{\mathrm{d}\phi} - y\frac{\mathrm{d}x}{\mathrm{d}\phi}}{x^2 + y^2} = \frac{x\frac{\mathrm{d}y}{\mathrm{d}\phi} - y\frac{\mathrm{d}x}{\mathrm{d}\phi}}{r^2} \tag{2-78}$$

$$= \frac{\left[(r_0 - b)\cos(\phi - \phi_0) - (r_0 - b)\cos\left(\frac{r_0 - b}{b}(\phi - \phi_0)\right)\right]\left[(r_0 - b)\cos(\phi - \phi_0) + b\cos\left(\frac{r_0 - b}{b}(\phi - \phi_0)\right)\right]}{r^2} \tag{2-79}$$

$$+ \frac{\left[(r_0 - b)\sin(\phi - \phi_0) + (r_0 - b)\sin\left(\frac{r_0 - b}{b}(\phi - \phi_0)\right)\right]\left[(r_0 - b)\sin(\phi - \phi_0) - b\sin\left(\frac{r_0 - b}{b}(\phi - \phi_0)\right)\right]}{r^2} \tag{2-80}$$

$$= \frac{(r_0 - b)^2(\cos^2(\phi - \phi_0) + \sin^2(\phi - \phi_0)) - b(r_0 - b)\left(\cos^2\left(\frac{r_0 - b}{b}(\phi - \phi_0)\right) + \sin^2\left(\frac{r_0 - b}{b}(\phi - \phi_0)\right)\right)}{r^2} \tag{2-81}$$

$$+ \frac{(b(r_0 - b) - (r_0 - b)^2)\left[\cos(\phi - \phi_0)\cos\left(\frac{r_0 - b}{b}(\phi - \phi_0)\right) - \sin(\phi - \phi_0)\sin\left(\frac{r_0 - b}{b}(\phi - \phi_0)\right)\right]}{r^2}. \tag{2-82}$$

Applying the cosine subtraction identity yields

$$\frac{\mathrm{d}\theta}{\mathrm{d}\phi} = \frac{(r_0-b)^2 - b(r_0-b)}{r^2}\left[1-\cos\left((\phi-\phi_0)+\frac{r_0-b}{b}(\phi-\phi_0)\right)\right] \tag{2-83}$$

$$= \frac{(r_0-b)^2 - b(r_0-b)}{r^2}\left[1-\cos\left(\frac{r_0}{b}(\phi-\phi_0)\right)\right] \tag{2-84}$$

$$= \frac{(r_0-b)^2 - b(r_0-b)}{r^2}\left[1-\cos(x)\right], \tag{2-85}$$

where I have introduced the parameter $x := \frac{r_0}{b}(\phi-\phi_0)$ for convenience. Next, I have

$$\frac{\mathrm{d}r}{\mathrm{d}\phi} = \frac{2(b^2-r_0b)\frac{r_0}{b}\sin\left(\frac{r_0}{b}(\phi-\phi_0)\right)}{2r} = \frac{(r_0^2-r_0b)\sin(x)}{r} \tag{2-86}$$

so that

$$\dot{r} = \frac{\mathrm{d}r}{\mathrm{d}\phi}\frac{\mathrm{d}\phi}{\mathrm{d}\theta} = \frac{(r_0^2-r_0b)\sin(x)}{r}\frac{r^2}{(1-\cos(x))((r_0-b)^2-b(r_0-b))} \tag{2-87}$$

$$= \frac{r_0(r_0-b)}{(r_0-b)(r_0-2b)}\frac{r\sin(x)}{1-\cos(x)} \tag{2-88}$$

$$= A\frac{\sin(x)}{1-\cos(x)}r, \tag{2-89}$$

where $A := \frac{r_0}{r_0-2b}$. Next, I have

$$C = \frac{r^2}{\sqrt{\dot{r}^2+r^2}\sqrt{r_0^2-r^2}} = \frac{1}{\sqrt{\left(\frac{\dot{r}}{r}\right)^2+1}\sqrt{\left(\frac{r_0}{r}\right)^2-1}}. \tag{2-90}$$

To elcudiate further algebra, I compute the inverse $D := 1/C$ by

$$D = \sqrt{\left(\frac{\dot{r}}{r}\right)^2+1}\sqrt{\left(\frac{r_0}{r}\right)^2-1} = \sqrt{A^2\frac{\sin^2(x)}{(1-\cos(x))^2}+1}\sqrt{\frac{r_0^2}{r_0^2+2(b^2-r_0b)(1-\cos(x))}-1} \tag{2-91}$$

$$= \sqrt{\frac{A^2\sin^2(x)+(1-\cos(x))^2}{(1-\cos(x))^2}}\sqrt{\frac{2(r_0b-b^2)(1-\cos(x))}{r_0^2+2(b^2-r_0b)(1-\cos(x))}} \tag{2-92}$$

$$= \sqrt{\frac{A^2\sin^2(x)+(1-\cos(x))^2}{1-\cos(x)}}\sqrt{\frac{2(r_0b-b^2)}{r_0^2+2(b^2-r_0b)(1-\cos(x))}} \tag{2-93}$$

$$= \sqrt{\frac{2A^2(r_0b-b^2)\sin^2(x)+2(r_0b-b^2)(1-\cos(x))^2}{r_0^2(1-\cos(x))-2(r_0b-b^2)(1-\cos(x))^2}}. \tag{2-94}$$

I can now start factoring out some terms:

$$D = \sqrt{\frac{2A^2\sin^2(x) + 2(1-\cos(x))^2}{\frac{r_0^2}{r_0 b - b^2}(1-\cos(x)) - 2(1-\cos(x))^2}} \tag{2-95}$$

$$= \sqrt{\frac{\frac{2r_0^2\sin^2(x)}{(r_0-2b)^2} + 2(1-\cos(x))^2}{\frac{r_0^2}{r_0 b - b^2}(1-\cos(x)) - 2(1-\cos(x))^2}} \tag{2-96}$$

$$= \sqrt{\frac{\frac{2r_0^2(1+\cos(x))}{(r_0-2b)^2} + 2(1-\cos(x))}{\frac{r_0^2}{r_0 b - b^2} - 2(1-\cos(x))}} \tag{2-97}$$

$$= \frac{\sqrt{2(r_0 b - b^2)}}{r_0 - 2b}\sqrt{\frac{r_0^2(1+\cos(x)) + (r_0-2b)^2(1-\cos(x))}{r_0^2 - 2(r_0 b - b^2)(1-\cos(x))}}. \tag{2-98}$$

Expanding yields

$$D = \frac{\sqrt{2(r_0 b - b^2)}}{r_0 - 2b}\sqrt{\frac{2r_0^2 - 4r_0 b + 4r_0 b\cos(x) + 4b^2 - 4b^2\cos(x)}{r_0^2 - 2r_0 b + 2r_0 b\cos(x) + 2b^2 - 2b^2\cos(x)}} \tag{2-99}$$

$$= 2\frac{\sqrt{r_0 b - b^2}}{r_0 - 2b} \tag{2-100}$$

and hence

$$C = \frac{r_0 - 2b}{2\sqrt{b(r_0 - b)}} \tag{2-101}$$

which is constant and determined by the initial conditions, as deisred. Note that in our construction various singularities have arisen; these correspond to the nodes of the hypocycloid and are inevitable but do not hinder the applicability of the curve to points between the nodes, which is the path of interest. To compute the time taken to travel between two points on the surface of the Earth,

$$T = \sqrt{\frac{r_0^3}{GM_0}}\int_1^2 \sqrt{\frac{\dot{r}^2 + r^2}{r_0^2 - r^2}}\,\mathrm{d}\theta = C\sqrt{\frac{r_0^3}{GM_0}}\int_1^2 1 + \frac{A^2\sin^2(x)}{(1-\cos(x))^2}\,\mathrm{d}\theta. \tag{2-102}$$

I can change my variable of integration to ϕ such that

$$T = C\sqrt{\frac{r_0^3}{GM_0}}\int_1^2 \left(1 + \frac{A^2\sin^2(x)}{(1-\cos(x))^2}\right)\frac{B(1-\cos(x))}{r^2}\,\mathrm{d}\phi \tag{2-103}$$

$$= BC\sqrt{\frac{r_0^3}{GM_0}}\int_1^2 \frac{1-\cos(x) + \frac{A^2\sin^2(x)}{1-\cos(x)}}{r^2}\,\mathrm{d}\phi \tag{2-104}$$

$$= BC\sqrt{\frac{r_0^3}{GM_0}}\int_1^2 \frac{1-\cos(x) + A^2(1+\cos(x))}{r^2}\,\mathrm{d}\phi \tag{2-105}$$

$$= BC\sqrt{\frac{r_0^3}{GM_0}}\int_1^2 \frac{(A^2-1)\cos(x) + A^2 + 1}{r_0^2 + 2(b^2 - r_0 b)(1-\cos(x))}\,\mathrm{d}\phi \tag{2-106}$$

where $B := ((r_0 - b)^2 - b(r_0 - b))$. I next switch to x, with 2π corresponding to one cusp-to-cusp trip:

$$T = \frac{BCb}{r_0}\sqrt{\frac{r_0^3}{GM_0}}\int_0^{2\pi} \frac{(A^2-1)\cos(x) + A^2 + 1}{r_0^2 + 2b^2 - 2r_0 b + 2(r_0 b - b^2)\cos(x)}\,\mathrm{d}x \tag{2-107}$$

To compute the integrand, I employ tangent half-angle substitution. In general, for constants A, B, C, and D (not our aforementioned constants – I apologize for the abuse of notation), I first have

$$\frac{A + B\cos(x)}{C + D\cos(x)} = \frac{A}{C + D\cos(x)} + \frac{B\cos(x)}{C + D\cos(x)} = \frac{A}{C + D\cos(x)} + \frac{B}{D}\left(\frac{C + D\cos(x)}{C + D\cos(x)} - \frac{C}{C + D\cos(x)}\right) \tag{2-108}$$

$$= \frac{B}{D} + \left(A - \frac{BC}{D}\right)\frac{1}{C + D\cos(x)}. \tag{2-109}$$

Thus,

$$\int_0^{2\pi} \frac{A + B\cos(x)}{C + D\cos(x)}\,\mathrm{d}x = \int_0^{2\pi} \frac{B}{D} + \left(A - \frac{BC}{D}\right)\frac{1}{C + D\cos(x)}\,\mathrm{d}x \tag{2-110}$$

$$= \frac{2\pi B}{D} + \int_0^{2\pi} \left(A - \frac{BC}{D}\right)\frac{1}{C + D\cos(x)}\,\mathrm{d}x \tag{2-111}$$

$$= \frac{2\pi B}{D} + \left(A - \frac{BC}{D}\right)\int_0^{2\pi} \frac{1}{C + D\cos(x)}\,\mathrm{d}x. \tag{2-112}$$

Then employing the standard subtitution $t = \tan(x/2)$, giving $\cos(x) = \frac{1-t^2}{1+t^2}$ and $\mathrm{d}x = \frac{2}{1+t^2}\,\mathrm{d}t$, and splitting the integral in two to avoid the singularity at $x = \pi$, I have

$$\int_0^{2\pi} \frac{1}{C + D\cos(x)}\,\mathrm{d}x = \int_0^{\infty} \frac{1}{C + D\frac{1-t^2}{1+t^2}}\frac{2}{1+t^2}\,\mathrm{d}t + \int_{-\infty}^{0} \frac{1}{C + D\frac{1-t^2}{1+t^2}}\frac{2}{1+t^2}\,\mathrm{d}t \tag{2-113}$$

$$= \int_{-\infty}^{\infty} \frac{1}{C + D\frac{1-t^2}{1+t^2}}\frac{2}{1+t^2}\,\mathrm{d}t \tag{2-114}$$

$$= \int_{-\infty}^{\infty} \frac{2}{C(1+t^2) + D(1-t^2)}\,\mathrm{d}t \tag{2-115}$$

$$= \int_{-\infty}^{\infty} \frac{2}{(C-D)t^2 + C + D}\,\mathrm{d}t \tag{2-116}$$

$$= \frac{2}{C+D}\int_{-\infty}^{\infty} \frac{1}{1 + \frac{C-D}{C+D}t^2}\,\mathrm{d}t. \tag{2-117}$$

I now introduce the parameter $u := \frac{C-D}{C+D}t$, giving

$$\int_0^{2\pi} \frac{1}{C + D\cos(x)}\,\mathrm{d}x = \frac{2}{C+D}\sqrt{\frac{C+D}{C-D}}\int_{-\infty}^{\infty} \frac{1}{1+u^2}\,\mathrm{d}u = \frac{2\pi}{\sqrt{(C+D)(C-D)}}. \tag{2-118}$$

Therefore,

$$\int_0^{2\pi} \frac{A + B\cos(x)}{C + D\cos(x)}\,\mathrm{d}x = \frac{2\pi B}{D} + \left(A - \frac{BC}{D}\right)\frac{2\pi}{\sqrt{(C+D)(C-D)}} \tag{2-119}$$

$$= 2\pi\left(\frac{B}{D} + \left(A - \frac{BC}{D}\right)\frac{1}{\sqrt{(C+D)(C-D)}}\right). \tag{2-120}$$

Now subtituting the relevant constants,

$$T = \frac{BCb}{r_0}\sqrt{\frac{r_0^3}{GM_0}}\int_0^{2\pi}\frac{A^2+1+(A^2-1)\cos(x)}{r_0^2+2b^2-2r_0b+2(r_0b-b^2)\cos(x)}\,\mathrm{d}x \tag{2-121}$$

$$= 2\pi\frac{BCb}{r_0}\sqrt{\frac{r_0^3}{GM_0}}\left(\frac{A^2-1}{2(r_0b-b^2)}+\left(A^2+1-\frac{(A^2-1)(r_0^2+2b^2-2r_0b)}{2(r_0b-b^2)}\right)\frac{1}{\sqrt{r_0^2(r_0^2+4b^2-4r_0b)}}\right). \tag{2-122}$$

This is what we desire – an integrated expression for T in terms of the constants r_0, b, G, and M_0. This reduces to

$$T = 2\pi\sqrt{\frac{r_0b(r_0-b)}{GM_0}}. \tag{2-123}$$

For the particular case considered, I need to find b. This can be done by identifying the value of b which yields $r = r_0$, i.e. $\frac{r_0}{b}\phi = 2\pi$, which yields $\phi_f = \frac{2\pi b}{r_0}$. I comptue directly that

$$\theta_f = \arctan\left[\frac{(r_0-b)\sin\left(\frac{2\pi b}{r_0}\right)-b\sin\left(\frac{r_0-b}{b}\frac{2\pi b}{r_0}\right)}{(r_0-b)\cos\left(\frac{2\pi b}{r_0}\right)+b\cos\left(\frac{r_0-b}{b}\frac{2\pi b}{r_0}\right)}\right] \tag{2-124}$$

$$= \arctan\left[\frac{(r_0-b)\sin\left(\frac{2\pi b}{r_0}\right)-b\sin\left(2\pi-\frac{2\pi b}{r_0}\right)}{(r_0-b)\cos\left(\frac{2\pi b}{r_0}\right)+b\cos\left(2\pi-\frac{2\pi b}{r_0}\right)}\right] \tag{2-125}$$

$$= \arctan\left[\frac{(r_0-b)\sin\left(\frac{2\pi b}{r_0}\right)+b\sin\left(\frac{2\pi b}{r_0}\right)}{(r_0-b)\cos\left(\frac{2\pi b}{r_0}\right)+b\cos\left(\frac{2\pi b}{r_0}\right)}\right] \tag{2-126}$$

$$= \arctan\left(\tan(\phi_f)\right) = \phi_f. \tag{2-127}$$

I know physically that θ_f corresponds to the arc subtended on the path from Los Angeles to New York; the arc is ∼3000 miles while the circumference of the Earth is ∼25,000 miles, so that

$$\theta_f \approx 2\pi\frac{3000}{25000} = 0.75, \tag{2-128}$$

and I can solve numerically for b with $r_0 = 6.371\cdot10^6$ m as

$$b = \frac{\phi_f r_0}{2\pi} = 7.64\cdot10^4 \text{ m}. \tag{2-129}$$

Finally, with $G = 6.67\cdot10^{-11}\mathrm{N\,m^2\,kg^{-1}}$ and $M_0 = 5.97\cdot10^{24}\,\mathrm{kg}$, the time taken to traverse the path is given as

$$T = 2\pi\sqrt{\frac{r_0b(r_0-b)}{GM_0}} = 1645\,\mathrm{s} \approx 27.4\,\mathrm{min} \tag{2-130}$$

The minimal value of r can be found when the cosine argument is minimized, i.e.

$$r_{\min} = \sqrt{r_0^2+4(b^2-r_0b)} = 4.84\cdot10^6\,\mathrm{m}, \tag{2-131}$$

which corresponds to a depth of $r_0 - r_{\min} \approx 1.5\cdot10^6\,\mathrm{m} = 1500\,\mathrm{km}$.

TODO: find simplification of Equation 2-121 w/o Mathematca. show that the only physically interpretable ϕ_f is the first cusp, i.e. it should not be the second cusp, etc.

Problem 2.5

Problem Statement

In the problem of the minimum surface of revolution examine the symmetric case $x_1 = x_2$, $y_2 = -y_1 > 0$, and express the condition for the parameter a as a transcendental equation in terms of the dimensionless quantities $k = x_2/a$, and $\alpha = y_2/x_2$ [a]. Show that for α less than a certain value α_0 two values of k are possible, for $\alpha = \alpha_0$ only one value of k is possible, while if $\alpha > \alpha_0$ no real value of k (or a) can be found, so that no catenary solution exists in this region. [b]. Find the value of α_0, numerically if necessary.

[a] Note than [1] swtiches x and y. That is not a solveable problem and was fixed in the third ediiton to reflect the version presented here.

[b] The ordering on α is flipped – this is a recognized error from Golstein. (https://math.stackexchange.com/questions/3812386/help-with-minimal-surface-of-revolution-symmetric-case)

The initial conditions yield

$$x_2 = a\cosh\left(\frac{y_2 - b}{a}\right) = a\cosh\left(\frac{-y_2 - b}{a}\right) = x_2. \tag{2-132}$$

Equality in the cosh evaluation requires the magnitude of the arguments to be identical; this yields

$$\left|\frac{y_2 - b}{a}\right| = \left|\frac{-y_2 - b}{a}\right| \implies |y_2 - b| = |-y_2 - b|\,, \tag{2-133}$$

which, since $y_2 \neq 0$, implies $b = 0$. Therefore,

$$x_2 = a\cosh\frac{y_2}{a}. \tag{2-134}$$

Introducing the relevant parameters gives

$$k = \cosh(\alpha k), \tag{2-135}$$

the desired transcendental form. Now consider the infinitely differentiable function $f(k) = \cosh(\alpha k) - k$. The allowed values of k correspond precisely to $f(k) = 0$. I first compute the first and second derivatives:

$$f'(k) = \alpha\sinh(\alpha k) - 1, \tag{2-136}$$
$$f''(k) = \alpha^2\cosh^2(\alpha k). \tag{2-137}$$

Note that $f''(k) < 0$ for all $k \in \mathbb{R}$. I now consider $k = 0$: I have

$$f(0) = 1, \tag{2-138}$$
$$f'(0) = -1. \tag{2-139}$$

I will show that there exists some $k_1 > 0 \in \mathbb{R}$ such that $f(k_1) < 1$. To start, I will identify a k_1 with $f'(k_1) < -1$. Since $f'(k)$ is continuous,

$$\lim_{k\to 0} f'(k) = -1. \tag{2-140}$$

By the definition of the limit, taking $\epsilon = 1/2$, there exists some $\delta > 0$ such that for all $|k| < \delta$, I have $|f'(k) + 1| < 1/2$, in particular $f'(k) < 0$. Taking $k_1 = \delta/2$, I thus have $f'(k) < 0$ on the interval $[0, k_1]$. Therefore, clearly $f(k_1) < f(0) = 1$. Next, I will find a $k_2 > 0$ such that $f(k_2) > 1$. That is, I need to find a k_2 such that

$$\frac{e^{\alpha k} + e^{-\alpha k}}{2} - k > 1 \iff \frac{e^{\alpha k} + e^{-\alpha k} - 2k - 2}{2} > 0 \iff e^{\alpha k} + e^{-\alpha k} - 2k - 2 > 0. \tag{2-141}$$

A sufficient condition is then

$$e^{\alpha k} > \text{Max}(4, 4k). \tag{2-142}$$

Now fix some arbitrary α. Then it is readily observable that

$$k > \frac{\ln(4)}{\alpha} \tag{2-143}$$

establishes the first half of Equation 2-142. The second half proceeds as follows, restricting ourselves to $k > \frac{1}{4}$:

$$e^{\alpha k} > 4k \iff \alpha k > \ln(4k) \iff \frac{\alpha k}{\ln(4k)} \iff \frac{\alpha}{4}\frac{u}{\ln(u)} > 1 \iff \frac{u}{\ln(u)} > \frac{4}{\alpha} \tag{2-144}$$

where $u = 4k$. Since

$$\lim_{u\to\infty} \frac{u}{\ln(u)} = \infty, \tag{2-145}$$

I can thus choose u (and hence k) such that $\frac{u}{\ln(u)}$ is arbitrarily large and in particular greater than $\frac{4}{\alpha}$. I choose such a k value as k_2. By the Intermediate Value Theorem, there exists a k_3 on the interval $[k_1, k_2]$ such that $f(k_3) = 1$, since $f(k_1) < 1$ and $f(k_2) > 1$. By Rolle's Theorem, there exists some $k_4 \in [0, k_3]$ such that $f'(k_4) = 0$; since $f''(k) > 0$, this is the only point at which $f'(k) = 0$. I wish to examine the examine the conditions on α owing to the parity of $f(k_4)$; I define the value α_0 as that which yields $f(k_4) = 0$, that is

$$f(k_4) = \cos(\alpha_0 k_4) - k_4 = 0 \implies \alpha_0 k_4 = \text{arcCosh}(k_4) \implies \alpha_0 = \frac{\text{arcCosh}(k_4)}{k_4}. \tag{2-146}$$

First, consider the case where $f(k_4) < 0$. This corresponds to

$$\cosh(\alpha k_4) - k_4 < 0 \implies \cosh(\alpha k_4) < k_4 \implies \alpha k_4 < \text{arcCosh}(k_4) \implies \alpha < \frac{k_4}{\text{arcCosh}\, k_4} = \alpha_0. \tag{2-147}$$

By the Intermediate Value Theorem there exists some earlier k_5 at which $f(k_5) = 0$. This point is unique since $f'(k) < 0$ on the interval $[0, k_4)$. Moreover, there exists some $k_6 \in (k_4, k_2]$ such that $f(k_6) = 0$ by the Intermediate Value Theorem, which is again unique since $f'(k) > 0$ on (k_4, ∞). Therefore, there are two permissable values of k for $\alpha < \alpha_0$.

Next, consider $f(k_4) = 0$. We have already seen that this corresponds to $\alpha = \alpha_0$. Since $f'(k_4) = 0$ and $f''(k) > 0$, it must be that $f'(k) < 0$ for $k < k_4$ and in particular $f(k) \neq 0$ for $k < k_4$. However, for $k > k_4$ I have by a similar argument that $f'(k) > 0$, so that $f(k) \neq 0$ for $k > k_4$. Therefore, k_4 is the only permissable k value.

Finally, consider the case where $f(k_4) > 0$. This corresponds to

$$\cosh(\alpha k_4) - k_4 > 0 \implies \cosh(\alpha k_4) > k_4 \implies \alpha k_4 > \text{arcCosh}(k_4) \implies \alpha > \frac{k_4}{\text{arcCosh}\, k_4} = \alpha_0. \tag{2-148}$$

Then just as in the previous case $f(k) \neq 0$ for $k < k_4$. Moreover, I apply the same argument to conclude $f(k) \neq 0$ for $k > k_4$. But since $f(k_4) \neq 0$, it must be that $f(k) \neq 0$ for all $k \in \mathbb{R}$ and in particular no permissable k values and no catenary solution.

I numerically solve $\alpha_0 \approx 0.663$ by setting $f(k) = 0$ and $f'(k) = 0$.

Problem 2.6

Problem Statement

For the symmetric situation, obtain an expression for the ratio of the area generated by the catenary solution to that given by the Goldschmidt solution. Your result should be a function only of the parameters k and α. Show that for sufficiently small values of α at least one of the catenaries gives an area below that of the Goldschmidt solution. On the other hand show that if $\alpha = \alpha_0$, the Goldschmidt solution gives a lower area than the catenary.

I can compute the area A_c of the catenary solution via

$$A_c = 2\pi \int_{-x_2}^{x_2} x\sqrt{1+\dot{y}^2}\,\mathrm{d}x = 4\pi \int_{a}^{x_2} x\sqrt{1+\dot{y}^2}\,\mathrm{d}x \tag{2-149}$$

upon recognition of the symmetric nature of the integral about the x-axis and from the integrated form that $y(0) = a$. Substituting $\dot{y}$ from the text yields

$$A_c = 4\pi \int_{a}^{x_2} x\sqrt{1+\frac{a^2}{x^2-a^2}}\,\mathrm{d}x = 4\pi \int_{a}^{x_2} x\sqrt{\frac{x^2}{x^2-a^2}}\,\mathrm{d}x = 4\pi \int_{a}^{x_2} \frac{x^2}{\sqrt{x^2-a^2}}\,\mathrm{d}x \tag{2-150}$$

The change of variables $u = \mathrm{arcCosh}(x/a)$ yields

$$A_c = 4\pi \int_0^{\alpha k} a^2\cosh^2(u)\,\mathrm{d}u = 2\pi a^2 \int_0^{\alpha k} 1+\cosh(2u)\,\mathrm{d}u = \pi a^2\left(2u+\sinh(2u)\right)\Big|_0^{\alpha k} \tag{2-151}$$

$$= \pi a^2(2\alpha k + \sinh(2\alpha k)). \tag{2-152}$$

The Goldschmidt solution area A_g is given as

$$A_g = \pi x_1^2 + \pi x_2^2 = 2\pi x_2^2, \tag{2-153}$$

and the ratio is given as

$$\frac{A_c}{A_g} = \frac{2\alpha k + \sinh(2\alpha k)}{2k^2}. \tag{2-154}$$

Now consider $f := 2k^2 - 2\alpha k - \sinh(2\alpha k)$. I have

$$\lim_{\alpha\to 0} f(k) = 2k^2 > 0, \tag{2-155}$$

which yields that $A_c > A_g$ at sufficiently low α. This argument works because the less of the catenary solutions k (where $\alpha < \alpha_0$ and hence there are two catenary solutions) is bounded by $k_0 \approx 1.81$, the k solution corresponding to α_0; we therefore can fix a k value and reduce α arbirtrarily small and do not have to deal with an unbounded k.

For $\alpha = \alpha_0$, I find $f(\alpha_0, k_0) = -1.3 < 0$, which yields the Goldschmidt solution with lesser area.

Problem 2.7

Problem Statement

A chain or rope of indefinite length passes freely over pulleys at heights y_1 and y_2 above the plane surface of the earth, with a horizontal distance $x_2 - x_1$ between them. If the chain or rope has a uniform linear mass density show that the problem of finding the curve assumed between the pulleys is identical with that of the problem of minimum surface of revolution.

The system will tend to the state of lowest energy. Therefore, I seek the stationary value of the energy E, and in particular the potential energy V since any kinetic term will vanish once the curve has

stabilized. Taking the zero of the gravitational potential to be the plane surface of the earth, and using the form $V = mgh$, I have

$$V = V_1 + V_c + V_2 = \frac{\lambda g y_1^2}{2} + V_c + \frac{\lambda g y_2^2}{2}, \tag{2-156}$$

where V_1 and V_2 describe the gravitational potential energy of the rope hanging off the pulley for the (x_1, y_1) and (x_2, y_2) pulleys, respectively, while V_c denotes the potential energy of the curved rope between the pulleys. Note that I have integrated over their heights to pick up the factor of $1/2$, and utilized the linear mass density λ to describe the mass. Note that V_1 and V_2 are invariant under transformation in the shape of the curve, so that the problem corresponds to minimizing V_c. I compute

$$V_c = \int_1^2 gy \, \mathrm{d}m = \int_1^2 \lambda g y \, \mathrm{d}s = \lambda g \int_1^2 y \sqrt{\mathrm{d}x^2 + \mathrm{d}y^2} = \lambda g \int_1^2 y \sqrt{1 + \dot{x}^2} \, \mathrm{d}y. \tag{2-157}$$

This is the exact problem of the minimum surface of revolution, but this time we revolve about the x-axis rather than the y-axis. Note that the fact that the rope can rest upon the plane surface of the Earth is essential in deriving this reuslt.

Problem 2.8

Problem Statement

Suppose that it was known experimentally that a particle fell a given distance y_0 in a time $t_0 = \sqrt{2y_0/g}$, but that the time of fall for distances other than y_0 were not known. Suppose further that the Lagrangian for the problem is known, but that instead of solving the equation of motion for y as a function of t, it is gussed that the functinoal form is

$$y = at + bt^2. \tag{2-158}$$

If the constants a and b are adjusted always so that the time to fall y_0 is correctly given by t_0, show directly that the integral

$$\int_0^{t_0} L \, \mathrm{d}t \tag{2-159}$$

is an extremum for real values of the coefficients only when $a = 0$ and $b = g/2$.

With the condition that $y(t_0) = y_0$, the parameter a can be expressed explicitly in terms of b. That is,

$$y_0 = at_0 + bt_0^2 = a\sqrt{\frac{2y_0}{g}} + \frac{2by_0}{g} \implies a = \sqrt{\frac{y_0}{2g}}(g - 2b) = \frac{t_0}{2}(g - 2b). \tag{2-160}$$

Using the functional form of $y(t)$, I can explicitly compute

$$L = T - V = \frac{m\dot{y}^2}{2} + mgy. \tag{2-161}$$

Therefore, we seek the staionary value of

$$\int_0^{t_0} \frac{m\dot{y}^2}{2} + mgy \, \mathrm{d}t = m \int_0^{t_0} \frac{\dot{y}^2}{2} + gy \, \mathrm{d}t. \tag{2-162}$$

With $\dot{y} = a + 2bt$, and neglecting the overall factor of m, I have

$$\int_0^{t_0} \frac{a^2 + 4abt + 4b^2t^2}{2} + g(at + bt^2)\,\mathrm{d}t = \int_0^{t_0} \frac{a^2}{2} + a(2b+g)t + b(2b+g)t^2\,\mathrm{d}t \tag{2-163}$$

$$= \int_0^{t_0} \frac{t_0^2}{8}(g-2b)^2 + \frac{t_0}{2}(2b+g)(g-2b)t + b(2b+g)t^2\,\mathrm{d}t \tag{2-164}$$

$$= \frac{t_0^3}{24}\left(9g^2 - 4bg + 4b^2\right) \tag{2-165}$$

Then the condition for the integral to be an extremum is given as

$$\frac{\mathrm{d}}{\mathrm{d}b}\left(9g^2 - 4bg + 4b^2\right) = 8b - 4g = 0, \tag{2-166}$$

requiring $b = g/2$. Substituting into a yields $a = 0$, as desired.

Problem 2.9

Problem Statement

When two billiard balls collide the instantaneous forces between them are very large but act only in an infinitesimal time Δt, in such a manner that the qantity

$$\int_{\Delta t} F\,\mathrm{d}t \tag{2-167}$$

remains finite. Such forces are described as *impulsive* forces, and the integral over Δt is known as the *impulse* of the force. Show that if impulsive forces are present Lagrange's equations may be transformed into

$$\left(\frac{\partial L}{\partial \dot{q}_j}\right)_f - \left(\frac{\partial L}{\partial \dot{q}_j}\right)_i = S_j, \tag{2-168}$$

where i and f refer to the state of the system before and after the impulse, S_j is the impulse of the generalized imulsive force corresponding to q_j, and L is the Lagrangian.

Lagrange's Equations can be written as

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{q}_j}\right) = Q_j + \frac{\partial T}{\partial q_j}. \tag{2-169}$$

Integrating over a time Δt yields

$$\int_{\Delta t} \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial T}{\partial \dot{q}_j}\right)\mathrm{d}t = \int_{\Delta t}\left(Q_j + \frac{\partial T}{\partial q_j}\right)\mathrm{d}t \tag{2-170}$$

$$\left(\frac{\partial T}{\partial \dot{q}_j}\right)_f - \left(\frac{\partial T}{\partial \dot{q}_j}\right)_i = S_j + \int_{\Delta t} \frac{\partial T}{\partial q_j}\,\mathrm{d}t, \tag{2-171}$$

where I take neglected the contributions of any other non-impulsive generalized force components present in formulating S_j. Note that the term $\frac{\partial T}{\partial q_j}$ is 0 in Cartesian coordinates, but in general is not 0. However, this term remains finite over the integration period and is vanishing relative to the impulsive force Q_j, so I neglect it here when integrating over an infinitesimal time Δt. This estalishes the desired equality:

$$\left(\frac{\partial T}{\partial \dot{q}_j}\right)_f - \left(\frac{\partial T}{\partial \dot{q}_j}\right)_i = S_j. \tag{2-172}$$

Problem 2.10

Problem Statement

The term *generalized mechanics* has come to designate a variety of classical mechanics in which the Lagrangian contains time derivatives of q_i higher than the first. By applying the methods of the calculus of variations, show that if there is a Lagrangian of the form $L(q, \dot{q}, \ddot{q}, t)$, and Hamilton's principle holds with the zero variation of both q_i and $\dot{q}_i$ at the end points, then the correspond Euler-Lagrange equations are

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}\left(\frac{\partial L}{\partial \ddot{q}_i}\right) - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}_i}\right) + \frac{\partial L}{\partial q_i} = 0, \qquad i = 1, 2, \ldots, n. \tag{2-173}$$

Apply this result to the Lagrangian

$$L = -\frac{m}{2}q\ddot{q} - \frac{k}{2}q^2. \tag{2-174}$$

Do you recognize the equations of motion?

I will solve this generically by finding the stationary condition for

$$J = \int_1^2 f(y_1(x), y_2(x), \ldots \dot{y}_1(x), \dot{y}_2(x), \ldots \ddot{y}_1(x), \ddot{y}_2(x), \ldots, x)\,\mathrm{d}x, \tag{2-175}$$

where, in the formulation

$$y_1(x, \alpha) = y_1(x, 0) + \alpha\eta_1(x), \tag{2-176}$$
$$y_2(x, \alpha) = y_2(x, 0) + \alpha\eta_2(x), \tag{2-177}$$
$$\vdots \qquad \vdots \qquad \vdots$$

I have the condition $\eta_i(x) = 0$ and $\eta_i'(x) = 0$ at x_1 and x_2 for $i = 1, 2, \ldots n$. I have

$$\frac{\mathrm{d}J}{\mathrm{d}\alpha} = \int_1^2 \sum_i \left(\frac{\partial f}{\partial y_i}\frac{\partial y_i}{\partial \alpha} + \frac{\partial f}{\partial \dot{y}_i}\frac{\partial \dot{y}_i}{\partial \alpha} + \frac{\partial f}{\partial \ddot{y}_i}\frac{\partial \ddot{y}_i}{\partial \alpha}\right)\mathrm{d}x. \tag{2-178}$$

Consider some fixed i. The relevant integrals are then

$$\int_1^2 \frac{\partial f}{\partial y_i}\frac{\partial y_i}{\partial \alpha}\,\mathrm{d}x + \int_1^2 \frac{\partial f}{\partial \dot{y}_i}\frac{\partial \dot{y}_i}{\partial \alpha}\,\mathrm{d}x + \int_1^2 \frac{\partial f}{\partial \ddot{y}_i}\frac{\partial \ddot{y}_i}{\partial \alpha}\,\mathrm{d}x. \tag{2-179}$$

The first two integrals have been treated in the text. The last integral can be rewritten as

$$\int_1^2 \frac{\partial f}{\partial \ddot{y}_i}\frac{\partial}{\partial \alpha}\left(\frac{\partial^2 y_i}{\partial x^2}\right)\mathrm{d}x = \int_1^2 \frac{\partial f}{\partial \ddot{y}_i}\frac{\partial}{\partial x}\left(\frac{\partial^2 y_i}{\partial x \partial \alpha}\right)\mathrm{d}x. \tag{2-180}$$

Integrating by parts gives

$$\int_1^2 \frac{\partial f}{\partial \ddot{y}_i}\frac{\partial}{\partial x}\left(\frac{\partial^2 y_i}{\partial x \partial \alpha}\right)\mathrm{d}x = \frac{\partial f}{\partial \ddot{y}_i}\frac{\partial^2 y}{\partial x \partial \alpha}\bigg|_1^2 - \int_1^2 \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial f}{\partial \ddot{y}_i}\right)\frac{\partial^2 y_i}{\partial x \partial \alpha}\,\mathrm{d}x \tag{2-181}$$
$$= \frac{\partial f}{\partial \ddot{y}_i}\frac{\partial \dot{y}}{\partial \alpha}\bigg|_1^2 - \int_1^2 \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial f}{\partial \ddot{y}_i}\right)\frac{\partial^2 y_i}{\partial x \partial \alpha}\,\mathrm{d}x \tag{2-182}$$
$$= -\int_1^2 \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial f}{\partial \ddot{y}_i}\right)\frac{\partial^2 y_i}{\partial x \partial \alpha}\,\mathrm{d}x \tag{2-183}$$
$$= -\int_1^2 \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial f}{\partial \ddot{y}_i}\right)\frac{\partial}{\partial x}\left(\frac{\partial y_i}{\partial \alpha}\right)\mathrm{d}x \tag{2-184}$$

where the boundary term vanishes owing to $\eta_i'(x_1) = 0 = \eta_i'(x_2)$. I can integrate by parts one more time to find

$$-\int_1^2 \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial f}{\partial \ddot{y}_i}\right)\frac{\partial}{\partial x}\left(\frac{\partial y_i}{\partial \alpha}\right)\mathrm{d}x = -\frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial f}{\partial \ddot{y}_i}\right)\frac{\partial y_i}{\partial \alpha}\bigg|_1^2 + \int_1^2 \frac{\mathrm{d}^2}{\mathrm{d}x^2}\left(\frac{\partial f}{\partial \ddot{y}_i}\right)\frac{\partial y_i}{\partial \alpha}\,\mathrm{d}x \tag{2-185}$$

$$= \int_1^2 \frac{\mathrm{d}^2}{\mathrm{d}x^2}\left(\frac{\partial f}{\partial \ddot{y}_i}\right)\frac{\partial y_i}{\partial \alpha}\,\mathrm{d}x, \tag{2-186}$$

with boundary terms vanishing due to the standard restriction $\eta_i(x) = 0$ at the boundaries. Therefore, substituting the results from the text, I can rewrite

$$\frac{\mathrm{d}J}{\mathrm{d}\alpha} = \int_1^2 \sum_i \left(\frac{\mathrm{d}^2}{\mathrm{d}x^2}\left(\frac{\partial f}{\partial \ddot{y}_i}\right) - \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial f}{\partial \dot{y}_i}\right) + \frac{\partial f}{\partial y_i}\right)\frac{\partial y_i}{\partial \alpha}\,\mathrm{d}x = 0 \tag{2-187}$$

Utilizing the δ notation, this is equivalent to

$$\delta J = \int_1^2 \sum_i \left(\frac{\mathrm{d}^2}{\mathrm{d}x^2}\left(\frac{\partial f}{\partial \ddot{y}_i}\right) - \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial f}{\partial \dot{y}_i}\right) + \frac{\partial f}{\partial y_i}\right)\delta y_i\,\mathrm{d}x = 0 \tag{2-188}$$

With each δy_i independent and arbitrary, the only way for this condition to hold is for

$$\frac{\mathrm{d}^2}{\mathrm{d}x^2}\left(\frac{\partial f}{\partial \ddot{y}_i}\right) - \frac{\mathrm{d}}{\mathrm{d}x}\left(\frac{\partial f}{\partial \dot{y}_i}\right) + \frac{\partial f}{\partial y_i} = 0, \qquad i = 1, 2, \ldots n. \tag{2-189}$$

Finally, with the substitutions $f \to L$, $y_i \to q_i$, and $x \to t$, the assumed version of Hamilton's principle implies that

$$\frac{\mathrm{d}^2}{\mathrm{d}t^2}\left(\frac{\partial L}{\partial \ddot{q}_i}\right) - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}_i}\right) + \frac{\partial L}{\partial q_i} = 0, \qquad i = 1, 2, \ldots, n, \tag{2-190}$$

as desired. With the given Lagrangian, the relevant partial deriatives are

$$\frac{\partial L}{\partial q} = -\frac{m}{2}\ddot{q} - kq, \tag{2-191}$$

$$\frac{\partial L}{\partial \dot{q}} = 0, \tag{2-192}$$

$$\frac{\partial L}{\partial \ddot{q}} = -\frac{m}{2}q. \tag{2-193}$$

Therefore,

$$0 = \frac{\mathrm{d}^2}{\mathrm{d}t^2}\left(\frac{\partial L}{\partial \ddot{q}}\right) - \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\partial L}{\partial \dot{q}}\right) + \frac{\partial L}{\partial q} \tag{2-194}$$

$$= \frac{\mathrm{d}^2}{\mathrm{d}t^2}\left(-\frac{m}{2}q\right) - \frac{\mathrm{d}}{\mathrm{d}t}(0) - \frac{m}{2}\ddot{q} - kq \tag{2-195}$$

$$= -m\ddot{q} - kq, \tag{2-196}$$

which is the simple harmonic oscillator.

Problem 2.11

Problem Statement

A heavy particle is placed at the top of a vertical hoop. Calculate the reaction of the hoop on the particle by means of Lagrange's undetermined multipliers and Lagrange's equations. Find the height at which the particle falls off.

The hoop is assumed to be fixed in place. The system can be treated entirely by a single generalized coordinate θ, where θ denotes the angle relative to the vertical line running through the center of the hoop and extending upwards. However, to understand the constraint forces at hand I also utilize the polar distance r. The Lagrangian can then be stated as

$$L = \frac{m\dot{r}^2}{2} + \frac{mr^2\dot{\theta}^2}{2} - mgr\cos(\theta). \tag{2-197}$$

The constraint can be written as

$$\mathrm{d}r = 0. \tag{2-198}$$

Lagrange's equations thus yield

$$m\ddot{r} - mr\dot{\theta}^2 + mg\cos(\theta) = \lambda, \tag{2-199}$$
$$2m\dot{r}\dot{\theta}r + mr^2\ddot{\theta} - mgr\sin(\theta) = 0, \tag{2-200}$$

where λ is the force responsible for the constraint $\mathrm{d}r = 0$, understood to be the normal force imparted by the hoop to retain the particle in circular motion. The constraint condition yields

$$\dot{r} = 0, \tag{2-201}$$

which, when differentiated, yields

$$\ddot{r} = 0. \tag{2-202}$$

Examing Equation 2-200 with this constraint yields

$$mr^2\ddot{\theta} = mgr\sin(\theta) \implies \ddot{\theta} = \frac{g}{r}\sin(\theta). \tag{2-203}$$

Note that $\ddot{\theta}$ can be re-written via the chain rule as

$$\ddot{\theta} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\frac{\mathrm{d}\theta}{\mathrm{d}t}\right) = \frac{\mathrm{d}}{\mathrm{d}\theta}\left(\frac{\mathrm{d}\theta}{\mathrm{d}t}\right)\frac{\mathrm{d}\theta}{\mathrm{d}t} = \dot{\theta}\frac{\mathrm{d}\dot{\theta}}{\mathrm{d}\theta}. \tag{2-204}$$

The resulting differential equation can be re-written and integrated using the initial condition that the particle starts at rest at the top of the hoop:

$$\int_0^{\dot{\theta}} \dot{\theta}\,\mathrm{d}\dot{\theta} = \int_0^{\theta} \frac{g}{r}\sin(\theta)\,\mathrm{d}\theta \tag{2-205}$$
$$\dot{\theta}^2(\theta) = \frac{2g}{r}(1 - \cos(\theta)). \tag{2-206}$$

Turning to the normal force N, I can obtain an expression

$$N(\theta) = mg\cos(\theta) = mr\dot{\theta}^2 = mg\cos(\theta) - 2mg(1 - \cos(\theta)) = mg(3\cos(\theta) - 2). \tag{2-207}$$

I see then that the reaction of the hoop is to push against the particle, initially at a magnitude equivalent to gravity but lineary decreasing with resepct to the angle ϕ, until finally at $\theta = \arccos(2/3)$ the normal force vanishes. At this point, the direction of the force (outside of gravity) required to continue circular motion switches direction, and the hoop can no longer help; the particle hence flies off. This value of θ corresponds to a height of $5r/3$ above the ground.

Problem 2.12

Problem Statement

A uniform hoop of mass m and radius r rolls without slipping on a fixed cylinder of radius R. The only external force is that of gravity. If the smaller cylinder starts rolling form rest on top of the bigger cylinder, find by the method of Lagrange multipliers the point at which the hoop falls off the cylinder.

The constraint force of interest is the normal force between the cylinder and the hoop. Therefore, I will implicitize the rolling without slipping condition within my generalized coordinates. I could, for instance, utilize the angle θ and distance R of the point of contact of the hoop with the cylinder relative to the center of the cylinder, with $\theta = 0$ defined vertically. Utilizing the rolling without slipping condition, and noting that rotaional translation kinetic energy are equal in the case of the hoop, the Lagrangian is given as

$$L = T - V = mR^2\dot{\theta}^2 - mg(R+r)\cos(\theta). \tag{2-208}$$

The constraint can be expressed as $\mathrm{d}R = 0$, which yields $\dot{R} = 0$ and moreover $\ddot{R} = 0$. The Euler-Lagrange equations can then be written as

$$mg\cos(\theta) - 2mR^2\dot{\theta}^2 = \lambda \tag{2-209}$$

$$4mR\dot{R}\dot{\theta} + 2mR^2\ddot{\theta} - mg(R+r)\sin(\theta) = 0 \tag{2-210}$$

The constraint on $\dot{R}$ transforms Equation 2-210 into

$$\ddot{\theta} = \frac{g(R+r)}{2R^2}\sin(\theta). \tag{2-211}$$

I can find $\dot{\theta}^2(\theta)$ in exact analogy to Problem 2.1 as

$$\dot{\theta}^2(\theta) = \frac{g(R+r)}{2R^2}(1 - \cos(\theta)). \tag{2-212}$$

Substituting into Equation 2-209 and some algebraic simplification gives

$$\cos(\theta) = \frac{2R+2r}{3R+2r} \tag{2-213}$$

as the condition that $\lambda = 0$, i.e. the normal force vanishes and the hoop falls off. This translates to a height

$$h = R + \frac{2R+2r}{3R+2r}. \tag{2-214}$$

Note that this is greater than $5R/3$, the height at which the point particle would fall off; in the limit $r \to 0$, I recover $h = 5R/3$, as expected.

Problem 2.14

Problem Statement

In certain situations, particularly one-dimensional systems, it is possible to incorporate frictional effects without introducing the dissipation function. As an example, find the equations of motion for the Lagrangian

$$L = e^{\gamma t}\left[\frac{m\dot{q}^2}{2} - \frac{kq^2}{2}\right]. \tag{2-215}$$

How would you describe the system? Are there any constants of motion? Suppose a point transformation is made of the form

$$s = e^{\gamma t/2}q.^{a} \tag{2-216}$$

What is the effective Lagrangian in terms of s. What do these results say about the conserved quantities for the system?

[a] Golstein does not have this factor of $1/2$, but the equations of motion to the transformation he presented are not nearly as enlightening as for this transformation.

Applying the Euler-Lagrange equation yields

$$\ddot{q} + \gamma\dot{q} + \frac{k}{m}q = 0, \tag{2-217}$$

i.e. a damped simple harmonic oscillator in q. No constants of motion are immediately evident, since the energy function h is not conserved owing to the explicit dependence of L on t, and no cyclic coordinates are evident. However, the point transformation $s = e^{\gamma t/2}q$ can be implemented explicitly in the Lagrangian via

$$q = e^{-\gamma t/2}s, \tag{2-218}$$

$$\dot{q} = e^{-\gamma t/2}(\dot{s} - \frac{\gamma}{2}s). \tag{2-219}$$

The Lagrangian is therefore

$$L = e^{\gamma t}\left[\frac{me^{-\gamma t}}{2}\left(\dot{s}^2 - \gamma\dot{s}s + \frac{\gamma^2 s^2}{4}\right) - e^{-\gamma t}\left(\frac{ks^2}{2}\right)\right] \tag{2-220}$$

$$= \frac{m}{2}\dot{s}^2 - \frac{m\gamma s}{2}\dot{s} - \frac{k - m\gamma^2/4}{2}s^2 + \frac{m\gamma^2}{8}. \tag{2-221}$$

The corresponding Euler-Lagrange equation is given as

$$m\ddot{s} = -(k - m\gamma^2/4)s, \tag{2-222}$$

which is the undamped simple harmonic oscillator with a modified spring constant $k - m\gamma^2/4$. The undamped nature of the oscillator suggests a conserved quantity relating to energy, and indeed the energy function h will be conserved owing to no explicit time-dependence in the Lagrangian. Explicitly,

$$h(s, \dot{s}, t) = \dot{s}\frac{\partial L}{\partial \dot{s}} - L = \frac{m}{2}\dot{s}^2 + \frac{(k - m\gamma^2/4)}{2}s^2 \tag{2-223}$$

is constant. With the transformation equations

$$s = e^{\gamma t/2}q, \tag{2-224}$$

$$\dot{s} = e^{\gamma t/2}(\dot{q} + \gamma q/2), \tag{2-225}$$

the energy function h can be expressed as

$$h(q, \dot{q}, t) = e^{\gamma t}\left(\frac{m}{2}\dot{q}^2 + m\gamma q\dot{q} + \frac{k}{2}q^2\right), \tag{2-226}$$

which naturally is a conserved quantity of the system. That is, the term in the parentheses decays exponentially at a rate fixed by γ, with a scale determined by initial conditions.

Problem 2.15

Problem Statement

Show that if the potential in the Lagrangian contains velocity-dependent terms, the canonical momentum corresponding to a coordinate of rotation θ of the entire system is no longer the mechanical angular momentum L_θ but is given by

$$p_\theta = L_\theta - \sum_i \mathbf{n}\cdot\mathbf{r}_i \times \nabla_{v_i} U, \tag{2-227}$$

where ∇_v is the gradient operator in which the derivatives are with respect to the velocity components and $\mathbf{n}$ is a unit vector in the direction of rotation. If the forces are electromagnetic in character the canonical momentum is therefore

$$p_\theta = L_\theta + \sum_i \mathbf{n}\cdot\mathbf{r}_i \times \frac{q_i}{c}\mathbf{A}_i. \tag{2-228}$$

The canonical momentum is given as

$$p_\theta = \frac{\partial L}{\partial \dot{\theta}} = \frac{\partial T}{\partial \dot{\theta}} - \frac{\partial U}{\partial \dot{\theta}}. \tag{2-229}$$

It has already been shown in the text that for a coordinate θ corresponding to a total rotation of the system of particles that $\frac{\partial T}{\partial \dot{\theta}} = L_\theta$. I can expand the partial derivative of the latter term in accordance with the chain rule as

$$\frac{\partial U}{\partial \dot{\theta}} = \frac{\partial U}{\partial \dot{x}}\frac{\partial \dot{x}}{\partial \dot{\theta}} + \frac{\partial U}{\partial \dot{y}}\frac{\partial \dot{y}}{\partial \dot{\theta}} + \frac{\partial U}{\partial \dot{z}}\frac{\partial \dot{z}}{\partial \dot{\theta}}, \tag{2-230}$$

since $U = U(\mathbf{r}, \mathbf{v})$ and terms such as $\frac{\partial x}{\partial \dot{\theta}}$ must vanish. This can be rewritten as

$$\frac{\partial U}{\partial \dot{\theta}} = \frac{\partial U}{\partial \dot{x}}\frac{\partial x}{\partial \theta} + \frac{\partial U}{\partial \dot{y}}\frac{\partial y}{\partial \theta} + \frac{\partial U}{\partial \dot{z}}\frac{\partial z}{\partial \theta} = \frac{\partial \mathbf{r}}{\partial \theta}\cdot\nabla_v U. \tag{2-231}$$

This scalar quantity will be evaluated at the location of each particle, so that

$$\frac{\partial U}{\partial \dot{\theta}} = \sum_i \frac{\partial \mathbf{r}_i}{\partial \theta}\cdot\nabla_{v_i} U, \tag{2-232}$$

where i is understood to indicate an evlaluation of the function at the state of the particle i. A further reduction and vector identity yields

$$\frac{\partial U}{\partial \dot{\theta}} = \sum_i \mathbf{n}\times\mathbf{r}_i\cdot\nabla_{v_i} U = \sum_i \mathbf{n}\cdot\mathbf{r}_i\times\nabla_{v_i} U. \tag{2-233}$$

Substitution into Equation 2-229 thereby gives

$$p_\theta = L_\theta - \sum_i \mathbf{n}\cdot\mathbf{r}_i \times \nabla_{v_i} U, \tag{2-234}$$

as desired.

Problem 2.16

Problem Statement

It sometimes occurs that the generalized coordinates appear separately in the kinetic energy and the potential energy in such a manner that T and V may be written in the form

$$T = \sum_i f_i(q_i) \qquad \text{and} \qquad V = \sum_i V_i(q_i). \tag{2-235}$$

Show that Lagrange's equations then separate, and that the problem can always be reduced to quadratures.

The Euler-Lagrange equations are given as

$$2f_i(q_i)\ddot{q}_i + \frac{\mathrm{d}f_i}{\mathrm{d}q_i}\dot{q}_i^2 = -\frac{\mathrm{d}V_i}{\mathrm{d}q_i} \qquad i = 1, 2, \ldots \tag{2-236}$$

That is, for each particle i we have the differential equation

$$2f_i\ddot{q}_i + f_i'(q_i)\dot{q}_i^2 + V_i'(q_i) = 0, \tag{2-237}$$

where the ' denotes differentiation with respect to q_i. By analogy to the simple harmonic oscillator and its conservation of energy, I may multiply through by $\dot{q}_i$ to find

$$2f_i\ddot{q}_i\dot{q}_i + f_i'(q_i)\dot{q}_i^3 + V_i'(q_i)\dot{q}_i = 0, \tag{2-238}$$

and recognize that the left-hand side can be expressed as a time derivative as follows:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(f_i(q_i)\dot{q}_i^2 + V_i(q_i)\right) = 0. \tag{2-239}$$

That is,

$$f_i(q_i)\dot{q}_i^2 + V_i(q_i) = C, \tag{2-240}$$

where C is a constant. That is, total energy is conserved for each individual particle within the system. This can be rearranged as

$$\dot{q}_i = \pm\sqrt{\frac{C - V_i(q_i)}{f_i(q_i)}} \tag{2-241}$$

and thus

$$\int_{q_{i_0}}^{q_i} \sqrt{\frac{f_i(q_i)}{C - V_i(q_i)}}\,\mathrm{d}q_i = \int_0^t \mathrm{d}t = t, \tag{2-242}$$

where the $\pm$ is absorbed into the constants of integration determined by the initial conditions of the system.

Problem 2.17

Problem Statement

A point mass is constrained to move on a massless hoop of radius a fixed in a vertical plane that is rotating about the vertical axis with constant angular speed ω. Obtain the Lagrange equations of motion assuming the only external forces arise from gravity. What are the constants of motion? Show that if ω is greater than a critical value ω_0, there can be a solution in which the particle remains stationary on the hoop at a point other than at the bottom, but that if $\omega < \omega_0$, the only stationary point for the particle is at the bottom of the hoop. What is the value of ω_0?

The location of the particle can be described by spherical polar coordinates ρ, ϕ, and θ with the constraints that $\rho = a$ and $\dot{\phi} = \omega$. The only independent generalized coordinate is therefore the zenith angle θ. The Lagrangian can thereby be formed as

$$L = T - V = \frac{m}{2}(\dot{\rho}^2 + \rho^2\dot{\theta}^2 + \rho^2\sin^2(\theta)\dot{\phi}^2) - mg\rho\cos(\theta) = \frac{ma^2}{2}\left(\dot{\theta}^2 + \omega^2\sin^2(\theta)\right) - mga\cos(\theta). \quad (2\text{-}243)$$

The Lagrange equation is given as

$$\ddot{\theta} = \sin(\theta)\left(\omega^2\cos(\theta) + \frac{g}{a}\right). \quad (2\text{-}244)$$

With no cyclic variables, the conserved quantity of interest is the energy function h owing to the Lagrangian not being explicitly time-dependent. The energy function is given as

$$h = \dot{\theta}\frac{\partial L}{\partial \dot{\theta}} - L = \frac{ma^2}{2}\left(\dot{\theta}^2 - \omega^2\sin^2(\theta)\right) + mga\cos(\theta). \quad (2\text{-}245)$$

Note in particluar that $h \neq T + V$, so that we do not have a conservation of total energy statement, owing to the time-dependent nature of the constraint. The requirement for a staionary point is that of a stable equilibirum. To achieve equilibrium at some point θ_0, we need $\ddot{\theta}$ to vanish upon insertion of the particle at the point with $\dot{\theta} = 0$. Note that Equation 2-244 is not dependent on $\dot{\theta}$, so that we achieve equilibirum when $\sin(\theta) = 0$ or $\omega^2\cos(\theta) + \frac{g}{a} = 0$, yielding solutions

$$\theta = 0,\ \pi,\ \arccos\left(-\frac{g}{\omega^2 a}\right). \quad (2\text{-}246)$$

I first consider the case $\theta = 0$. Consider some small perturbation ϵ away from the equilibirum point. We then have

$$\ddot{\theta}(\epsilon) = \sin(\epsilon)\left(\omega^2\cos(\epsilon) + \frac{g}{a}\right) \approx \epsilon\left(\omega^2 + \frac{g}{a}\right) > 0, \quad (2\text{-}247)$$

so that the system tends away from the equilibirum. That is, $\theta = 0$ is an unstable equilibirum point and therefore not a stationary point. Next, consider $\theta = \pi$. A small perturbation ϵ away from the equilibirum corresponds to an angle $\pi - \epsilon$ and therefore

$$\ddot{\theta}(\pi - \epsilon) = \sin(\pi - \epsilon)\left(\omega^2\cos(\pi - \epsilon) + \frac{g}{a}\right) \approx \epsilon\left(\frac{g}{a} - \omega^2\right). \quad (2\text{-}248)$$

The condition for stable equilibirum at the bottom of the hoop is therefore achieved when $\ddot{\theta}(\pi - \epsilon) > 0$ and thus exactly when $\omega < \omega_0$, where $\omega_0 \equiv \sqrt{\frac{g}{a}}$. Finally, we consider the final equilibirum point $\gamma \equiv \arccos\left(-\frac{g}{\omega^2 a}\right)$. We osberve that γ is well-defined only if $\omega > \omega_0$; a bifurcation therefore occurs at ω_0, with no equilibirum points for $\omega < \omega_0$. Observing that $\pi/2 < \gamma < \pi$ (discarding the non-robust degeneracy when $\gamma = \pi$), we have $\sin(\gamma) > 0$. A Taylor series expansion of the cosine function at γ yields

$$\ddot{\theta}(\gamma + \epsilon) = \sin(\gamma + \epsilon)\left(\omega^2\cos(\gamma + \epsilon) + \frac{g}{a}\right) \approx \sin(\gamma)\left(-\sin(\gamma)\epsilon\right) = -\epsilon\sin^2(\gamma), \quad (2\text{-}249)$$

which has the desired features of a stable equilibirum and hence a stationary point. In light of the nature of spherical polar coordinates we conclude a supercritical pitchfork bifurcation occurs at ω_0.

Problem 2.18

Problem Statement

A particle moves in a conservative field of force produced by various mass distributions. In each instance the force generated by a volume element of the distribution is derived from a potential that is proportional to the mass of the volume element and is a function only of the scalar distance from the volume element. For the following fixed, homogeneous mass distributions, state the conserved qunatities in the motion of the particle:

a) The mass is uniformly distributed in the plane $z = 0$.

b) The mass is uniformly distributed in the half-plane $z = 0$, $y > 0$.

c) The mass is uniformly distributed in a circular cylinder of infinite length, with axis along the z axis.

d) The mass is uniformly distributed in a circular cyclinder of finite length, with axis along the z axis.

e) The mass is uniformly distributed in a right cylinder of elliptical cross section and infinite length, with axis along the z axis.

f) The mass is uniformly distributed in a dumbbell whose axis is oriented along the z axis.

g) The mass is in the form of a uniform wire wound in the geometry of an infinite helical solenoid, with axis along the z axis.

a) Coordinate translation in the x and y directions, as well as rotation about the z axis, yield a formally identitical Lagrangian. Therefore, linear momentum in the x and y directions is conserved, as well as angular momentum in the z direction (about the origin). The potential is position-dependent, and no time-dependence is observed, so that h is the total energy; since h is conserved, total energy is also conserved.

b) The potential and hence the Lagrangian is invariant under a translation in the x direction, so that linear momentum in the x direction is conserved. Total energy is conserved once again.

c) The potential is invariant under a translation in the z direction, so that z linear momentum is conserved. The potential is also invariant under a rotation about the z axis, so that the z angular momentum is conserved. Total energy is conserved.

d) The potential is invariant under a rotation about the z axis, so that the z angular momentum is conserved. Total energy is conserved.

e) The potential is invariant under a translation in the z direction, so that z linear momentum is conserved. A general rotation does not conserve the Lagrangian, however, a rotation of $n\pi$ does (so that z angular momenutm is conserved in this case). Total energy is conserved.

f) The potential is invariant under a rotation in the z direction, so that z angular momentum is conserved. Total energy is conserved.

g) The potential is invariant under a translation in the z direction, so that z linear momentum is conserved. A sufficiently tightly wound solenoid also yields a potential invariant under a z translation so that z linear momentum is conserved. Total energy is conserved.

Problem 2.19

Problem Statement

A particle of mass m slides without friction on a wedge of angle α and mass M that can move without friction on a smooth horizontal surface. Treating the constraint of the particle on the wedge by the method of Lagrange multipliers, find the equations of motion for the particle and wedge. Also obtain an expression for the forces of constraint. Calculate the work done in time t by the forces of constraint acting on the particle and on the wedge. What are the constants of motion for the system? Contrast the results you have found with the situation when the wedge is fixed.

We will denote X as the horizontal position of the left-most part of the wedge. Relative to this corner, we will use polar coordinates r and θ to specify the position of the particle. The system is subject to the constraint that $\theta = \alpha$, and thus that $\mathrm{d}\theta = 0$; we will nonetheless compute the general Lagrangian with θ as a coordinate in accordance with the method of Lagrange multipliers. The velocity of the particle can be determined by components in the horizontal and vertical directions x and y. Explicitly,

$$v_x = \dot{X} + \dot{r}\cos(\theta) - \dot{\theta} r \sin(\theta), \tag{2-250}$$

$$v_y = \dot{r}\sin(\theta) + \dot{\theta} r \cos(\theta), \tag{2-251}$$

and $v^2 = v_x^2 + v_y^2$. The velocity of the wedge is given entirely as $\dot{X}$. Defining the potential energy to be 0 at the ground, we have $V_m = mgr\sin(\theta)$ while we neglect V_M since it is a compleletly additive term, with no dependence on r, θ, or their deriatives (it *is* dependent on α but not on θ since it is depdenent only on the angle of the incline, not on the angle formed by the particle). The Lagrangian can then be given as

$$L = \frac{M+m}{2}\dot{X}^2 + \frac{m}{2}\dot{r}^2 + \frac{m}{2}r^2\dot{\theta}^2 + m\dot{X}\dot{r}\cos(\theta) - m\dot{X}\dot{\theta}r\sin(\theta) - mgr\sin(\theta). \tag{2-252}$$

Computing the Lagrange equation for x and imposing the conditions $\theta = \alpha$, $\dot{\theta} = 0$, and $\ddot{\theta} = 0$ all stemming from the constraint yields

$$\ddot{X} = -\frac{m\cos(\theta)}{M+m}\ddot{r} \equiv -A\ddot{r}. \tag{2-253}$$

Similarly, for θ we derive

$$m\ddot{r} + m\ddot{X}\cos(\theta) + mg\sin(\theta) = 0, \tag{2-254}$$

which, upon insertion of Equation 2-253, yields

$$\ddot{r} = -\frac{(M+m)g\sin(\alpha)}{M + m(1+\cos^2(\alpha))} \equiv -B. \tag{2-255}$$

The expression for $\ddot{x}$ can then be explicitly treated as

$$\ddot{X} = AB. \tag{2-256}$$

Note that A and B are constants of the parameters of the system, and hence X and r are described entirely with constant acceleration. For θ, we include the force of constraint λ via the method of Lagrange multipliers and find

$$\lambda = mr\left(AB\sin(\alpha) - g\cos(\alpha)\right) \tag{2-257}$$

This can be interpreted as the torque necessary to keep the particle on the wedge, and since the normal force F_N is perpendicular to the distance vector $\mathbf{r}$, we have

$$F_N = \frac{\lambda}{r} = m(AB\sin(\alpha) - g\cos(\alpha)) = mg\cos(\alpha)\left(1 - \frac{m\sin^2(\alpha)}{M + m(1+\cos^2(\alpha))}\right). \tag{2-258}$$

This can be shown to go to 0 at the vertical wedge $\alpha_0 = \pi/2$, which is when the wedge can no longer constrain the particle. The normal force is constant over time so that the work done by the constraint on the wedge is given as

$$W = \int \mathbf{F}_N \cdot \mathrm{d}\mathbf{X} = mg\cos^2(\alpha)\left(1 - \frac{m\sin^2(\alpha)}{M + m(1+\cos^2(\alpha))}\right)\int_{X_0}^{X} \mathrm{d}X \tag{2-259}$$

$$= mg\cos^2(\alpha)\left(1 - \frac{m\sin^2(\alpha)}{M + m(1+\cos^2(\alpha))}\right)\int_0^t \dot{X}_0 + ABt\,\mathrm{d}t \tag{2-260}$$

$$= \frac{m^2g^2\sin(\alpha)\cos^2(\alpha)(M + m(1+\cos(2\alpha)))}{2(M + m(1+\cos^2(\alpha)))^2}t^2 \tag{2-261}$$

$$+ mg\dot{X}_0\cos(\alpha)\left(1 - \frac{m\sin^2(\alpha)}{M + m(1+\cos^2(\alpha))}\right)t. \tag{2-262}$$

In the case that $\dot{X}_0 = 0$, the work done by the constraint on the wedge simplifies to

$$W = \frac{m^2g^2\sin(\alpha)\cos^2(\alpha)(M + m(1+\cos(2\alpha)))}{2(M + m(1+\cos^2(\alpha)))^2}t^2, \tag{2-263}$$

scaling quadratically with t. There is no work done by the constraining force on the particle since it is at all times normal to the velocity vector. We observe that the energy function is conserved and is indeed the total energy:

$$h = T + V = \frac{M+m}{2}\dot{X}^2 + \frac{m}{2}\dot{r}^2 + m\dot{X}\dot{r}\cos(\theta) + mgr\sin(\theta). \tag{2-264}$$

The X coordinate is ignorable so that the generalized momentum conjugate $p_X = \frac{\partial L}{\partial \dot{X}}$ is constant. Since $\mathrm{d}X$ maps to a translation of the entire system in the X direction, we conclude that linear momentum in the X direction is conserved, as we would have derived using Newtonian methods. The case of the fixed wedge should be found in the limit $\frac{M}{m} \to \infty$. We observe that in this limit

$$\ddot{r} \to -g\sin(\alpha), \tag{2-265}$$

which is the standard result. This directly implies

$$\ddot{X} \to 0, \tag{2-266}$$

as is required. Similarly,

$$F_N \to mg\cos(\alpha), \tag{2-267}$$

and

$$W \to 0, \tag{2-268}$$

as we would expect.

Problem 2.20

Problem Statement

A carriage runs along rails on a rigid beam. The carriage is attached to one end of a spring, equilibirum length r_0 and force constant K, whose other end is fixed on the beam. On the carriage there is another set of rails perpendicular to the first along which a particle of mass m moves, held by a spring fixed on the beam, of force constant k and zero equilibirum length. Beams, rails, springs and carriage are assumed to have zero mass. The whole system is forced to move in a plane about the point of attachment of the first spring, with a constant angular speed ω. The length of the second spring is at all times considered small compared to r_0.

a) What is the energy of the system? Is it conserved?

b) Using generalized coordinates in the laboratory system what is the Jacobi integral for the system? Is it conserved?

c) In terms of generalized coordinates relative to a system rotating with the angular speed ω, what is the Lagrangian? What is the Jacobi integral? Is it conserved? Discuss the relationship between the two Jacobi integrals.

a) The kinetic energy is determined by that of the particle, while the potential is given as the sum of that of the two springs. Since the system is driven externally in a time-dependent fashion, I do not expect energy to be conserved.

b) I will use the generalized coordiante x_0 to denote the distance from the first spring to the carriage, and x_1 the distance from the particle from the carriage. Using polar coordinates, the radial velocity can be given as $\dot{x}_0$, while the perpendicular velocity as $\dot{x}_1 + \omega$; we have treated the particle as essentially localized at the carriage in applying this resolution, owing to the vanishing of x_1 relative to x_0. The kinetic energy is therefore

$$T = \frac{m}{2}(\dot{x}_0^2 + \dot{x}_1^2 + 2\omega x_0 \dot{x}_1 + \omega^2 x_0^2). \tag{2-269}$$

The potential is amenable to our choice of generalized coordinates and is given as

$$V = \frac{K}{2}(x_0 - r_0)^2 + \frac{k}{2}x_1^2, \tag{2-270}$$

so that

$$L = \frac{m}{2}(\dot{x}_0^2 + \dot{x}_1^2 + 2\omega x_0 \dot{x}_1 + \omega^2 x_0^2) - \frac{K}{2}(x_0 - r_0)^2 - \frac{k}{2}x_1^2. \tag{2-271}$$

The Jacobi integral is thereby given as

$$h = \frac{m}{2}(\dot{x}_0^2 + \dot{x}_1^2 - \omega^2 x_0^2) + \frac{K}{2}(x_0 - r_0)^2 + \frac{k}{2}x_1^2 \tag{2-272}$$

and is conserved since L does not depend explicitly upon t.

c) The kinetic energy is now independent of the angular velocity ω and is given as

$$T = \frac{m}{2}(\dot{x}_0^2 + \dot{x}_1^2). \tag{2-273}$$

With the potential unchanged, the Lagrangian is

$$L = \frac{m}{2}(\dot{x}_0^2 + \dot{x}_1^2) - \frac{K}{2}(x_0 - r_0)^2 - \frac{k}{2}x_1^2. \tag{2-274}$$

The Jacobi integral is thus

$$h = \frac{m}{2}(\dot{x}_0^2 + \dot{x}_1^2) + \frac{K}{2}(x_0 - r_0)^2 + \frac{k}{2}x_1^2 = T + V, \tag{2-275}$$

and is conserved since L is not explicitly dependent on t. This apparent mismatch emphasizes that the transformation to generalized coordinates has to this point been exclusively within the original reference frame. Clearly T is dependent on the refefrence frame present, however a translation to another inertial reference frame will not cause such disagreement as to the conservation of energy. The discrepancy arises because the modified reference frame considered here is *non-inertial*, and hence has an associated kinetic energy associated with the frame itself.

Problem 2.21

Problem Statement

Suppose that a particle moves in a space subject to a conservative potential $V(\mathbf{r})$ but is constrained always to move on a surface whose equation is $\sigma(\mathbf{r}, t) = 0$. The instantaneous force of constraint is taken as always perpendicular to the surface. Show analytically that the energy of the particle is not conserved if the surface moves in time. What physically is the reason for nonconservation of energy under this circumstance?

The truth of this statement can be most readily seen in the classical work-energy analysis framework. First, consider the case of a static surface. In the absence of the constraining force, the work done over a differential distance is given as

$$\Delta T = \mathbf{F} \cdot \mathrm{d}\mathbf{s} = -\frac{\partial V}{\partial s}\,\mathrm{d}s = -\Delta V \tag{2-276}$$

by the definition of the conservative V. The addition of the constraining force does not change the right-hand side, since any displacement on the surface is necessarily locally parallel to the surface (assuming it exhibits “nice" behavior) and therefore parallel to the constraining force (which is perpendicular to the surface), causing the dot product to vanish. Therefore, the total energy $T + V$ must be conserved. Now consider a surface which is moving over time. We emphasize that this need not be strictly translation, but also rotations or any sort of well-behaved mophological movement. The work-energy theorem yields that

$$\Delta T = \mathbf{F} \cdot \mathrm{d}\mathbf{s} = \mathbf{F}_{\text{conservative}} \cdot \mathrm{d}\mathbf{s} + \mathbf{F}_{\text{constraint}} \cdot \mathrm{d}\mathbf{s} = -\Delta V + \mathbf{F}_{\text{constraint}} \cdot \mathrm{d}\mathbf{s}. \tag{2-277}$$

But now note that the morphology of the surface constributes to the $\mathrm{d}\mathbf{s}$, so that it need no longer be strictly perpendicular to the constraining force; the dot product then in general is non-vanishing and

$$\Delta E = \Delta T + \Delta V = \mathbf{F}_{\text{constraint}} \cdot \mathrm{d}\mathbf{s} \neq 0. \tag{2-278}$$

The system could alternatively be modeled as having an additional potential term dependent explicitly both on position and time in which the morpohology of the surface adds a time-dependent potential term to the Lagrangian to derive the correct instantaneous forces acting on the particle, in which case the Lagrangian is explicitly time-dependent; this again shows that total energy is not conserved (since the energy function h is indeed the total energy in this case).

Acknolwedgements

2.4 To compute Equation 2-107 I first just asked an LLM to compute it, as I did not know how to deal with the cosine terms. The so-called Weirstrass substitution method was mentioned in the solution; I then decided to learn the basics behind this integration method and then carried out the integral without reference to the provided substitution. I used Mathematica to simplify Equation 2-121.

2.6 I inititally tried to solve the integral in Equation 2-150 via integration by parts, coming up with the change of variables $u = \text{arcCosh}(x/a)$. When trying to compare my result to the known result, LLM showed me I could just apply the change of variables to the original intergral, which I then did.

2.11 I initially derived the functional form of $\dot{\theta}^2(\theta)$ from conservation of energy. However, I was not satisfied by this and in looking for an alternative solution found the nifty trick to obtain $\dot{\theta}^2$ directly from $\ddot{\theta}$ (https://edubirdie.com/docs/eastern-kentucky-university/phy-201-university-physics-i/119534-homework-3-2-13-2-14). The rest of the derivation was carried out myself.

To-do

2.16 – figure out why we can absorb $\pm$ into integral

2.20 – revise - not true right now since $T \neq T_2$. I think the time-dependent part should be in the potential. Also think about extending to the more general case.

3 The Two-Body Central Force Problem

4 The Kinematics of Rigid Body Motions

Solutions

Problem 4.1

Problem Statement

Prove that matrix multiplication is associative. Show that the product of two orthogonal matrices is also orthogonal.

We aim to show that for arbitrary $m_1 \times m_2$ matrix $\mathbf{A}$, $m_2 \times m_3$ matrix $\mathbf{B}$, and $m_3 \times m_4$ matrix $\mathbf{C}$, where m_1, m_2,m_3, and m_4 are arbitrary integers, satisfies

$$(\mathbf{AB})\mathbf{C} = \mathbf{A}(\mathbf{BC}) \tag{4-1}$$

This can be shown readily by direct computation. We have

$$\mathbf{AB} = \begin{pmatrix} a_{11} & \dots & a_{1m_2} \\ \vdots & \ddots & \vdots \\ a_{m_1 1} & \dots & a_{m_1 m_2} \end{pmatrix} \begin{pmatrix} b_{11} & \dots & b_{1m_3} \\ \vdots & \ddots & \vdots \\ b_{m_2 1} & \dots & b_{m_2 m_3} \end{pmatrix} \tag{4-2}$$

$$= \begin{pmatrix} a_{1i}b_{i1} & \dots & a_{1i}b_{im_3} \\ \vdots & \ddots & \vdots \\ a_{m_1 i}b_{i1} & \dots & a_{m_1 i}b_{im_3} \end{pmatrix} \tag{4-3}$$

where the summation over i is implicitly assumed, ranging from 1 to m_2. Right multiplication by $\mathbf{C}$ yields

$$(\mathbf{AB})\mathbf{C} = \begin{pmatrix} a_{1i}b_{i1} & \dots & a_{1i}b_{im_3} \\ \vdots & \ddots & \vdots \\ a_{m_1 i}b_{i1} & \dots & a_{m_1 i}b_{im_3} \end{pmatrix} \begin{pmatrix} c_{11} & \dots & c_{1m_4} \\ \vdots & \ddots & \vdots \\ c_{m_3 1} & \dots & c_{m_3 m_4} \end{pmatrix} \tag{4-4}$$

$$= \begin{pmatrix} a_{1i}b_{ij}c_{j1} & \dots & a_{1i}b_{ij}c_{jm_4} \\ \vdots & \ddots & \vdots \\ a_{m_1 i}b_{ij}c_{j1} & \dots & a_{m_1 i}b_{ij}c_{jm_4} \end{pmatrix} \tag{4-5}$$

where we have picked up an additional summation index j from 1 to m_3. Note that Einstein's summation notation is flexible enough to readily incorporate this additional index, which can be

verified by writing out the summation explicitly and absorbing the two summatino indices into the form of $\Sigma_{i,j}$. On the other hand,

$$\mathbf{BC} = \begin{pmatrix} b_{11} & \dots & b_{1m_3} \\ \vdots & \ddots & \vdots \\ b_{m_2 1} & \dots & b_{m_2 m_3} \end{pmatrix} \begin{pmatrix} c_{11} & \dots & c_{1m_4} \\ \vdots & \ddots & \vdots \\ c_{m_3 1} & \dots & c_{m_3 m_4} \end{pmatrix} \tag{4-6}$$

$$= \begin{pmatrix} b_{1j}c_{j1} & \dots & b_{1j}c_{jm_4} \\ \vdots & \ddots & \vdots \\ b_{m_2 j}c_{j1} & \dots & b_{m_2 j}c_{jm_4} \end{pmatrix} \tag{4-7}$$

where I have labeled the summation index j taken from 1 to m_3 for clarity in later comparisons. Thus,

$$\mathbf{A}(\mathbf{BC}) = \begin{pmatrix} a_{11} & \dots & a_{1m_2} \\ \vdots & \ddots & \vdots \\ a_{m_1 1} & \dots & a_{m_1 m_2} \end{pmatrix} \begin{pmatrix} b_{1j}c_{j1} & \dots & b_{1j}c_{jm_4} \\ \vdots & \ddots & \vdots \\ b_{m_2 j}c_{j1} & \dots & b_{m_2 j}c_{jm_4} \end{pmatrix} \tag{4-8}$$

$$= \begin{pmatrix} a_{1i}b_{ij}c_{j1} & \dots & a_{1i}b_{ij}c_{jm_4} \\ \vdots & \ddots & \vdots \\ a_{m_1 i}b_{ij}c_{j1} & \dots & a_{m_1 i}b_{ij}c_{jm_4} \end{pmatrix} = (\mathbf{AB})\mathbf{C} \tag{4-9}$$

where i is taken from 1 to m_2, completing the proof. For orthogonality, we will consider only $m \times m$ matrices for this is context in which the orthogonal transform functions. It will be of interest primarily for the 3×3 case but we will show it in general. We have

$$\mathbf{AB} = \begin{pmatrix} a_{1i}b_{i1} & \dots & a_{1i}b_{im} \\ \vdots & \ddots & \vdots \\ a_{mi}b_{i1} & \dots & a_{mi}b_{im} \end{pmatrix} \begin{pmatrix} c_{11} & \dots & c_{1m} \\ \vdots & \ddots & \vdots \\ c_{m1} & \dots & c_{mm} \end{pmatrix} = \mathbf{C} \tag{4-10}$$

where we will call our product matrix $\mathbf{C}$. If $\mathbf{A}$ and $\mathbf{B}$ are orthogonal, we want to show that $\mathbf{C}$ is orthogonal as well. We can expand the orthogonality condition as

$$c_{ij}c_{ik} = a_{tl}b_{lj}a_{tp}b_{pk} \tag{4-11}$$

where j and k are fixed and all other indices vary from 1 to m. By commutivity of scalars,

$$c_{ij}c_{ik} = a_{tl}a_{tp}b_{pk}b_{lj} = \delta_{pl}b_{pk}b_{lj} = b_{pk}b_{pj} = \delta_{jk} \tag{4-12}$$

where we have used the orthogonality of $\mathbf{A}$ to construct the δ function, and then use that to collapse the remaining sum into the orthognality condition for $\mathbf{B}$. Therefore, $\mathbf{C}$ is orthogonal as well.

Problem 4.2

Problem Statement

Prove the following properties of the transposed and adjoint matrices:

$$\widetilde{\mathbf{AB}} = \widetilde{\mathbf{B}}\widetilde{\mathbf{A}}, \tag{4-13}$$

$$(\mathbf{AB})^\dagger = \mathbf{B}^\dagger\mathbf{A}^\dagger \tag{4-14}$$

The properties can also readily be seen by direct computation. We first note that the transpose only acts on an $m \times m$ matrix. We can compute the transpose directly by

$$\widetilde{\mathbf{AB}} = \begin{pmatrix} a_{1i}b_{i1} & \widetilde{\cdots} & a_{1i}b_{im} \\ \vdots & \ddots & \vdots \\ a_{mi}b_{i1} & \cdots & a_{mi}b_{im} \end{pmatrix} = \begin{pmatrix} a_{1i}b_{i1} & \cdots & a_{mi}b_{i1} \\ \vdots & \ddots & \vdots \\ a_{1i}b_{im} & \cdots & a_{mi}b_{im} \end{pmatrix} \tag{4-15}$$

Now considering $\widetilde{\mathbf{B}}\widetilde{\mathbf{A}}$, we have

$$\widetilde{\mathbf{B}}\widetilde{\mathbf{A}} = \begin{pmatrix} a_{11} & \widetilde{\cdots} & a_{1m} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mm} \end{pmatrix} \begin{pmatrix} b_{11} & \widetilde{\cdots} & b_{1m} \\ \vdots & \ddots & \vdots \\ b_{m1} & \cdots & b_{mm} \end{pmatrix} = \begin{pmatrix} a_{11} & \cdots & a_{m1} \\ \vdots & \ddots & \vdots \\ a_{1m} & \cdots & a_{mm} \end{pmatrix} \begin{pmatrix} b_{11} & \cdots & b_{m1} \\ \vdots & \ddots & \vdots \\ b_{1m} & \cdots & b_{mm} \end{pmatrix} \tag{4-16}$$

$$= \begin{pmatrix} a_{1i}b_{i1} & \cdots & a_{mi}b_{i1} \\ \vdots & \ddots & \vdots \\ a_{1i}b_{im} & \cdots & a_{mi}b_{im} \end{pmatrix} \tag{4-17}$$

which completes the proof of the first property. Next, we can compute

$$(\mathbf{AB})^\dagger = \begin{pmatrix} a_{1i}b_{i1} & \cdots & a_{1i}b_{im} \\ \vdots & \ddots & \vdots \\ a_{mi}b_{i1} & \cdots & a_{mi}b_{im} \end{pmatrix}^\dagger = \begin{pmatrix} a^*_{1i}b^*_{i1} & \cdots & a^*_{mi}b^*_{i1} \\ \vdots & \ddots & \vdots \\ a^*_{1i}b^*_{im} & \cdots & a^*_{mi}b^*_{im} \end{pmatrix} \tag{4-18}$$

and accordingly

$$\mathbf{B}^\dagger\mathbf{A}^\dagger = \begin{pmatrix} b_{11} & \cdots & b_{1m} \\ \vdots & \ddots & \vdots \\ b_{m1} & \cdots & b_{mm} \end{pmatrix}^\dagger \begin{pmatrix} a_{11} & \cdots & a_{1m} \\ \vdots & \ddots & \vdots \\ a_{m1} & \cdots & a_{mm} \end{pmatrix}^\dagger = \begin{pmatrix} b^*_{11} & \cdots & b^*_{m1} \\ \vdots & \ddots & \vdots \\ b^*_{1m} & \cdots & b^*_{mm} \end{pmatrix} \begin{pmatrix} a^*_{11} & \cdots & a^*_{m1} \\ \vdots & \ddots & \vdots \\ a^*_{1m} & \cdots & a^*_{mm} \end{pmatrix} \tag{4-19}$$

$$= \begin{pmatrix} a^*_{1i}b^*_{i1} & \cdots & a^*_{mi}b^*_{i1} \\ \vdots & \ddots & \vdots \\ a^*_{1i}b^*_{im} & \cdots & a^*_{mi}b^*_{im} \end{pmatrix} \tag{4-20}$$

which completes the proof of the second property.

Problem 4.3

Problem Statement

Show that the trace of a matrix is invariant under any similarity transformation. Show also that the antisymmetry property of a matrix is preserved under an orthogonal similarity transformation, while the hermitean property is invariant under any unitary similarity transformation.

For the first property, we would like to perform trace-preserving actions on the objcet of interest. Unfortunately, the trace is distributive only under matrix addition, so that in general

$$\mathrm{Tr}(\mathbf{AB}) \neq \mathrm{Tr}(\mathbf{A})\,\mathrm{Tr}(\mathbf{B}) \tag{4-21}$$

The trace does admit the cyclic permutation property, which emerges from the group structure of matrices. The property can be stated as

$$\mathrm{Tr}(\mathbf{ABC}) = \mathrm{Tr}(\mathbf{CAB}) = \mathrm{Tr}(\mathbf{BCA}) \tag{4-22}$$

To prove this, it suffices to show that given some $\mathbf{AB}$, we can left or right multiply and get the same trace. We know that $\mathbf{AB}$ is just some matrix which we will call $\mathbf{D}$, so really the content of the proof follows from showing that

$$\mathrm{Tr}(\mathbf{CD}) = \mathrm{Tr}(\mathbf{DC}) \tag{4-23}$$

This can readily be shown explicitly by

$$\mathrm{Tr}(\mathbf{CD}) = \mathrm{Tr}\left[\begin{pmatrix} c_{11} & \dots & c_{1m} \\ \vdots & \ddots & \vdots \\ c_{m1} & \vdots & c_{mm} \end{pmatrix}\begin{pmatrix} d_{11} & \dots & d_{1m} \\ \vdots & \ddots & \vdots \\ d_{m1} & \vdots & d_{mm} \end{pmatrix}\right] \tag{4-24}$$

$$= \mathrm{Tr}\left[\begin{pmatrix} c_{1i}d_{i1} \dots & c_{1i}d_{im} \\ \vdots & \ddots & \vdots \\ c_{mi}d_{i1} & \vdots & c_{mi}d_{im} \end{pmatrix}\right] \tag{4-25}$$

$$= c_{ji}d_{ij} \tag{4-26}$$

and

$$\mathrm{Tr}(\mathbf{DC}) = \mathrm{Tr}\left[\begin{pmatrix} d_{11} & \dots & d_{1m} \\ \vdots & \ddots & \vdots \\ d_{m1} & \vdots & d_{mm} \end{pmatrix}\begin{pmatrix} c_{11} & \dots & c_{1m} \\ \vdots & \ddots & \vdots \\ c_{m1} & \vdots & c_{mm} \end{pmatrix}\right] \tag{4-27}$$

$$= \mathrm{Tr}\left[\begin{pmatrix} d_{1i}c_{i1} \dots & d_{1i}c_{im} \\ \vdots & \ddots & \vdots \\ d_{mi}c_{i1} & \vdots & d_{mi}c_{im} \end{pmatrix}\right] \tag{4-28}$$

$$= d_{ji}c_{ij} \tag{4-29}$$

since the labellings of the indices i and j is arbitrary. Therefore, given a matrix $\mathbf{A}$, the trace of a similarity transform facilitated by some $\mathbf{B}$ is given as

$$\mathrm{Tr}(\mathbf{A}') = \mathrm{Tr}(\mathbf{BAB}^{-1}) = \mathrm{Tr}(\mathbf{B}^{-1}\mathbf{BA}) = \mathrm{Tr}(\mathbf{A}) \tag{4-30}$$

Now consider an anti-symmetric matrix $\mathbf{A}$ under some similarity transformation facilitated by an orthogonal matrix $\mathbf{B}$. We can compute

$$\widetilde{\mathbf{A}'} = \widetilde{\mathbf{BAB}^{-1}} = \widetilde{\mathbf{B}^{-1}}\widetilde{\mathbf{A}}\widetilde{\mathbf{B}} = -\mathbf{BAB}^{-1} = -\mathbf{A}' \tag{4-31}$$

since $\widetilde{\mathbf{A}} = -\mathbf{A}$ and $\widetilde{\mathbf{B}} = \mathbf{B}^{-1}$. We can apply the transpose to this last identity to find

$$\widetilde{\widetilde{\mathbf{B}}} = \mathbf{B} = \widetilde{\mathbf{B}^{-1}} \tag{4-32}$$

since applying the transpose applied twice is simply the identity, hence verifying the identity. Given a hermitean $\mathbf{A}$, a similarlity transformation facilitated by a unitary $\mathbf{B}$ admits

$$\mathbf{A}'^{\dagger} = (\mathbf{BAB}^{-1})^{\dagger} = (\mathbf{B}^{-1})^{\dagger}\mathbf{A}^{\dagger}\mathbf{B}^{\dagger} = \mathbf{BAB}^{-1} = \mathbf{A}' \tag{4-33}$$

since $\mathbf{A}^{\dagger} = \mathbf{A}$, $\mathbf{B}^{\dagger} = \mathbf{B}^{-1}$, and $(\mathbf{B}^{-1})^{\dagger} = (\mathbf{B}^{-1})^{-1} = \mathbf{B}$ since $\mathbf{B}^{-1}$ is also unitary. This completes the proof.

Problem 4.4

Problem Statement

a) By examining the eigenvalues of an antisymmetric real 3×3 matrix $\mathbf{A}$, show that $\mathbf{1} \pm \mathbf{A}$ is nonsingular.

b) Show then that under the same conditions the matrix

$$\mathbf{B} = (\mathbf{1} + \mathbf{A})(\mathbf{1} - \mathbf{A})^{-1} \tag{4-34}$$

is orthogonal.

a) The most general form of $\mathbf{A}$ is given as

$$\mathbf{A} = \begin{pmatrix} 0 & a_1 & a_2 \\ -a_1 & 0 & a_3 \\ -a_2 & -a_3 & 0 \end{pmatrix} \tag{4-35}$$

The characteristic equation is

$$\det(\mathbf{A}-\lambda\mathbf{1})\begin{vmatrix} -\lambda & a_1 & a_2 \\ -a_1 & -\lambda & a_3 \\ -a_2 & -a_3 & -\lambda \end{vmatrix} = -\lambda(\lambda^2+a_3^2)+a_1(-a_2a_3-a_1\lambda)+a_2(a_1a_3-a_2\lambda) = -\lambda(\lambda^2+a_1^2+a_2^2+a_3^2) = 0, \tag{4-36}$$

which admits only the solution $\lambda = 0$, apparently with a multiplicity of 3. We can now evaluate the matrix $\mathbf{1} + \mathbf{A}$ which has the characteristic equation

$$\det(\mathbf{1} + \mathbf{A} - \lambda\mathbf{1})\begin{vmatrix} 1-\lambda & a_1 & a_2 \\ -a_1 & 1-\lambda & a_3 \\ -a_2 & -a_3 & 1-\lambda \end{vmatrix} = \begin{vmatrix} -\delta & a_1 & a_2 \\ -a_1 & -\delta & a_3 \\ -a_2 & -a_3 & -\delta \end{vmatrix} = 0, \tag{4-37}$$

where $\delta \equiv \lambda - 1$. This is exactly the same matrix to solve as in Equation 4-36, only with $\lambda \mapsto \delta$ and hence the solution is $\delta = 0$ and thus $\lambda = 1$ with a multiplicity of three. Therefore $\mathbf{1}+\mathbf{A}$ has a determinant of $\lambda_1\lambda_2\lambda_3 = 1$ and is nonsingular. The exact same argument can be made for $\mathbf{1}-\mathbf{A}$, where we can simply redefine $a_1 \mapsto -a_1$, $a_2 \mapsto -a_2$, and $a_3 \mapsto -a_3$ since they were arbitrary constants to begin with and as we will see we will not need to include a comparison between the two parity cases. The situation is hence identical and we have $\lambda = 1$ with a multiplicity of 3, and a total determinant of 1.

b) We want to show that $\mathbf{B}\widetilde{\mathbf{B}} = \mathbf{1}$, or equivalently (and as we will approach it) $\widetilde{\mathbf{B}}\mathbf{B} = \mathbf{1}$. The form can be given as

$$\widetilde{\mathbf{B}}\mathbf{B} = \widetilde{(\mathbf{1} - \mathbf{A})^{-1}}\widetilde{(\mathbf{1} + \mathbf{A})}(\mathbf{1} + \mathbf{A})(\mathbf{1} - \mathbf{A})^{-1} \tag{4-38}$$

Several lemmas need to be established before progress can be made. First, it can been from the matrix forms that $\widetilde{\mathbf{1} - \mathbf{A}} = \mathbf{1} + \mathbf{A}$, and analogously $\widetilde{\mathbf{1} + \mathbf{A}} = \mathbf{1} - \mathbf{A}$. We also note that for any matrix $\mathbf{C}$, it must be that $\widetilde{(\mathbf{C})^{-1}} = (\widetilde{\mathbf{C}})^{-1}$, for

$$\widetilde{(\mathbf{C})^{-1}}\widetilde{\mathbf{C}} = \widetilde{\mathbf{C}\mathbf{C}^{-1}} = \widetilde{\mathbf{1}} = \mathbf{1}. \tag{4-39}$$

We can use these to reduce as

$$\widetilde{\mathbf{B}}\mathbf{B} = \widetilde{(\mathbf{1}-\mathbf{A})}^{-1}(\mathbf{1}+\mathbf{A})(\mathbf{1}+\mathbf{A})(\mathbf{1}-\mathbf{A})^{-1} \tag{4-40}$$

$$= (\mathbf{1}+\mathbf{A})^{-1}(\mathbf{1}-\mathbf{A})(\mathbf{1}+\mathbf{A})(\mathbf{1}-\mathbf{A})^{-1} \tag{4-41}$$

It will now be shown that the matrices $(\mathbf{1}-\mathbf{A})$ and $(\mathbf{1}+\mathbf{A})$ are commutative. This is seen by direct computation as

$$(\mathbf{1}-\mathbf{A})(\mathbf{1}+\mathbf{A}) = \begin{pmatrix} 1+a_1^2+a_2^2 & a_2a_3 & -a_1a_3 \\ a_2a_3 & 1+a_1^2+a_3^2 & a_1a_2 \\ -a_1a_3 & a_1a_2 & 1+a_2^2+a_3^2 \end{pmatrix} = (\mathbf{1}+\mathbf{A})(\mathbf{1}-\mathbf{A}) \tag{4-42}$$

Thus,

$$\widetilde{\mathbf{B}}\mathbf{B} = (\mathbf{1}+\mathbf{A})^{-1}(\mathbf{1}+\mathbf{A})(\mathbf{1}-\mathbf{A})(\mathbf{1}-\mathbf{A})^{-1} = \mathbf{1} \tag{4-43}$$

which satisfies the orthogonality condition and hence completes the proof.

Problem 4.6

Problem Statement

a) Find the vector equation describing the reflection of $\mathbf{r}$ in a plane whose unit normal vector is $\mathbf{n}$.

b) Show that if l_i, $i = 1, 2, 3$, are the direction cosines of $\hat{\mathbf{n}}$, then the matrix of transformation has the elements

$$A_{ij} = \delta_{ij} - 2l_il_j, \tag{4-44}$$

and verify that $\mathbf{A}$ is an improper orthogonal matrix.

a) Consider the action of reflection on $\mathbf{r}$. This can be broken into two equal steps – the projection onto the plane, and then the reflection process itself, which can be seen as the repeated action of the projection. Therefore, the initial projection is all which needs to be accounted for, and then applied again.

The projection action is equivalent to the displacement vector of shortest length from the plane. That vector is therefore proportional to $\hat{\mathbf{n}}$, and with the plane equation $n_xx + n_yy + n_zz = 0$, the magntiude is given as

$$d = \frac{n_xr_x + n_yr_y + n_zr_z}{\sqrt{n_x^2+n_y^2+n_z^2}} = \hat{\mathbf{n}}\cdot\mathbf{r} \tag{4-45}$$

The transformation is then

$$\mathbf{r} \mapsto \mathbf{r} - (\hat{\mathbf{n}}\cdot\mathbf{r})\hat{\mathbf{n}} \tag{4-46}$$

where the dot product $\hat{\mathbf{n}}\cdot\mathbf{r}$ automatically accounts for the desired direction of transformation. Applying this twice thereby yields

$$\mathbf{r} \mapsto \mathbf{r} - 2(\hat{\mathbf{n}}\cdot\mathbf{r})\hat{\mathbf{n}} \tag{4-47}$$

which is the desired vector form.

b) In explicit form the transformations can be written as

$$x \mapsto x - 2(n_x^2 x + n_x n_y y + n_x n_z z), \tag{4-48}$$
$$y \mapsto y - 2(n_x n_y x + n_y^2 y + n_y n_z z), \tag{4-49}$$
$$z \mapsto z - 2(n_x n_z x + n_y n_z y + n_z^2 x), \tag{4-50}$$

Equivalently, this can be written as

$$\mathbf{r} \mapsto \mathbf{A}\mathbf{r}, \tag{4-51}$$

where

$$\mathbf{A}_{ij} := \delta_{ij} - 2n_i n_j \tag{4-52}$$

where i, j iterate over x, y, and z in an ordered fashion. Noting that n_x, n_y, and n_z may just as well defined as direction cosines of $\hat{\mathbf{n}}$ on the standard coordinate system, we have shown the desired transformation matrix. By direction computation, and the useful property $n_x^2+n_y^2+n_z^2 = 1$, it can be shown that

$$\mathbf{A}\widetilde{\mathbf{A}} = \mathbf{1} \tag{4-53}$$

and hence that $\mathbf{A}$ is orthogonal. By direct computation again, and with the substitution $n_x \mapsto a$, $n_y \mapsto b$, $n_z \mapsto c$ for notational convenience,

$$\begin{aligned}
\det \mathbf{A} &= (1-2a^2)\left[(1-2b^2)(1-2c^2) - 4b^2c^2\right] && (4\text{-}54)\\
&\quad - 2ab\left[4abc^2 + (2ab)(1-2c^2)\right] && (4\text{-}55)\\
&\quad - 2ac\left[4ab^2c + 2ac(1-2b^2)\right] && (4\text{-}56)\\
&= (1-2a^2)(1-2b^2-2c^2) - (2ab)(2ab) - (2ac)(2ac) && (4\text{-}57)\\
&= 1 - 2b^2 - 2c^2 - 2a^2 + 4a^2b^2 + 4a^2c^2 - 4a^2b^2 - 4a^2c^2 && (4\text{-}58)\\
&= -1 && (4\text{-}59)
\end{aligned}$$

which demonstrates that it is improper in nature.

Problem 4.7

Problem Statement

The body set of axes can be related to the space set in terms of Euler's angles by the following set of rotations:

1) Rotation about the x-axis by an angle θ
2) Rotation about the z'-axis by an angle ψ
3) Rotation about the *old* z-axis by an angle ϕ

Show that this sequence leads to the same elements of the matrix of transformation as the sequence of rotations given in the book.

Denoting a rotation $\mathbf{Z}(\tau)$ of τ radians about a fixed z-axis by $\mathbf{Z}_{\text{og}}(\tau)$, we will first show more broadly that for any $\mathbf{O} \in \mathbf{SO}(3)$, it follows that $\mathbf{Z}_{\text{og}}(\tau)\mathbf{O} = \mathbf{O}\mathbf{Z}_{\text{og}}(\tau)$, i.e. that they commute. To see this, we first consider $\mathbf{Z}_{\text{og}}(\tau)\mathbf{O}$. Defining $\mathbf{O}$ generically as

$$\mathbf{O} = \begin{pmatrix} O_{11} & O_{12} & O_{13} \\ O_{21} & O_{22} & O_{23} \\ O_{31} & O_{32} & O_{33} \end{pmatrix}, \tag{4-60}$$

the action of $\mathbf{O}$ can be understood as defining new unit vectors

$$\hat{\mathbf{x}}' := O_{11}\hat{\mathbf{x}} + O_{12}\hat{\mathbf{y}} + O_{13}\hat{\mathbf{z}}, \tag{4-61}$$
$$\hat{\mathbf{y}}' := O_{21}\hat{\mathbf{x}} + O_{22}\hat{\mathbf{y}} + O_{23}\hat{\mathbf{z}}, \tag{4-62}$$
$$\hat{\mathbf{z}}' := O_{31}\hat{\mathbf{x}} + O_{32}\hat{\mathbf{y}} + O_{33}\hat{\mathbf{z}}. \tag{4-63}$$

Now the action of $\mathbf{Z}_{\text{og}}(\tau)$ is specified on the unit vectors $\hat{\mathbf{x}}$, $\hat{\mathbf{y}}$, $\hat{\mathbf{z}}$ via

$$\hat{\mathbf{x}} \mapsto \cos(\tau)\hat{\mathbf{x}} + \sin(\tau)\hat{\mathbf{y}}, \tag{4-64}$$
$$\hat{\mathbf{y}} \mapsto -\sin(\tau)\hat{\mathbf{x}} + \cos(\tau)\hat{\mathbf{y}}, \tag{4-65}$$
$$\hat{\mathbf{z}} \mapsto \hat{\mathbf{z}}, \tag{4-66}$$

so that the full action of $\mathbf{Z}_{\text{og}}(\tau)$ can be understood as defining new unit vectors

$$\hat{\mathbf{x}}\text{"} := O_{11}(\cos(\tau)\hat{\mathbf{x}} + \sin(\tau)\hat{\mathbf{y}}) + O_{12}(-\sin(\tau)\hat{\mathbf{x}} + \cos(\tau)\hat{\mathbf{y}}) + O_{13}\hat{\mathbf{z}}, \tag{4-67}$$
$$\hat{\mathbf{y}}\text{"} := O_{21}(\cos(\tau)\hat{\mathbf{x}} + \sin(\tau)\hat{\mathbf{y}}) + O_{22}(-\sin(\tau)\hat{\mathbf{x}} + \cos(\tau)\hat{\mathbf{y}}) + O_{23}\hat{\mathbf{z}}, \tag{4-68}$$
$$\hat{\mathbf{z}}\text{"} := O_{31}(\cos(\tau)\hat{\mathbf{x}} + \sin(\tau)\hat{\mathbf{y}}) + O_{32}(-\sin(\tau)\hat{\mathbf{x}} + \cos(\tau)\hat{\mathbf{y}}) + O_{33}\hat{\mathbf{z}}. \tag{4-69}$$

which finally can be put into a matrix form as

$$\begin{pmatrix} O_{11}\cos(\tau) - O_{12}\sin(\tau) & O_{11}\sin(\tau) + O_{12}\cos(\tau) & O_{13} \\ O_{21}\cos(\tau) - O_{22}\sin(\tau) & O_{21}\sin(\tau) + O_{22}\cos(\tau) & O_{13} \\ O_{31}\cos(\tau) - O_{32}\sin(\tau) & O_{31}\sin(\tau) + O_{32}\cos(\tau) & O_{33} \end{pmatrix}. \tag{4-70}$$

We now consider $\mathbf{O}\mathbf{Z}_{\text{og}}(\tau)$. We note that since here the standard matrix form of $\mathbf{Z}(\tau)$ may be applied since it acts on the original state anyways, i.e. $\mathbf{O}\mathbf{Z}_{\text{og}}(\tau) = \mathbf{O}\mathbf{Z}(\tau)$. The matrix form may now be explicitly computed as

$$\mathbf{O}\mathbf{Z}(\tau) = \begin{pmatrix} O_{11} & O_{12} & O_{13} \\ O_{21} & O_{22} & O_{23} \\ O_{31} & O_{32} & O_{33} \end{pmatrix} \begin{pmatrix} \cos(\tau) & \sin(\tau) & 0 \\ -\sin(\tau) & \cos(\tau) & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{4-71}$$
$$= \begin{pmatrix} O_{11}\cos(\tau) - O_{12}\sin(\tau) & O_{11}\sin(\tau) + O_{12}\cos(\tau) & O_{13} \\ O_{21}\cos(\tau) - O_{22}\sin(\tau) & O_{21}\sin(\tau) + O_{22}\cos(\tau) & O_{13} \\ O_{31}\cos(\tau) - O_{32}\sin(\tau) & O_{31}\sin(\tau) + O_{32}\cos(\tau) & O_{33} \end{pmatrix} \tag{4-72}$$

in agreement with Equation 4-70. Although not essential for the problem at hand, this proof can be easily extended from the specific $\mathbf{Z}_{\text{og}}(\tau)$ to any arbitrary rotation $\mathbf{O}_{\text{og}} \in \mathbf{SO}(3)$ for nothing specific about the functional form of $\mathbf{Z}_{\text{og}}(\tau)$ was essential to the proof. We have $\mathbf{Z}(\psi)\mathbf{X}(\theta) \in \mathbf{SO}(3)$ by the closure of $\mathbf{SO}(3)$ and hence

$$\mathbf{Z}_{\text{og}}(\phi)\mathbf{Z}(\psi)\mathbf{X}(\theta) = \mathbf{Z}(\psi)\mathbf{X}(\theta)\mathbf{Z}_{\text{og}}(\phi) = \mathbf{Z}(\psi)\mathbf{X}(\theta)\mathbf{Z}(\phi) \tag{4-73}$$

which is exactly the property we wished to show.

Problem 4.8

Problem Statement

If $\mathbf{A}$ is the matrix of a rotation through 180° about any axis, show that if

$$\mathbf{P}_{\pm} = \frac{1}{2}(\mathbf{1} \pm \mathbf{A}), \tag{4-74}$$

then $\mathbf{P}_{\pm}^2 = \mathbf{P}_{\pm}$. Obtain the elements of $\mathbf{P}_{\pm}$ in any suitable system, and find a geometric interpretation of the operation $\mathbf{P}_+$ and $\mathbf{P}_-$ on any vector $\mathbf{F}$.

By the distributivity of matrix multipplication under addition, and the commutation property of $\mathbf{1}$, we have

$$\mathbf{P}_{\pm}^2 = \frac{1}{4}(\mathbf{1} \pm \mathbf{A})(\mathbf{1} \pm \mathbf{A}) = \frac{1}{4}(2\mathbf{1} \pm 2\mathbf{A}) = \mathbf{P}_{pm} \tag{4-75}$$

Taking $\mathbf{A}$ to rotate about the z-axis, the matrix form is

$$\mathbf{A} = \begin{pmatrix} -1 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{4-76}$$

by the standard formula. We thus have by direct computation

$$\mathbf{P}_+ = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{4-77}$$

and

$$\mathbf{P}_- = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{4-78}$$

Therefore, $\mathbf{P}_+$ can be interpreted as a projection onto the axis of rotation, while $\mathbf{P}_-$ can be interpreted as a projection on the plane which is rotating (which includes the origin).

References

[1] Herbert Goldstein. *Classical Mechanics*. Addison-Wesley, 2nd edition, 1980.

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